

1 **Title: Genomic analyses reveal distinct demographic history and identify**
2 **genetic changes underlying adaption, speciation and domestication in ducks**

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19 **Abstract (250 words)**

20 **Background:** Domestic ducks was considered to being tamed from the mallard or the
21 descendant of the mallard and the spot-billed duck. Domestic ducks showed remarkable
22 phenotypic variation in morphology, physiology and behavior. However, molecular
23 genetics for ducks' origin and their phenotypic variation are still poorly studied.

24 **Results:** Here we presented the mallard and the spot-billed genomes and performed
25 whole-genome sequencing on eight domestic duck breeds and eight wild duck species.
26 Surprisingly, Analyses on these data supports a model that domestic ducks diverged
27 from their closely wild lineage (mallard duck and spot-billed duck) at the last glacial
28 period (LGP, 100~200 kilo years ago (Kyr)). Wild lineage further speciated into mallard
29 duck and spot-billed duck about 70 Kyr, whereas domestic lineage decreased their
30 population size through the LGP. A scan of wild duck genomes by comparing with
31 domestic duck genomes identified a number of locus that may have been affected by
32 positive selection in ancestral wild ducks after their divergence from domestic lineages.
33 Function analyses suggested that those genes usually affecting organs development and
34 energy metabolism that may involving long-distance flight ability. Further selective
35 sweep analyses identified two genes associated with egg production and three genes
36 related to feeding modulation being under selection in domestic ducks.

37 **Conclusions:** These analyses unravel a distinct evolutionary pattern of ducks and two
38 wild duck *de novo* genomes also provide a novel resource for speciation study.

39 **Key Words:** *De novo* assembly; Genomic analysis; Origin, speciation and
40 domestication; Duck

Background

As two major concerns of evolutionary study, both domestication and speciation (or population divergence) benefited from the advance of sequencing technology(1-9). The genome assembly and characterizing of genetic variation in many domestic and wild animals have revealed their complex demographic history, which provides profound knowledges for both domestication and speciation studies (6-8, 10). We previously drew the draft of a Beijing duck genome and identified genomic variation within this assembly(11). However, as the most economical important waterfowl and one of the principal natural hosts of influenza A viruses, genetic variations of duck were not well characterized. And debates about domestic ducks being domesticated from the mallard duck or the descendant of mallard duck and spot-billed duck are still continuing.

Until now there are more than 140 wild duck species across the world and most of them can hybridize each other(12, 13). Phylogeographic structure of dabbling ducks using 5'-end sequencing of the mtDNA control region and ODC-6 of nuclear DNA demonstrated that the mallard duck diverged from its closely population (spot-billed duck) recently, implying they are good resource for speciation studies (14). Domestic ducks are considered to originated from East Asian and Europe and show broad phenotypic difference from these wild duck species, such as loss of flight and migration ability, reducing of brain size, increasing of body size and fertility(15-19). Previous studies based on similar outlooks habits and distributions provide two hypotheses about domestic ducks' origin. One hypothesis suggested that domestic ducks were originated from the mallards, while the other argued that they were tamed from descents of the

63 mallard (*Anas platyrhynchos*) and spot-billed (*Anas zonorhynchos*)(20). Until now,
64 evolutionary patterns of wild and domestic ducks in genome-wide scale are poorly
65 understand.

66 Here, we report high-quality drafts of the mallard duck and spot-billed duck
67 respectively, and compare them to the genome of Beijing ducks and other birds. We
68 also generate a total of 1,219 Gb whole-genome resequencing data of thirty-five wild
69 ducks from eight species (gadwall (n=1), baikal teal (n=1), one pintail (n=1), falcated
70 teal (n=1), common teal (n=1), wigeons (n=3), spot-billed duck (n=13) and mallard
71 duck (n=14)), ninety-four domesticated ducks from six Chinese breeds, twenty ducks
72 from two commercial breeds (Cherry and Campebell ducks), together with twenty
73 individuals from two outgroups (ten Muscovy ducks and ten Taihu geese) to study
74 genetic diversity, effective population size (N_e) and genomic regions under selection as
75 well as to compare evolutionary patterns of wild ducks and domestic ducks. Our study
76 will give a comprehensive understanding of the evolutionary history of ducks, that not
77 only illustrates duck domestication processes but also would help to expose some
78 central issues of animal domestication process and speciation process.

79 **Results**

80 We generated 262.45 Gb (~209-fold) pair-end reads of a female mallard duck and
81 275.20 Gb pair-end reads (~220-fold) of a female spot-billed duck, respectively (Fig.
82 S2, S3, Table S1). Using methods similar to those applied to the Beijing duck
83 genome(11), we generated a high-quality mallard assembly consisting of 61,591
84 scaffolds and covering 1.27 Gb, and estimated this assembly having a heterozygosity

rate 3.61×10^{-3} higher than the corresponding in Beijing duck genome (2.40×10^{-3}). The contig N50 and scaffold N50 values of the mallard genome assembly were 38.27 Kb and 2.49 Mb, respectively (Table 1; Table S2). We also generated a high-quality spot-billed genome assembly containing 67,685 scaffolds and covering 1.31 Gb. Similarly, we found that spot-billed assembly had a high heterozygosity rate (3.54×10^{-3}). The contig N50 and scaffold N50 values of the spot-billed assembly were 33.06 Kb and 2.05 Mb, respectively (Table 1; Table S2). We then estimated coverage of the mallard duck and spot-billed duck assemblies with alignment to seven finished BACs (completed independently using Sanger sequencing technology) and 319,996 duck ESTs assembled in our previous duck genome project using BLAT (SI Appendix)(11, 21). This analysis suggested that 7 BACs covering 640 Kb on chromosomes 1, 3 and 4 were aligned over more than 92% of their lengths (Fig. S4, S5, Table S3), and greater than 98% of 319,996 ESTs were aligned to both the mallard duck and spot-billed duck assemblies (Table S4, S5). These data, together with our previous study, suggest that two wild duck assemblies presented here are comparative to our previous one domestic duck assembly (version 1.0) in coverage and provide a good resource for further analysis in this study.

More repeats in wild duck than Beijing duck

We annotated transposable elements (TE) of the mallard duck and spot-billed duck assemblies using a combined pipeline (SI Appendix). This effort found that the mallard duck and spot-billed duck assemblies contained ~136 Mb and ~139 Mb TE sequences, respectively, accounting for about 10% of their assemblies, respectively (Table S6).

Among those TE elements, the LINE (long interspersed elements) type CR1 (chicken repeat 1) transposons are of the most abundance, a situation similar as ones in Beijing duck and chicken. Phylogenetic analysis of CR1 in mallard duck, spot-billed duck, Beijing duck and chicken revealed two duck-specific CR1 clades (clade 1 and 3), in which large variation of copy numbers were shown between the two wild ducks and Beijing duck. This implies that, as a main class of TE, CR1 evolved rapidly among different avian species and distinct duck lineages. (Fig. 1a; Fig. S6).

Evolution of gene families in domestic and wild ducks

We predicted 21,056 and 21,123 protein-coding genes, which constitutes approximately 2.21% of the mallard genome and 2.13% of the spot-billed genome, using the BGI pipelines (SI Appendix; Table S7). These numbers are slightly larger than the corresponding in the domestic duck (Beijing duck) genome (20,629 protein-coding genes) (Table 1). Of the 21,056 mallard genes, 12,817 were mapped to categories established by the Gene Ontology (GO) project, 15,878 had orthologs in the Kyoto Encyclopedia of Genes and Genomes (KEGG) database, and 14,894 were supported by the duck ESTs produced in our previous duck genome project (Fig. S7). These numbers are similar to these of the spot-billed, where 12,815 genes were mapped to GO project, 15,803 had orthologs in KEGG database and 14,827 were supported by the duck ESTs (Fig. S8).

We then constructed gene families using the above mallard duck and spot-billed reference gene sets, our previous Beijing duck gene set, together with a combined outgroup gene sets from chicken, turkey, flycatcher, ostrich, mouse and human (Fig. S9,

Table S8). Interestingly, two wild ducks were similar and their ancestors were clustered to the domestic (Beijing) duck (Fig. 1b). We further examined large-scale differences in gene complements between one domestic duck and two wild ducks. We found that 1,305 gene families were specific to wild duck and this number was larger than the corresponding of Beijing duck (497 gene families) (Fig. S10). Among 19,405 gene families detected in the above three duck genomes, 25 were significantly expanded in wild duck. GO enrichment analysis showed that those expanded genes were significantly over represented in biological processes like chromatin organization, cell-cell adhesion, ATP synthesis and immune response. This implies enhanced genome regulation, energy metabolic activity and response to stimulation along with the speciation of the two wild ducks, which might be beneficial to their adaption to new environments (Table S9; Fig. 1c-d).

Domestic duck shared a comparative number of SNPs with the mallard duck and spot-billed duck

We performed whole genome sequences of 114 individuals from eight domestic duck breeds and 35 individuals from eight wild duck populations (Fig. S1). We generated 828.5 Gb 125-bp paired-end reads for 27 Beijing, 13 spot-billed and 14 mallard ducks, with an average of 14-fold genome coverage, and 372.4 Gb 100-bp paired-end reads of six pooled genomic DNA libraries, each representing one breed. We then detected SNPs using these 1,201 Gb reads together with 95.1 Gb 100-bp paired-end reads of 27 Shaoxing ducks (unpublished data). This effort totally detected 44,852,612 single nucleotide polymorphisms (SNPs) in those 10 duck populations (eight domestic duck

breeds and two wild duck populations). Among them, 20,676,377 and 42,012,846 SNPs were detected in domestic ducks and two wild duck populations (the mallard duck and spot-billed duck), respectively (Table S10). Detailed analyses suggested that domestic ducks shared large numbers of SNPs (18,093,115) with wild duck, and hold comparative numbers of common SNPs with the mallard (15,904,435) and spot-billed (16,062,465) ducks. Moreover, we found that the numbers of mallard specific SNPs (heterozygous in mallard and homozygous in spot-billed) are comparative to the numbers of spot-billed specific SNPs (homozygous in mallard and heterozygous in spot-billed) in each of eight tested domestic duck breeds (Fig. S11).

Domestic duck originated a lineage distinct from the mallard duck and spot-billed duck

To investigate the evolutionary relationship between domestic and wild ducks, we first performed a principal component analysis (PCA) using genome wide SNPs from 27 Shaoxing, 27 Beijing ducks representing domestic ducks and 35 wild ducks (gadwall (n=1), pintail (n=1), falcated teal (n=1), baikal teal (n=1), common teal (n=1), wigeon (n=3), spot-billed duck (n=13) and mallard duck (n=14)). Expectedly, the domestic ducks (Shaoxing and Beijing ducks) clustered to the mallard duck and spot-billed ducks, while separated clearly from other wild species (Fig. 2a). Such genetic relationships were also observed when we performed the PCA analysis with a genome wide SNPs set from eight domestic duck breeds and three wild duck species (mallard, spot-billed and wigeon) having population data, together with muscovy ducks (Fig. S12). Similarly, A neighbor-joining tree that based on the IBS (identical by state) distance of those 89

individuals supported that mallard duck and spot-billed duck had a more closely genetic relationship with domestic populations than other wild species (Fig. 2b). Such phylogenetic relations among the domestic ducks and wild ducks, together with their relationships estimated on PCA analyses and comparisons in SNPs' distribution, demonstrated that domestic ducks aren't offspring of the mallards (SI Appendix).

We then asked whether domestic ducks were originated from descents of the mallards and spot-billed ducks (20). We performed a 3-population test for admixture by taking the mallard duck and spot-billed duck as ancestral populations, Beijing and Shaoxing as target populations through a Block Jackknife analysis that corrected for linkage disequilibrium among SNPs (22, 23). Surprisingly, this effort obtained significantly positive scores for these four groups ($f_3 = 0.007279$, Z score = 23.834; Table S11) and did not support that domestic ducks were descents of the mallard duck and spot-billed ducks.

We further performed genetic structure analysis using the ADMIXTURE to estimate ancestry among individuals of the four populations (Beijing, Shaoxing, the mallard duck and spot-billed duck) with K (the number of ancestry components) ranging from two to seven(24). This analysis indicated that domestic ducks had a homogeneous genetic background and were subdivided into two populations (Beijing and Shaoxing, $K = 3$) following they separated from the wild ducks (the mallard duck and spot-billed duck) with low cross-validation error ($K = 2$). In contrast, the wild ducks showed a heterogeneous genetic background being consistent with PCA results, started to divide into the mallard duck and spot-billed when K were increased to four, and distinguished

each other clearly until K were increased to seven (Fig. 2d; SI Appendix; Fig. S14). The difference in allele frequencies between the wild lineage and the domestic lineage was larger than the corresponding between two wild duck species (the mallard duck and spot-billed duck). These observations neither support domestic ducks were offspring of the mallard and spot-billed duck.

To eliminate the effect of population drift within each domestic population which may also lead the separation between domestic and wild ducks previous than the two wild ducks' split in genetic structure analysis, and further confirm the tree topology of domestic duck and the two wild ducks, we randomly selected and polarized 20,333 unlinked SNPs from these 10 duck populations with two outgroups (Muscovy ducks) for a diffusion approach demographic simulation ($\partial a \partial i$)(Materials and Methods)(25). We estimated the divergent time of the domestic ducks (quantize all eight domestic breeds into one population) and two wild duck populations (the mallard duck and spot-billed duck) using two simple but informative models, which one took and the other did not take migration into account (Materials and Methods, Fig. 3a). One hundred bootstrap simulations of the two models support that both the divergent times between two wild duck populations and domestic duck were almost equal (100~300 Kyr) and longer than the corresponding one between the mallard duck and spot-billed duck (~70 Kyr, Fig. 3b-c; SI Appendix; Table S12, Fig. S15-16). These results further supported that the ancestor of domestic duck split from the ancestor of mallard duck and spot-billed duck before the two wild ducks' divergence.

Moreover, we performed a haploid pairwise sequential Markovian coalescent (PSMC)

analysis to reconstruct the demographic history and estimated the effective population size (N_e) change along time among the mallard, spot-billed, Beijing and Shaoxing duck. This analysis suggested that ancestors of these four populations hold a comparative level in effective population size before 110 Kyr ago (beginning of LGP). However, ancestors of the wild ducks and domestic ducks seemed to be under differently demographic histories after LGP. During this period, the effective populations of Beijing and Shaoxing duck ancestors decreased dramatically and continually. It is contrast to the case in the mallard and spot-billed duck ancestors, where their effective populations didn't decrease or even experienced once or twice times of dramatic increasing (Fig. 3d; SI Appendix; Fig. S17). PSMC can also be informally used to infer divergent times when N_e trajectories that are overlapping diverge moving forward in time towards the present (26, 27). Interestingly, the divergent time was roughly coincident with the one inferred from $\partial a \partial i$. Such consistency further encouraged us to infer that the ancestor of domestic duck diverged from the ancestor of mallard duck and spot-billed duck at about the beginning of the LGP. Similar to N_e estimated by the PSMC analysis, we found that populations size of domestic population decreased whereas these of both the two wild duck species increased after split between wild and domestic population when using $\partial a \partial i$ to simulated a more complex model (i.e. populations' size change exponentially after split) of divergence between populations (SI Appendix).

The early mallard duck and spot-billed duck ancestors might benefit from adaptive selections

It is interesting that different tendency of population size change occurred after divergence between wild lineage and domestic lineage during LGP. One possibility might be that nature selection granted the mallard duck and spot-billed duck ancestors **but not domestic duck ancestors** new advantage traits to environment change after their split. Consequently, environment changes lead the domestic ancestor population size decreased, whereas wild ancestor population size even increased (Fig. 3d; SI Appendix; Fig. S17). We detected positive selections in the early ancestors of these two wild populations and identified a total of 625 putative selective sweep regions, with an average size of 25,888 bp, using thresholds of 4 and 1.5 standard deviations away from the mean for autosomes and sex-chromosome, respectively (Fig. 4a-b; Fig. S18; Materials and Methods)(28, 29). Among them, 246 selective sweeps harbored 336 protein coding genes, while the remained ones don't contain annotated protein coding genes (SI Appendix; Table S13). Recent studies indicated that positive selections in intergenic and intragenic regions were associated with phenotypic variation (4, 7, 28). We then extended 180 kb in both upstream and downstream of the above 379 regions containing no annotated genes. This effort identified 309 closest genes in 228 of these extended regions (Table S13).

After that, we performed gene ontology (GO) analysis of 441 genes (48 of them detected in domestic selection were removed, Materials and Methods) in or closest (extended 180 kb in both upstream and downstream) to selective sweeps regions (SI Appendix) and highlighted 288 enriched categories related to 211 biological processes (Table S14, S15). Interestingly, the most remarked categories were related to a series of

organs and system, such as morphogenesis and development of lung, heart, metanephros, aortic and cardiac muscle, and functions of blood circulatory system, respiratory system, peripheral nervous system and ventricular system development, rhythmic excitation (Table 2). Moreover, an overrepresented term of positive selection related to dorsal/ventral pattern formation was detected. Since long distance flight need adaptive changes in body shape, energy metabolism and spatial navigation, we proposed that selection on those genes associated with above functions might enhance the ability to fly long distance in the ancestor of the mallard duck and spot-billed duck. We further investigated functions of genes in putative selective sweep regions by surveying published literature and listed several genes might be related to the flight activity. For example, Kinase Insert Domain Receptor (KDR) gene associated with lymph vessel development were required for lung inflation at birth and enhance the function of respiratory system(30). Selection of two genes including *DNAJA1* stimulates ATP hydrolysis(31) and *MRPS27* help in protein synthesis within the mitochondrion(32) might contribute to supply high energy for sustained flight. We also noted that *PARK2* were under selection when use a lower threshold ($S=-3$). *PARK2* promotes the autophagic degradation of dysfunctional depolarized mitochondria which might suggest an adaptation to facilitate degradation of damaged mitochondria caused by products of elevated metabolic rate(33). Previous studies found some genes play important roles in energetic metabolism as a ferrous iron transporter (i.e *SLC39A14*) or positive regulation of fever generation (i.e. *PTGS2*)(33, 34). Positive selection on these genes in the early mallard duck and spot-billed duck ancestor might enhance their

ability to transport oxygen in blood and adapt to cold and hypoxic environments of the
 high altitude during flying. Some cryptochrome-containing cells within the retina are
 active at night when the birds perform magnetic orientation. In some migratory birds, a
 distinct part of the forebrain, where primarily processes input from the eye, is highly
 active at night(35). We also found three genes associated with retinal ganglion cell axon
 guidance (*EFNA5*, *ISL1* and *EPHA7* (S=-3)) and two genes (*CHRD* and *WNT7B*)
 related to forebrain regionalization or development were under selection in the early
 mallard duck and spot-billed duck ancestor (Fig. 4d; Fig. S19)(36, 37). Gene family
 that function on short wavelength-sensitive opsin (Cryptochromes, which have been
 suggested to form the basis of light-dependent magnetic compass in birds, is excited by
 blue (short wavelength) light) expanded in wild ducks may also indicated an
 enhancement in magnetic navigation ability (Table S9)(35).

In addition, we found that at least 32 genes (e.g. *IL11RA*, *PRKAA1*, *LPAR1*, and *NS1BP*)
 involved into immune response to infections and three of them (i.e, *ADAP*, *RIG-I* and
CX3CR1) played critical roles against influenza A virus infections (Table S15; Fig.
 S20)(38, 39). Therefore, it seemed that positive selections on immune genes might help
 the early mallard and spot-billed duck ancestor keep evolutionary equilibrium with
 influenza A viruses. In addition, we examined functional activity of four mutations
 locating in intron, UTR or downstream of Ephrin-A5 (*EFNA5*,2), *ISL1* (1) and
 Mitochondrial Ribosomal Protein S27 (*MRPS27*,1), respectively. Three of the four sites
 with wild type showed significantly higher gene expression acidity than domestic
 genotype in luciferase activity assay (Fig. 4d; Table S22; Fig. S28).

The genomic islands of mallard duck and spot-billed duck divergence

We estimated that the mallard ducks diverged from the spot-billed ducks about 70 Kysr using two split models (Fig. 3a-b). This divergent time was consistent with that one counted using mtDNA sequences, and even less than the corresponding one (about 340 Kysr~2 Myrs) of two flycatcher species, which were studied extensively in speciation divergence(5, 10, 14). To elucidate genomic properties and speciation of these two wild duck species, we firstly calculated how many sites all mallard ducks homozygous for one allele and all spot-billed ducks homozygous for another (which we referred the density of these fixed difference sites as d_f) in 40kb sliding windows with 20kb step size along duck scaffolds (Materials and Methods). This effort found that a small number of d_f s (4,386) and only 26 SNPs of these located within protein-coding regions distributed in small genomic regions (758 of totally 52,215 windows) between mallard and spot-billed. We refer the windows showing highly elevated divergence up to 50 times higher than the genomic mean as “divergent islands”. Divergent islands contain most (3,522, 80%) of these d_f s that distribute in 348 (0.6% of the whole duck genome) windows. This is sharp contrast to the case in *Ficedula* flycatchers, where using the same criterion, that 25% (59,936) d_f s were distributed in the large genomic divergent islands covering 2.7% of the flycatcher genome. Such large difference in number of d_f s and the pattern of genomic divergent islands in two these species might due to that the wild duck (the mallard and spot-billed duck) has a shorter divergence time and is in an earlier stage than the *Ficedula* flycatcher(10). Surprisingly, detailed analysis of these genomic divergent islands suggested that all of them were homologous to the chicken

“Z” chromosome (Discussion Section; Fig. S21).

In order to study the genomic divergence in duck autosomes, we then estimated the population differentiation, nucleotide diversity through calculating F_{ST} (fixation index), dxy (sequence divergence), π (nucleotide diversity) and Tajima’s D values in sliding window that homologous to chicken autosomes. (Materials and Methods; Fig. 5a; Table S16). Scaffolds homologous to chicken Z chromosome showed higher level of F_{ST} (5.5-fold) and Tajima’s D (1.5-fold), but smaller number of SNP density (0.5-fold) and lower level of π (0.6-fold) than autosomes (5, 41)(Table S16, Fig. S22). These observations are consistent with change of F_{ST} , Tajima’s D and SNP density in flycatcher and wild ducks (Mexican and mallard duck)(5, 40). Analysis on autosomal differentiation regions shows that the distribution of F_{ST} between the mallard duck and spot-billed ranged from -0.0157 to 0.7785 with an average of 0.04930346 and distributed in a narrow peak, lower than F_{ST} between the wild ducks and domestic ducks ranged from 0.0880 to 0.8912 with an average of 0.2100 and distributed in a wide peak (Fig. 5b). Detailed analysis indicates that small part of genomic regions of the mallard duck and spot-billed duck have F_{ST} values being away from the mean F_{ST} at least 4 standard deviations (we referred these regions as high differentiation islands). These higher F_{ST} regions showed lower level of nucleotide diversity (π mallard: 0.0024 ± 0.0022 ; π spot-billed: 0.0023 ± 0.0020) and SNP density (360 ± 305 and 334 ± 280 SNP per window in the mallard duck and spot-billed duck respectively) than genomic backgrounds (π mallard: 0.0076 ± 0.0042 ; π spot-billed: 0.0077 ± 0.0043 . 1134 ± 569 and 1161 ± 597 SNP per window in the mallard duck and spot-billed duck respectively) in both mallard

duck and spot-billed duck (Fig. 5c-f; Fig. S22; Table S17). This is similar to the case in flycatchers, Darwin's finches and ducks, where high differentiation islands are characterized by reduced levels of nucleotide diversity and polymorphisms(5, 28, 40, 41). A noteworthy consequence of these coinciding features was that both Tajima's D of these high differentiation islands was not reduced in both mallard duck and spot-billed, whereas it is different with the case in flycatchers that Tajima's D of high differentiation islands was significantly reduced (Fig 5f; Table S17). Moreover, sequence divergence between mallard duck and spot-billed didn't exceed background levels in high differentiation islands that is consistent with the observation in flycatchers, whereas Darwin's finches' study show elevated dxy in high differentiation islands (Discussion Section; Fig. S22)(41).

Identification of selective sweep regions in domestic duck

To detect genomic regions under domestic selection, we searched the duck genome for regions with reduced heterozygosity (H_p) and/or increased fixation index (F_{ST}) using methods similar to the chicken and dog genome (Materials and Methods)(3, 4). We Z-transformed the H_p ($Z(H_p)$) and F_{ST} ($Z(F_{ST})$) distributions of autosomes and sex-chromosomes separately, and focused on putatively selective sweep regions falling at least four/two standard deviations away from the mean of the genome-wide $Z(H_p)$ and $Z(F_{ST})$ of autosomes and sex-chromosomes separately (Fig. S23-S25; Materials and Methods). This effort identified 137 regions with extremely low levels of H_p (average length = 63.6230 kb, average H_p = 0.2100 for autosomes; H_p = 0.2470 for sex-chromosomes), and 145 regions with significantly elevated F_{ST} values (average length

371 = 58.7670 kb, average F_{ST} = 0.5390 for autosomes, average F_{ST} = 0.394 for sex-
 372 chromosomes). However, only a small proportion (17 regions covering 7.4 Mb and
 373 containing 36 genes; 9 of these regions located in autosomes) putative selective sweep
 374 regions were predicted using both extremely low levels of H_p and significantly elevated
 375 F_{ST} values (Fig. S26). Such observation might be partly attributed to that the domestic
 376 lineage and wild lineage were under different evolutionary patterns after they had
 377 diverged about 100 ~200 Kys. For example, the wild lineage (mallard duck and spot-
 378 billed ancestors) seemed to benefit from positive selection and dramatically increased
 379 their effective population sizes once or twice times, while the domestic lineage
 380 dramatically declined their effective population sizes during LGP (Fig. 3; Fig. S17).
 381 This complex demographic history of the domestic and wild lineage might affect the
 382 F_{ST} distribution. We therefore focused on these 137 candidate domestic regions (CDRs)
 383 showing extremely lower levels of heterozygosity (SI Appendix; Fig. S18). We found
 384 that 131 genes and 289 miRNAs contained in 80 of these CDRs, while 47 genes closest
 385 to one of 33 CDRs when we extended 180 kb in both upstream and downstream of each
 386 these 57 CDRs containing none annotated protein coding genes (Table S18, S20).
 387 Further gene ontology analysis indicated that 162 genes in or closest to these 137 CDRs
 388 were related to 145 biological process categories (p-value <0.05) representing
 389 developmental signaling and nervous system, lipid, carbohydrate and protein metabolic
 390 process, cell division and immune response (Table S21). Among them, two genes
 391 (*NLGN1* and *PER2*) involve into circadian sleep/wake cycle or is associated with eggs
 392 production(42), three genes (*MRAP2*, *MC4R* and *RMII*) are associated with the

modulation of feeding behavior or reducing of food intake in response to dietary excess, and four genes (*HSD17B4*, *HEXB*, *B4GALT1* and *CYP19A1*) are related to sexual characteristics, male courtship behavior or androgen metabolic process (43, 44)(Fig. S26).

Among these candidate domestication genes, we examined functional activity of two mutations locating in introns of *GRIKI* and *LYST*, respectively. Domestic genotype of *LYST* mutation showed significantly higher gene expression acidity than wild genotype in luciferase activity assay (Fig. 4d; Table S21)

Reanalysis provide further evidence on evolutionary relationship of domestic duck, mallard wild duck and spot-billed wild duck

As the genomic average genetic distance between mallard and spot-billed (mean F_{ST} : 0.0637) is significantly lower than it in mallard/spot-billed vs domestic pairs (mean F_{ST} : 0.2832, 0.2498)(Fig. 5b). To further investigate whether this higher genomic "similarity" between mallard and spot-billed is caused by hybridization or phylogeny as we inferred above. We stratified the genomic divergence values of each windows by ΔDAF (or absolute ΔDAF) (derived allele frequency difference between the two populations) bins. According to recent research on introgression issue in chimpanzee, the chance of derive alleles being introduced through gene flow increases with the frequency in the donor population. Gene flow should introduce into receptor populations at low frequencies (45). So, we may expect lower divergence value in these higher ΔDAF (absolute ΔDAF) bins as it represented more probability of introgression. We did find decreased genetic distance (F_{ST} , d_{xy}) between mallard/spot-billed and domestic ducks in higher ΔDAF

bins (Fig. S29 b). However, both F_{ST} and d_{xy} between mallard and spot-billed is increased with the ΔDAF (Fig. S29 a). This observation suggested that this higher genomic “similarity” between the two wild duck species maybe not caused by introgression. Furthermore, we found that the mean ΔDAF value didn’t show significant difference between high differentiation regions and genomic background, which is consistent with recent research that gene flow is not the major factor in the formation of the genomic islands (Table S23) (41).

To further deciphering the introgression effect and It’s influence on evolutionary relationship inference of domestic and the two wild ducks, we performed ABBA-BABA test (assume that mallard, spot-billed, domestic duck is P1, P2, P3 respectively) using Muscovy duck as outgroup. It also suggested that genetic relationship between mallard and spot-billed duck are closer than it between the two wild species and domestic. We also found evidence of gene flow between domestic and wild ducks which is in consistent with above analysis. (D-statistics: -0.2972; jackknife std: 0.4189) (Fig. S30 a). In order to characterized the gene flow effect on the phylogenetic tree reconstruction, we used the same idea, ΔDAF , to test is there evidence that these derived allele sharing is caused by introgression (Fig. S30 b). We stratified the number of BBAA (ABBA, BABA) sites by the ΔDAF bins between mallard and spot-billed duck populations (spot-billed and domestic, mallard and domestic, respectively). The most majority proportion of BBAA sites were distributed in lower ΔDAF bins (-0.4 ~ 0.4), imply the major reason for excess derived allele sharing due to phylogeny rather than introgression. As the majority number of SNPs (polymorphic in the compared two

populations. i.e. Polymorphic sites in spot-billed and domestic, mallard and domestic, mallard and spot-billed for ABBA, BABA, BBAA, respectively) distributed at lower Δ DAF bins. We further measured the proportion distribution of the BBAA sites among all SNPs (i.e. polymorphic in the compared two populations). If the introgression exists between mallard and spot-billed (mallard and domestic, spot-billed and domestic), we may expect the proportion of BBAA sites would increase with the Δ DAF (or absolute Δ DAF in negative). We did find that the proportion of BABA, ABBA sites increased with Δ DAF, which indicate that the derived allele sharing between wild (mallard or spot-billed) and domestic populations is partly due to the introgression. Whereas the distribution of BBAA proportion across Δ DAF bins showed a different pattern that the BBAA rate has only little difference across Δ DAF bins, which indicate that absent evidence of introgression between mallard and spot-billed.

Discussion

How domestic animals originate and lead to striking morphological and behavioral changes are hot topics of evolutionary biology. Recently, the rapid development of high-throughput sequencing technologies has significantly boosted knowledges for origination and domestic process in dog, pig and rabbit, as well as chicken (6, 7, 26, 46, 47). Here we focused on duck origination and domestication using genome-wide resequencing data sets of domestic ducks and eight wild duck species, together with *de novo* genome sequences of two wild ducks (a mallard duck and a spot-billed duck). Surprisingly, these efforts made a conclusion that domestic ducks were neither originated from the mallard ducks nor tamed from descents of mallard duck and spot-

billed duck. Instead, we inferred that ducks were domesticated from an ancient wild
 duck lineage, which was close to the ancestor of mallard duck and spot-billed ducks.
 Such inference was supported by both phylogenetic comparative analysis and
 population genetic analysis. Despite the incomplete lineage sorting (ILS) and gene flow
 may confound the analyses, the different SNP sets used in genetic structure analysis,
 f3-statistics, SNPs distributions analysis and tree construction that was sampled from
 genome-wide would avoid these confounding effects in phylogenic inference(48).
 Furthermore, Genetic structure show each individual of both the mallard duck and spot-
 billed duck almost consistent of purely one ancestry component. This is different from
 the case of admixture, where individuals of one population with components of other
 populations' ancestry (45, 49, 50), thus indicating no evidence of introgression between
 the mallard and spot-billed duck (Fig. 2d; Fig. S14). Subsequent $\partial a \partial i$ and PSMC
 simulations that rely on different data sets and theory reconstructed the perfect matched
 demographic history further support the robustness of our inference. In addition, the
 subsequent observations of F_{ST} on domestication and speciation analyses are consistent
 with or could be best explained by this phylogenic inference (Fig. 5b).
 Accumulated studies have revealed that extant wild animal may not be the wild
 progenitors of its domestic forms (dog, horse, pig)(6, 51-53). Traditional domestication
 hypothesis, which is assumed to be initiated by humans and involve reproductive
 isolation and strong domestication bottlenecks, is not compatible with genetic data so
 well, implying hypothesis may need a rethink (6). Domestication process of duck
 supplied a new case to domestication studies. Our analysis may help to provide new

481 ideas to resolve the problems in current domestication studies.

482 Detecting genetic variation underlying phenotypic diversity between domestic duck
483 and wild duck could help us comprehensive understanding how duck was domesticated
484 and genetic changes of phenotype different between wild and domestic duck. Detail
485 analyses on ancestor of wild duck showed adaptive selection might contribute to the
486 enhancement of long-distance flight ability in wild duck. Meanwhile, we found that
487 duck domestication was accompanied by selection at genes affecting neuronal activities
488 (e.g. *SH3GL2* (47)), lekking behavior (e.g. *HEXB*, *B4GALT1* and *HSD17B4*). Moreover,
489 Genes associated with lipid, carbohydrate, protein metabolic process and food intake
490 control were under selection (e.g. *RMII*, *MRAP2* and *MC4R* (8)). In summary, an
491 adaption to food-rich and artificial feeding environment with development of
492 agriculture may contribute to duck domestication.

493 Highly heterogeneous patterns of genetic differentiation across the genome have been
494 found and drawn a great attention in speciation studies (41, 48, 54). However, which
495 process drive the evolution of divergent islands, how divergent islands are related to
496 speciation are still unclear (9). We have shown strong evidence suggesting that mallard
497 duck and spot-billed duck are in very early stage of divergence. We have also detected
498 divergent islands distributed in small part genome, which should harbor critical
499 information in decoding how these divergent islands occurred, thus making those genes
500 and mutations in divergent islands especially worthy to pay attention to. Interestingly,
501 all divergent islands (elevated dfs) were located in ‘Z’ chromosome when we align
502 scaffolds to chicken genome which is the most closely species that genome were

sequenced (Fig. S22). Studies on flycatchers suggest that both species-specific male
 plumage traits and species recognition are located on the Z chromosome(55, 56).
 Similar with flycatchers, male plumage traits of mallard duck and spot-billed duck are
 quite different. Therefore, we guessed that difference in male-plumage traits between
 two wild duck species might be associated to these divergent islands on Z chromosome
 and their speciation. Moreover, we note that Tajima's D is not reduced in high
 differentiation regions, which is different with the cases of flycatchers and Darwin's
 finches(9, 41). Tajima's D is used to detect selection of new mutations and correlated
 to difference between π and the expectation of π which is related to number of SNPs
 (57). After selection, new mutations increase diversity slowly and in a low frequency,
 thus results in allele frequency spectra to be skewed towards rare alleles (new mutations
 mainly increase population mutation rate (θ_w) rather than nucleotide diversity (π)).
 Whereas mallard duck and spot-billed duck are in earlier species divergent stage and
 having shorter divergent time (about 70 Kyr) than collared and pied flycatcher (about
 340 Kyr~2 Myr)(10). Thus, the mallard and spot-billed duck did not deposit much new
 mutations to increase (θ_w) and reduce Tajima's D in their genomic divergent islands
 like flycatchers and Darwin's finches
 In the end, we argue that our article has presented distinct genetic resource between
 wild and domestic duck. Genetic resource of wild duck may have been significantly
 shaped through the global climate change during LGP, which making wild duck evolved
 new property to adaption to environment change, also consist of the stability of the
 ecology system. We also note evidence in genetic structure showed that mallard duck

and spot-billed duck (especially spot-billed duck) have interspecific assignment to domestic ducks (Fig. 2d; Fig. S14. Especially when $K=3,4$). This may pose a significant threat to genetic diversity of wild stocks.

Methods

Sequence assembly and annotation

Sequencing was carried out with Illumina technology using genomic DNA of a wild female mallard and a wild female spot-billed collected by Aoji Duck Farm (Ningbo, Zhejiang) with hunting permission granted by the local government. Sequences from paired-end reads of 17 libraries for each individual with insert size ranging from 250 bp to 20,000 bp were assembled with SOAPdenovo and SSPACE software (Table S1). The super-scaffolds were positioned along the chromosomes based on the duck cytogenetic map and the comparative genomic map between duck and chicken (11, 58). A reference gene set for the mallard genome and the spot-billed genome were retrieved through merging the homologous set, the transcript set (predicted with 21 duck transcriptomes) and the *de novo* set(11).

Whole-genome resequencing and SNP detection

Genomic DNA libraries were prepared according to the manufacturer's instructions (Illumina). One DNA sample or pools of DNA samples from 10 individuals for each six duck breeds were used to generated pair-end reads with the Illumina technology (Table S1). Pair-end reads were mapped to the Beijing duck assembly (BGI_duck_1.0) using the bowtie2 with default parameters. After removing the reads which map to multiple place, we called SNPs using the Genome Analysis Toolkit (GATK) and filtered

SNPs according to the coverage of resequencing data with thresholds of $\leq$ one third of the mean or $\geq$ three-fold of the mean and MQ value ≤ 28 (59).

Estimation of nucleotide diversity and population mutation rate

We estimated nucleotide diversity (π) and Watterson's estimator Θ using the PoPoolation package, which corrects biases causing by pooling and sequencing errors, in 40 kb non-overlapping windows(60). We required per position a minimum coverage of 13,13,13,6 and 6 a maximum coverage of 150,150,500,100 and 100 for mallard, spot-billed, Beijing, Shaoxing, and other 8 domestic population each, and set min-count and min-covered-fraction to be 3 and 0.6 for all.

Analysis of linkage disequilibrium (LD), identical by state (IBS), principal component analysis (PCA), f3-statistics and population structure

LD level in Beijing, Shaoxing, mallard duck and spot-billed duck populations were measured through calculating the correlation coefficient (r^2) of alleles using the PLINK(61). The parameters were set as follows: --ld-window-kb 500 --ld-window 99999 --ld-window-r2 0 --r2. To reduce effect of LD, SNPs were pruning by LD and subsequently used for the PCA, f3-statistics and population structure analyses (Fig. S11). We constructed phylogenetic trees by using neighbor-joining method with MEGA based on IBS distance matrix data of all individuals calculated by the PLINK using the above sampling SNPs (62). Principal component analysis (PCA) was performed using the GCTA64 (63). The f3-statistics were calculated using the ADMIXTOOLS and genetic structure was inferred using the admixture program.

0a0i simulation and PSMC analysis

For $\partial a \partial i$ simulation analysis, SNPs were sampled randomly each per 100 kb from the SNPs identified in wild populations (mallard and spot-billed) and domestic populations (Cherry Valley, Campbell and 6 Chinese duck breeds). SNPs were further filtered according to the coverage of resequencing data at their position with thresholds of 20-fold for pooled data and total number of genotyped alleles more than 50, 36, 26, 28 for Beijing, Shaoxing, mallard duck and spot-billed duck populations, respectively. We then inferred ancestral alleles of the above sampling filtered SNP using Muscovy ducks as outgroups using the following thresholds: (1) more than 30-fold coverage two outgroups; (2) the allele was fixed and same in outgroups. After that, we quantized the frequency of each SNP of 8 domestic duck populations into one population and simulated the divergence among spot-billed, mallard and “domestic” using split without migration, split with migration and IM models 50 bootstraps each (Fig. 3a; SI Appendix). N_{ref} was calculated with the following format:

$$N_{ref} = \frac{\text{Watterson's estimator } \Theta}{4 * \text{mutation rate} * L};$$
 L was length of the SNPs from and diluted by rate of SNP sampling.

For the PSMC analysis(64, 65), SNPs were filtered according to the coverage of resequencing data at their position with thresholds of 10- to 100-fold for Beijing, spot-billed and mallard, or 5- to 80-fold for Shaoxing. Effective population sizes were assessed by PSMC model analysis. We assumed that mutation rate per year to be 9.97×10^{-10} and that the generation time to be one year for both $\partial a \partial i$ simulation and PSMC analysis(66).

Selective sweep analysis

591 For the wild duck lineage (mallard duck and spot-billed duck), we selected SNPs
592 according to the inferred ancestral alleles and using the thresholds: (1) SNPs were fixed
593 in the tested Muscovy ducks with at least 20-fold coverage; (2) derived alleles were
594 detected in both domestic lineage and wild lineage; (3) wasn't in CpG sites. We then
595 calculated the derived allele frequency of 18,083,967 selected SNPs in both wild
596 lineage and domestic lineage, and defined selective sweep regions of the early mallard
597 duck and spot-billed duck ancestors using the thresholds of -0.2819 (4-fold standard
598 deviations away from mean) for autosomes and -0.3089 for sex-chromosome (1.5-fold
599 standard deviations away from mean), respectively.

600 As expected, the derived allele frequencies in both lineages are highly correlated (Fig.
601 S18). This correlation was used to calculate the expected derived allele frequency in
602 the domestic lineage as a function of the derived allele frequency in wild lineage. We
603 then calculated the observed derived allele frequency in domestic lineage, and the log-
604 transformed value of the ratio for the observed/expected derived allele frequency in
605 sliding 20-kb windows with a step size of 10kb. After Z-transform the genome-wide
606 distributions to arrive at a measure S. As mentioned previous, this method naturally
607 disregards regions of recent selection or strong purifying selection in wild lineage.
608 However, recent selective sweeps regions in domestic lineage could also reduce S value,
609 thus we exclude regions that show lower zHp (CDRs) in domestic lineage.

610 For the domestic ducks, we counted the heterozygosity of these populations (Hp) in 40-
611 kb windows (windows containing less than 10 SNPs were not included) in eight duck
612 breeds respectively. We further calculated the Hp mean of these breeds for each window.

Similarly, we calculated fixation index (F_{ST}) between each domestic duck breeds and the wild duck population (combine spot-billed and mallard) using an estimator introduced in dog in 40-kb windows, and counted the F_{ST} of eight domestic duck breeds for each window(67). After that, we Z-transformed the H_p and F_{ST} to $Z(H_p)$ and $Z(F_{ST})$, and defined regions with $Z(H_p)$ or $Z(F_{ST})$ being four standards deviations for autosomes or two standard deviations for sex-chromosomes from the mean of the genome-wide ($Z(H_p)$) as putative selective sweep regions.

Estimation of genetic diversity and distance

Nucleotide diversity (π) was calculated using the vcfTools(68). Tajima's D was calculated using the Popgenome R package(69). Pairwise F_{ST} was calculated in 40-kb windows with a sliding size of 20-kb. Neighbor-joining tree based on pairwise F_{ST} values were constructed with phylip-3.696 "neighbor" function by taking the distance matrix of pairwise F_{ST} and integrating neighbor-joining trees of 27,374 windows.

Availability of supporting data and materials

Data availability

The genome assembly has been deposited in GenBank under the accession number PRJNA392350. The duck, Taihu goose and Muscovy duck genome resequencing data under the bioproject PRJNA315043.

Declarations

Competing interests

The authors declare no completing interests.

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Author contributions

Y.H.H and N.L. designed the project. Z.L.L, H.F.L., L.C., H.B.C., W.T.S. and E.G.R collected and purified DNA samples. Q.W.L., Y.Z, G.Y.F., H.F.L., Y.S.S., W.B.C., X.L. and X.X. performed genome assembly, gene annotation and gene family evolution of the spot-billed and mallard genome. R.L performed phylogenetic evolution, selective sweep and speciation analysis. E.G.R. and X.X.W. carried out luciferase activity analysis. Y.H.H., R.L. and Q.W.L. wrote the manuscript. Y.H.H. and R.L. revised the manuscript.

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Figure legends

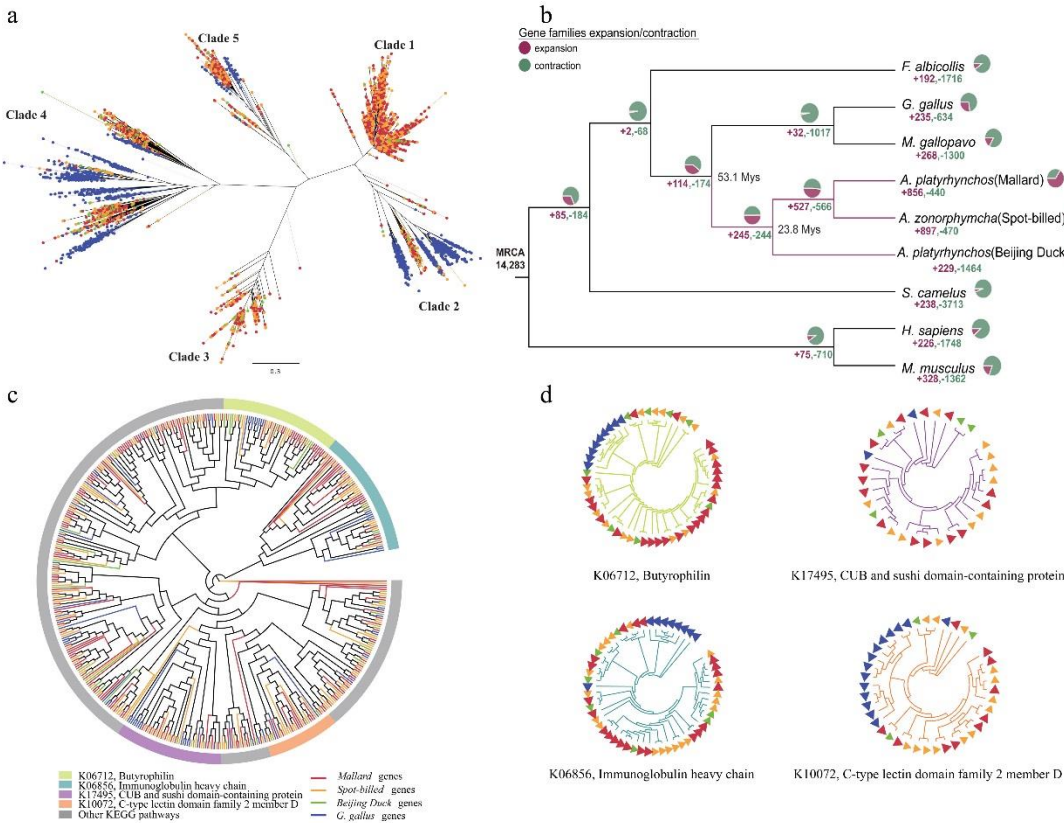


Figure 1: Expansion and contraction of transportable element and gene in mallard, spot-billed and Beijing duck genome. Green, red, yellow and blue in a and c panel represent Beijing duck, mallard, spot-billed and chicken, respectively. (a) Phylogeny of CR1 retrotransposons in four bird genomes. (b) Numbers of gene family losses and gains across ten vertebrates. Pie chart and numbers nearby indicate the number of expanded and contracted gene families in each branch. (c) Phylogeny of gene families expanded in spot-billed and mallard. Top four families ranked by the number of gene gained are indicated in color arcs. KEGG pathway annotation of those four families are listed in the bottom. Other families are indicated in grey color. (d) Detailed phylogeny of four gene families expanded in spot-billed and mallard duck as indicated in (c).

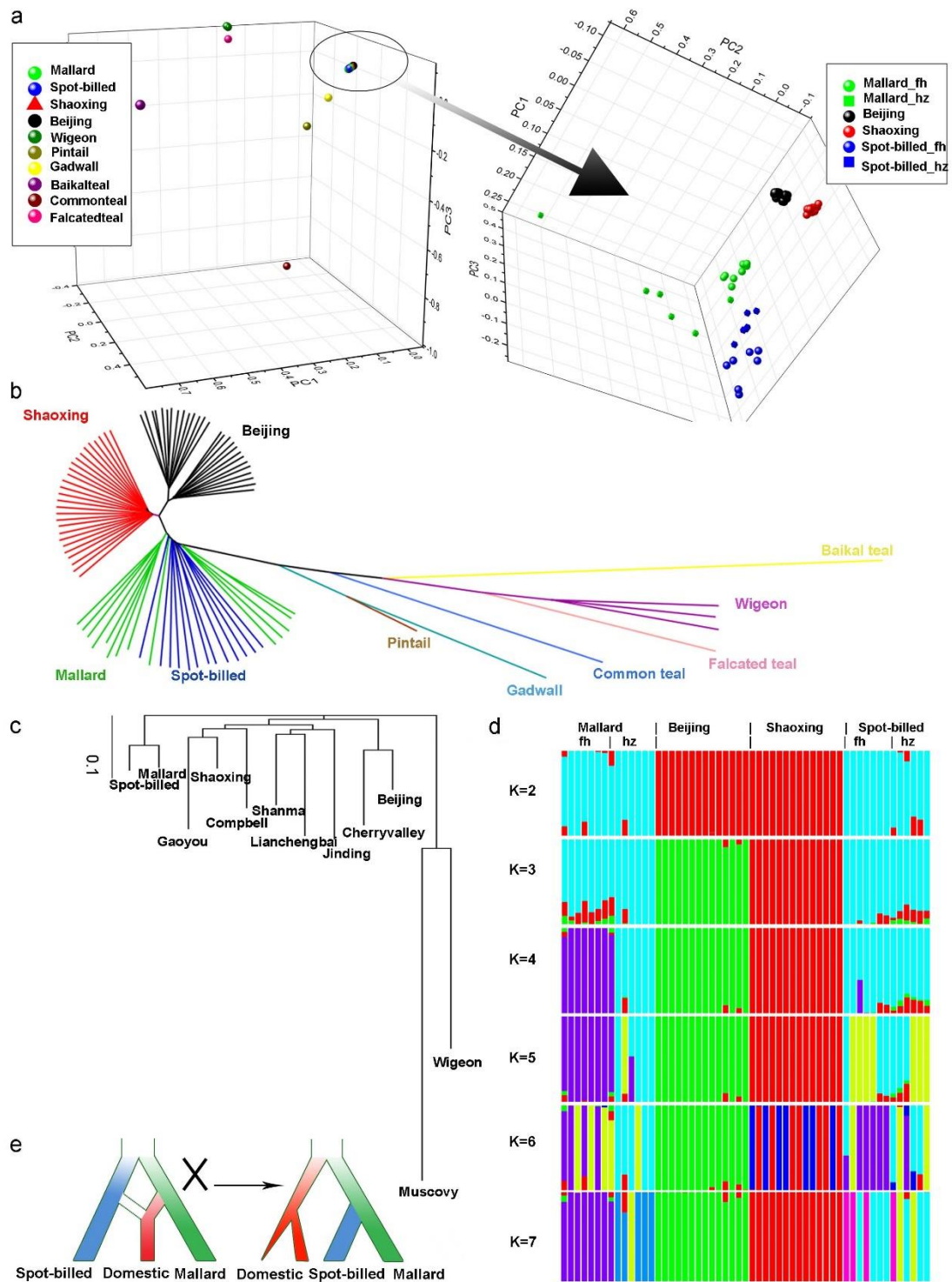


Figure 2: Inference of phylogenetic topology in wild and domestic ducks. Beijing and Shaoxing ducks representing domestic ducks were used the Principal Component Analysis (PCA), phylogenetic analysis based on identity-by-state (IBS) and genetic

818 structure analyses. “_fh” and “_hz” mean spot-billed and mallard collected from
 819 Fenghua and Hangzhou respectively. **(a)** PCA plot constructed for Principal
 820 Component 1 (PC1), PC2 and PC3 using 10,525 high-quality filtered SNPs from 27
 821 Beijing and 27 Shaoxing ducks and 35 wild ducks (one bailkal teal, one gadwall, one
 822 pintail one falcated teal, common teal, three wigeon, 13 spot-billeds and 14 mallards)
 823 shows that domestic ducks are close to spot-billed and mallard wild ducks. **(b)**
 824 Neighbor-joining tree based on IBS distance of 80,505,576 SNPs. **(c)** Neighbor-joining
 825 tree based on genome wide pairwise fixation index (F_{ST}) of 44,852,612 SNP. **(d)**
 826 Genetic structure analyses of 13 spot-billed and 14 mallard wild ducks, together with
 827 14 Beijing and 14 Shaoxing domestic ducks, using admixture ranging K from two to
 828 seven. **(e)** The inferred demographic models for wild and domestic ducks inferred based
 829 on genetic analysis of **a-d** (right). The traditional demographic model, hypothesized
 830 that domestic ducks were originated from mallard or the descendant of a mixture of
 831 spot-billed and mallard (left) are incompatible with observations in neighbor-joining
 832 trees **(b-c)** and genetic data**(d)**.

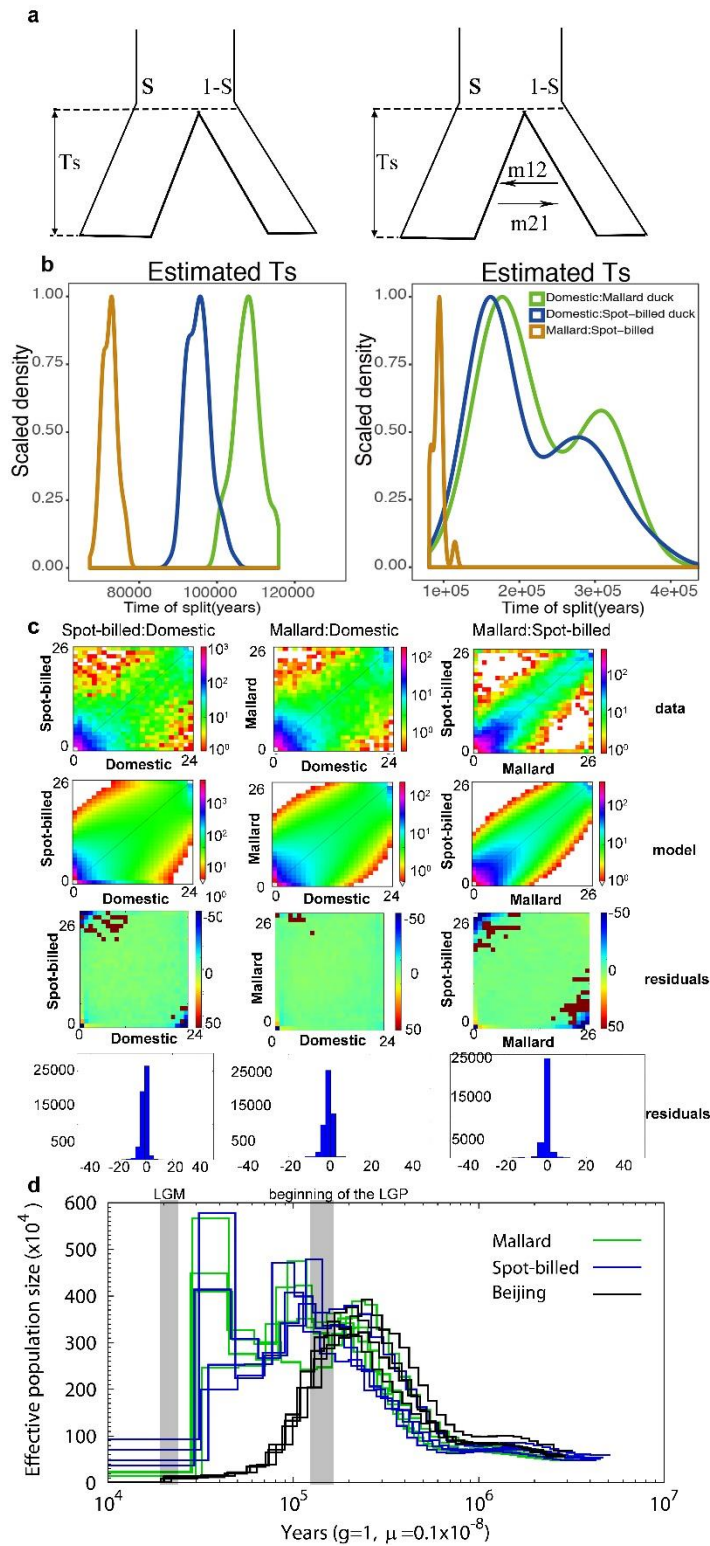


Figure 3: Inference of joint demographic history of ducks. Wild ducks included spot-billed and mallard ducks. Domestic ducks included Beijing, Cherryvalley, Compbel, Gaoyou, Jingding, Lianchengbai, Shaoxing and Shanma. (a) Illustration of

837 the two models (with/without migration) used in the history inference through the $\partial a \partial i$
838 analysis. Parameters corresponding to the code that were provided in supporting
839 information. **(b)** Distribution of estimate T_s (divergence time) in 100 simulations
840 between spot-billed and mallard (orange), mallard and domestic lineage (green), spot-
841 billed and domestic lineage (blue), respectively. Corresponding model of each panel
842 were shown above in **(a)**. **(c)** Comparison between the observed, the model joint Site
843 Frequency Spectrum (SFS) for divergence and its residuals between spot-billed and
844 mallard, domestic lineage and mallard/spot-billed, respectively. **(d)** Demographic
845 history inference of mallard duck, spot-billed duck and beijing duck through the PSMC
846 analysis.

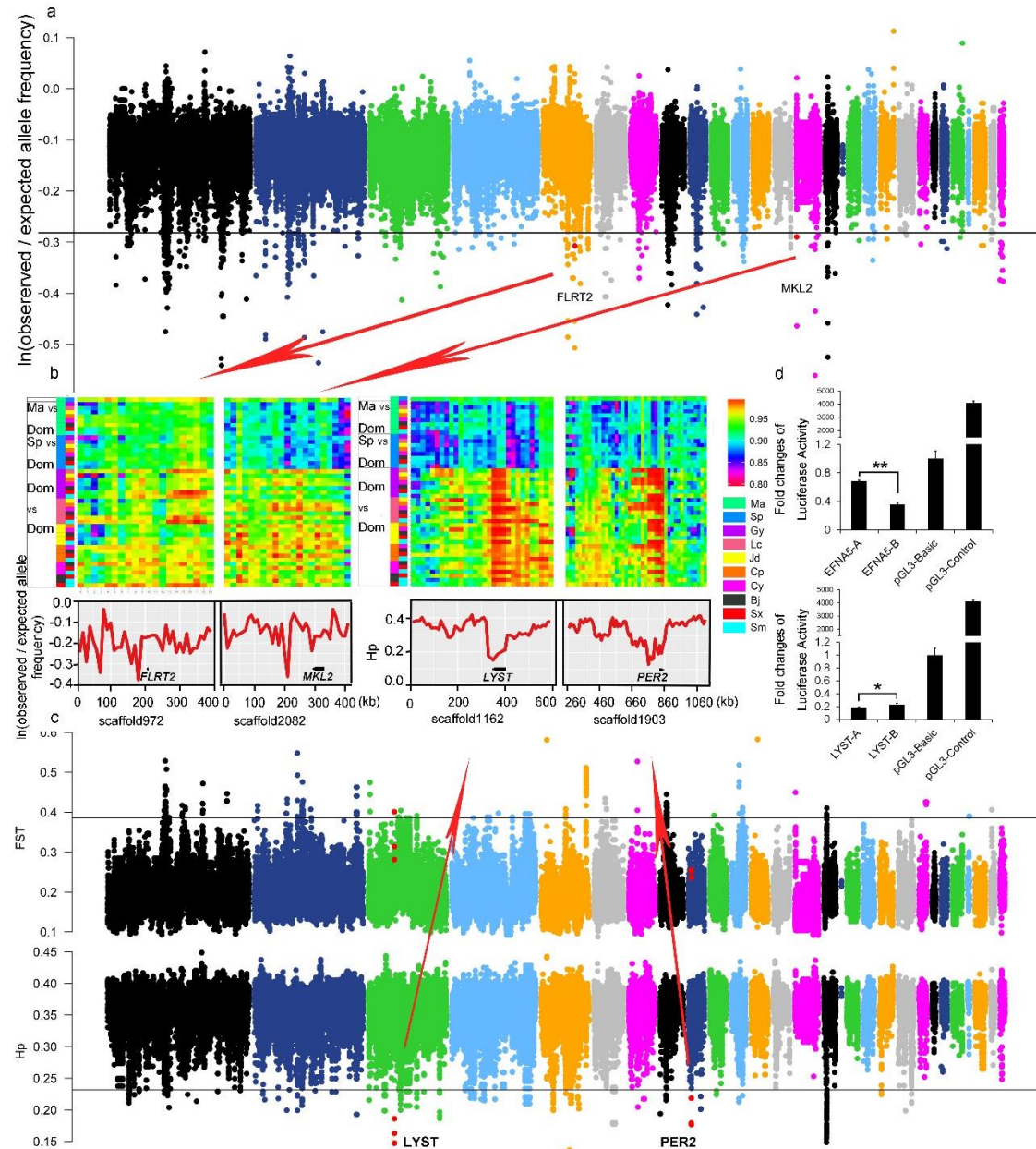


Figure 4: Summary of selective sweep analyses of the early wild duck lineage and domestic duck. (a) Distribution of $\ln(\text{observed}/\text{expected derived allele frequency})$ that is calculated in 20 kb window with a sliding size of 10 kb along scaffolds that is arranged according to their alignment to the chicken autosomes. The horizontal black line indicates the threshold at $\ln(\text{observed}/\text{expected derived allele frequency}) = -0.3136$ (mean of $\ln$ values - 4 standard deviations). Outlier regions that were detail visualized

854 by identity scores (IS) and re-computing values in more smaller window size in **(b)**
 855 were highlight with red dots. **(b)** Details of four putative selective sweep regions around
 856 the *FLRT2*, *MKL2*, *LYST* and *PER2* genes. Upside of each panel shown degree of
 857 haplotype sharing in pairwise comparisons among populations. Color boxes (left)
 858 indicate the comparison performed on that row. “Ma”, “Sp”, “Dom”, represent mallard,
 859 spot-billed and domestic duck respectively. Y-axis (downside) of *FLRT2*, *MKL2*, which
 860 is associated with retina/heart morphogenesis that were under selection in wild ancestor,
 861 are $\ln(\text{observed/expected derived allele frequency})$, whereas y-axis of *LYST*, *PER2* that
 862 were under selection in domestic duck are h_p . **(c)** Distribution of F_{ST} between the wild
 863 and the domestic duck, and h_p in the domestic duck that is calculated in 40 kb with a
 864 sliding size 20 kb along scaffolds that is arranged according their alignment to the
 865 chicken autosomes. The horizontal black line indicates the threshold at $F_{ST} = 0.3858$ or
 866 $h_p = 0.2319$ (mean of $\ln$ values - 4 standard deviations). Red dots indicate the location
 867 of outlier genes (4 standard deviations away from h_p means) that were detail visualized
 868 in **(b)**. **(d)** Comparison in luciferase activity of major alleles of *MRPS27* and *EFNA5*
 869 in wild (spot-billed and mallard) and domestic (Beijing) duck in DF1 (chicken
 870 embryonic fibroblast) cells. “A” and “B” represent wild duck and domestic duck,
 871 respectively. “*” $P \leq 0.05$. DF1 cells were transfected with recombinant pGL3
 872 luciferase reporter plasmids and luciferase activity was measured at 24 hours after
 873 transfection.

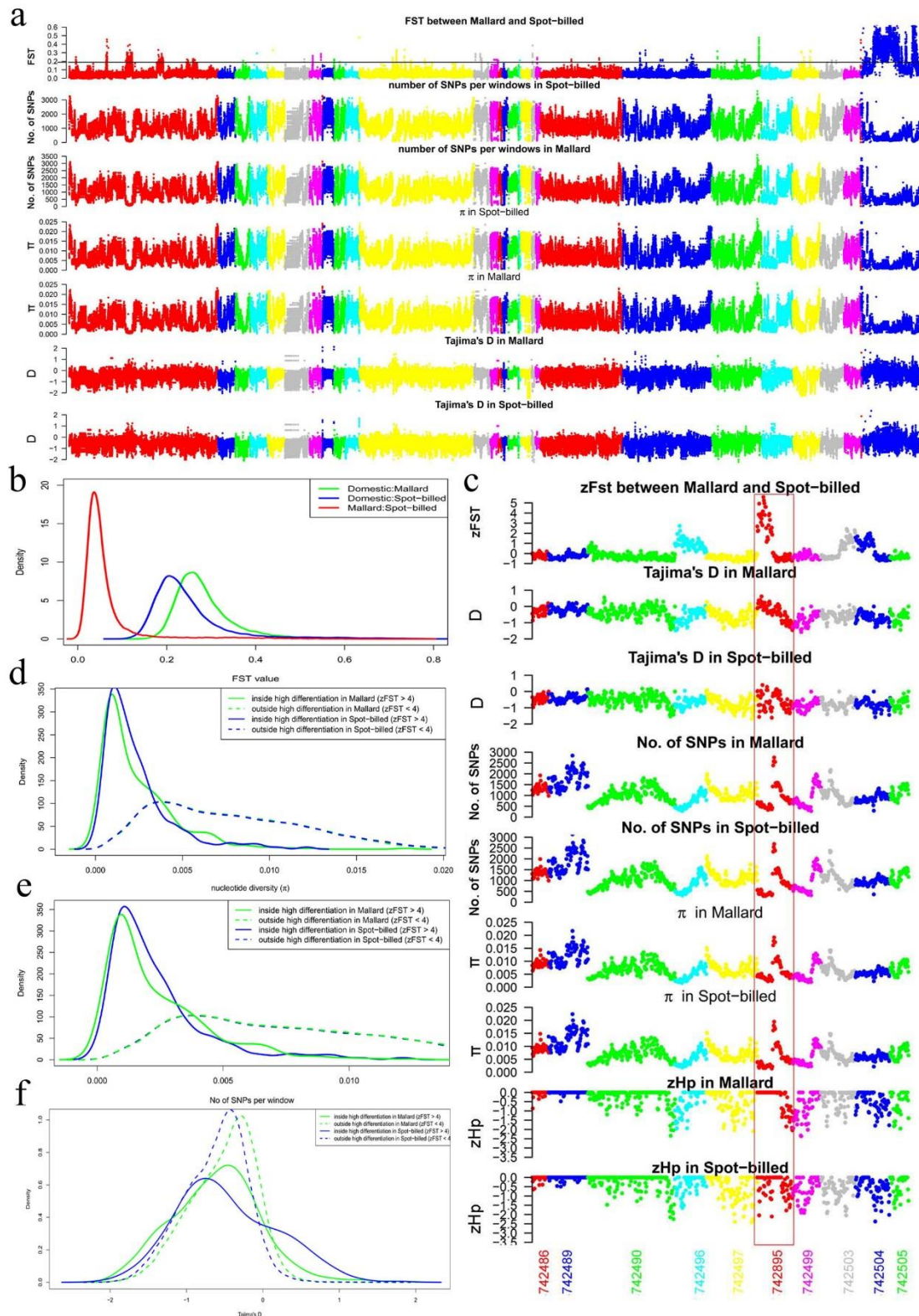


Figure 5: The genomic landscape of species divergence between spot-billed and mallard ducks. (a) Distribution of population genomic parameters of the Beijing duck assembly, in 40 kb windows sliding 20 kb steps. The blue scaffolds in the right were

878 referred as duck Z scaffolds. The black horizontal line indicates the threshold at $F_{ST} =$
879 0.1888 (mean of $F_{ST} + 4$ standard deviations, referred it as $Z(F_{ST}) = 4$). The plots show
880 the F_{ST} between spot-billed and mallard, number of SNPs per window, nucleotide
881 diversity (π), Tajima's D for spot-billed and mallard respectively. **(b)** Distribution of
882 pairwise F_{ST} for 40 kb windows sliding 20 kb steps. The red line shows the plot in the
883 first row of **(a)**. The blue and green line show the pairwise F_{ST} between spot-
884 billed/mallard and domestic ducks (combination of 8 domestic breeds). **(c)** Distribution
885 of population genomic parameters along ten scaffolds. **(d-f)** Distribution of nucleotide
886 diversity (π), number of SNPs per window and Tajima's D inside/outside high
887 differentiation regions in Beijing duck assembly (version 1.0).

Table 1. Comparison in features of the mallard, spot-billed and Beijing duck genomes

Genomic features	Mallard	Spot-billed	Beijing duck
Assembled genome size (Gb)	1.27	1.31	1.11
Contig N50 (bp)	38,271	33,061	26,114
Scaffold N50 (bp)	2,489,142	2,054,876	1,233,631
GC content (%)	41.90	41.90	40.90
Repeat rate (%)	10.71	10.55	7.31
Heterozygosity rate ($\times 10^{-3}$)	3.60	3.54	2.40
Predicted protein-coding genes	21,056	21,123	20,629

Table 2. Enriched gene ontology related to biological process terms for genes under positive selection in the early spot-billed and mallard ancestors

Gene ontology term	P-value	Number	Size
Synapse organization	0.0031	4	22
Fatty acid transport	0.0041	3	12
Blood vessel morphogenesis	0.0042	4	24
Protein polyubiquitination	0.0044	6	55
Metanephros morphogenesis	0.0046	2	4
Epithelial cell migration	0.0046	2	4
Nerve growth factor signaling pathway	0.0046	2	4
L-glutamate import	0.0046	2	4
T cell receptor signaling pathway	0.0057	4	26
Head development	0.0064	3	14
Positive regulation of MAPK cascade	0.0073	6	61
Respiratory system development	0.0075	2	5
epithelial cell differentiation	0.0084	5	45
positive regulation of protein ubiquitination	0.0084	5	45
Smooth muscle cell differentiation	0.0095	3	16
negative regulation of extrinsic apoptotic signaling pathway in absence of ligand	0.0095	3	16
negative regulation of peptidyl-threonine phosphorylation	0.0110	2	6

serotonin metabolic process	0.0110	2	6
Lung epithelium development	0.0110	2	6
Lung morphogenesis	0.0113	3	17
mitotic cytokinesis	0.0133	3	18
Ventricular cardiac muscle tissue morphogenesis	0.0133	3	18
Cardiac right ventricle morphogenesis	0.0152	2	7
Positive regulation of heart rate	0.01526	2	7
Negative regulation of programmed cell death	0.0152	2	7
Bud elongation involved in lung branching	0.0152	2	7
Positive regulation of mitotic cell cycle	0.0154	3	19
Nucleotide-excision repair	0.0178	3	20
Neutrophil chemotaxis	0.0178	3	20
Dorsal/ventral pattern formation	0.0180	4	36

892 Full lists of enriched terms are presented in Supplemental Table 15. Terms that considered to be
893 associated with enhancement of long range flight ability were in bold. Size represents total number genes
894 of this term.
895

**Supporting information for: Genomic analyses reveal distinct demographic
history and identify
genetic changes underlying adaption, speciation and domestication in ducks (SI
Appendix)**

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Supplementary text

1. Genome sequencing and assembly

1.1 Genome sequencing

We extracted genomic DNA with standard molecular biology techniques from blood and random fragmented DNA for library construction. For short insert size libraries, 5 µg of DNA were fragmented to 250-800bp, end-repaired, A-tailed and ligated to Illumina paired-end adapters (Illumina). The ligated fragments were size selected at 250, 350 and 800 bp on agarose gel and amplified by PCR to yield the corresponding short insert size libraries. For mate-pair library construction, 20-40µg DNA were sheared to the desired insert size using nebulization for 2 kb or HydroShear (Covaris) for 5 kb, 10 kb and 20 kb. Then the DNA fragments were end-repaired using biotinylated nucleotide analogues and circularized by intramolecular ligation. Circular DNA molecules were sheared with Adaptive Focused Acoustic (Covaris) to an average size of 500 bp. Biotinylated fragments were purified with Dynabeads® M-280 Streptavidin beads (Invitrogen), end-repaired, A-tailed and ligated to Illumina paired-end adapters, PCR amplification and agarose gel electrophoresis for size-selection. Then sequence all these libraries with Illumina HiSeq2000 platform.

1.2 Raw data filtering and genome size estimation

To reduce the effect of sequencing error or low-quality reads for genome assembly, we took a series of checking and filtering steps on raw reads. We filtered: 1) Reads having an 'N' over 10% of their length; 2) Reads with more than 40 bp low quality base (quality score ≤ 7); 3) Reads with more than 10 bp adapter sequences (allowing ≤ 3 bp mismatches); 4) Short insert size paired-end reads that were overlapped (≥ 10 bp); 5) Redundant duplicated reads generated by PCR amplification in the library construction (Read1 and Read2 of two pairs of paired-end reads were identical, respectively). In total, 275.20 and 262.45 Gb of data were kept for D2B and D2L *de novo* genome assembly respectively.

Genome size can be estimated using K-mer analysis method, the principle is briefly depicted as follows: A K-mer is an artificial sequence division of K bp iteratively from sequencing reads. A read with L bp in length composes (L-K+1) K-mers. The frequency of each K-mer can then be calculated from the input data. The resulted K-mer frequencies should generally follow a Poisson distribution, except for a high proportion of low frequencies due to sequencing errors. The genome size G can be estimated as $G = K_num / K_expect$, where K_num is the total number of K-mer, and K_expect is the expected value of the poisson distribution. We performed K-mer analysis (K=17) using about 21-27X filtered data, and estimated the mallard and spot-billed genome size to be about 1.15Gb and 1.14Gb, respectively (Figure S1).

1.3 Genome assembly and super-scaffold construction

At first, we assembled the two genomes using SOAP *de novo* [1] for contig and scaffold construction. Detailed assembly steps and the data used in each step: 1) Constructed *de*

Bruijn graph with reads from the short insert size libraries using K-mer size 29 bp, which means the parameter *-K* is set as 29. 2) Simplified the *de Bruijn* graph for contigs construction by removal tips, merging bubbles, removal the low coverage of the connection and resolve the small repeats. 3) Constructed scaffolds with short pair-end and mate-pair libraries step by step. 4) Filled in gaps inside the constructed scaffolds with gap closure step.

1.4 Quality evaluation of the genome assembly

We made use of known EST and BAC sequences to evaluate the accuracy of genome assembly. We mapped 7 BACs[2] and 319,996 ESTs of Beijing duck onto the two assemblies using BLAT [3] with default parameters. We found that, for those 7 BACs totally spanning 640 Kb, more than 92% of the spanning were aligned well to the assemblies (Figure S3 and S4). In addition, around 94% of the ESTs were mapped to individual scaffold of the assemblies (Table S6).

Excessive heterozygosity at regional homologous chromosomes usually lead to failure of collapsing the two haplotype sequences, and thus results in redundant sequences in the assembly. To determine whether there is any case of redundant sequences in our assemblies, we performed a self-to-self alignment of the assemblies. Then found the total length of the scaffolds with the coverage rate is more than 50% is 11.3Mb and 10.3Mb in female mallard and a female spot-billed genome, respectively. That analysis indicated there was almost no redundant sequence in the two assemblies and we masked these scaffolds with coverage rate more than 50% for further deep analysis.

2. Repeat prediction, gene structure and gene functional annotation

2.1 Transposable elements annotation

We annotated transposable elements (TE) using the following steps:

1) We first scanned for repeat elements using LTR_FINDER[4] with the default parameters. The resulted TE sequences were used as TE database for RepeatMasker[5] search in step 2.

2) We then used RepeatMasker to identify repeat elements against the Repbase database[6] as well as the TE sequences founded in step1.

3) We further searched the assembly against the protein repetitive database provided in RepeatMasker using RepeatProteinMask.

4) At last, we combined all the three set of predictions (two from step2 and one from step3) to generated an integrated repeat annotation result for each genome assembly.

5) Tandem repeats were identified by Tandem repeat finder[7] using the defaults of "Match=2, Mismatch=7, Delta=7, PM=80, PI=10, Minscore=50, and MaxPeriod=12".

2.2 Variation of CR1 (chicken repeat 1) transposon among ducks

From the result of repetitive elements annotation in section 2.1, we found that CR1 (chick repeat 1), a subtype of LINE retrotransposon, accounts for both the majority of both TE content and variation of TE content among mallard, spot-billed duck, Beijing duck and chicken assembly. In fact, we found there were more than 90Mb (7.4%) of

CR1 subtype repeat in two wild ducks, but only approximately 60Mb (5.5%) in Beijing duck. Therefore, we further studied the variation of CR1 among those four species in a perspective of phylogeny. We downloaded non-LTR retrovirus reverse transcriptase (RT) of chicken retrotransposon CR1, complete consensus sequence from NCBI. We extracted the 252 amino acids (aa) long RT_nLTR_like protein sequences from the consensus, and then mapped it to the four genome assemblies using tblastn. We identified 2579, 2294, 617, 1737 RT_nLTR_like domains in the four genome assemblies (evalue < 1e-5, identity >50%, alignment length >200bp), respectively. We then performed multiple sequence alignment of all the identified RT_nLTR_like domains using MUSCLE[8] (version 3.8.31). Based on the alignment, a phylogenetic tree was subsequently reconstructed using FastTree.

2.3 Gene structural annotation

First, we applied the homolog based gene prediction method using protein datasets of *Anas platyrhynchos*, *Gallus gallus*, *Struthio camelus*, *Meleagris gallopavo*, and *Ficedula albico*. All the protein sequences were firstly mapped to the assembly using BLASTN with the parameters as “-e 1e-5 -F F -m 8”. Then the aligned protein sequences were realigned to the assembly using GeneWise[9] for accurate spliced alignments.

Second, we used Beijing Duck transcriptome reads to polish our target gene annotation. We mapped the transcriptome reads to the two wild duck genome assemblies using TopHat[10] with its default settings and then constructed transcripts using Cufflinks. The process is the same as one preformed in the Beijing Duck genome project[11]. Finally, we combined all the gene models (five homologous gene models sets and the transcript gene models set) by GLEAN[12] to generate an consensus gene set.

2.4 Gene functional annotation

We aligned protein sequences of the final gene set to various protein databases with known functional knowledge, including InterPro[13] (iprscan_4.7) (profilescan, blastprodom, hmmsmart, hmmpanther, hmmpfam, fprintscan, patternScan), Gene Ontology[14], Swiss-Prot[15], TrEMBL[15] and KEGG[16] (release76), to identify conserved domains, putative biological functions and molecular pathways they may involve. In total, there are more than 87% of the annotated protein-coding gene models had at least one hit with known function (Table S5).

3. Evolutionary analysis

Mallard and spot-billed ducks, together with five other avian species (*Gallus gallus*, *Struthio camelus*, *Meleagris gallopavo*, *Ficedula albico* and *Anas platyrhynchos*, which is the Beijing duck) and two mammals (*Homo sapiens*, *Mus musculus*), were used in evolutionary analysis.

3.1 Gene family clustering

Protein-coding genes for *Gallus gallus*, *Struthio camelus*, *Meleagris gallopavo*, and

Ficedula albico were downloaded from Ensembl release 80. The gene sets for *Anas platyrhynchos* was obtained from the BGI inner database. For gene loci with alternative splicing isoforms, only the transcript with the longest translation product was retained. We carried out an all-to-all alignment of all the collected protein sequences using BLASTP (E-value < 1E-7), and conjoined fragmental alignments using Solar[17]. Then a simplified version of Treefam[18] methodology was used to cluster individual sequence into families based on the conjoined alignment (Table S11).

3.2 Phylogenetic tree reconstruction and divergent time estimation

After gene family clustering, single copy genes were selected to reconstruct phylogenetic tree of those nine species. Multiple sequence alignment of each gene family was performed by MUSCLE[8] (version 3.8.31). Phase1-base degenerate sites were extracted and concatenated to generate a super alignment matrix. We built phylogenetic tree using MrBayes[19] which takes advantage of both codon-based and amino acid-based algorithms and adjusts them to the topology of the species tree, to form a more accurate consensus tree according to phase1-base degenerate site. Divergence time of species were estimated using molecular clock model implemented by PAML mcmctree[20].

3.3 Segmental duplications

Segmental duplications (SDs) are duplicated blocks of genomic DNA typically ranging in size from 1–200 kb[21]. To explore any difference in the content of SDs among two wild duck and Beijing duck genome, we detected SDs in their genome assembly using a whole-genome-alignment based method. We performed self-to-self whole genome alignment of these three assemblies using BLASTZ with the parameters: T=2 C=2 H=2000 Y=3400 L=6000 K=2200. Repetitive elements were masked in advance. We then filtered out self-alignment results (>0.85 similarity and >1000 aligned bp) and obtained more credible align block pairs. Finally, we realigned and filtered these block pairs (>0.9 similarity and >1000 aligned bp), producing a final version of SD regions in each genome. In summary, we identified 59.64 Mb (coving 4,367 genes) and 66.05 Mb (coving 4,864 genes) segmental duplications in mallard and spot-billed genome respectively, which are much larger than Beijing duck's 8.39 Mb (coving 1,133 genes). We further did pair-wise alignment among SDs detected in all these three assemblies to identify SDs specific to each of them.

4. Resequencing and population genomic analyses

4.1 Polymorphism identification, accuracy verification and its combine between populations

We produce BAM file for each individual sample or population pool according its sequencing strategy (Table S11). VCF files for each population were using “UnifiedGenotyper” of GATK by taken all processed BAM files of individual samples as input, or taken pool BAM file of a population as input. And all BAM files were processed by “REMOVE_DUPLICATES”, “use unique mapped reads” and

“IndelRealigner”. In the functional validation section, 10 SNP sites were selected to be used to experimentally verify its functional activity. 9 of these 10 were true SNPs but one was INDEL instead. We think that these results could reflect the high accuracy of SNP calling in some extent. In the following selection analyses or $\partial a \partial i$ simulation, all SNPs are from regions that homologous to chicken autosomes. We combine the VCF files by using a way like outer join of SQL (Structured Query Language) (<https://github.com/vitogump/lifelit>). For the variations that present in one VCF but absent in the another one, we fill it as it fixed as ref allele when the coverage (information from BAM) of this site reached the threshold.

4.2 PCA, genetic structure and PSMC analysis

SNPs used in both PCA and genetic structure analysis were filtered using PLINK[22] (--indep-pairwise 500 50 0.4 --maf 0.05). First, we want to know the genetic relationship of all 8 wild duck species and these domestic populations as mentioned in main text Fig. 2a, which provided evidence that mallard duck and spot-billed duck having a far more closely genetic relationship with domestic duck than other wild species. To eliminate the unequal sample size effect, assembly difference effect and autosomes/sex difference effect, we supplemented PCA analysis by using same sample size for each population, with both mallard duck assembly and Beijing duck assembly (Figure S13). These results further supported our conclusion.

Second, in our genetic structure analysis, it is interesting that domestic lineage separated from wild duck first, and even subdivided into Beijing and Shaoxing previous than the mallard duck and spot-billed duck’s subdivision. To further investigate this phenomenon and eliminate the coverage effect. We perform genetic structure analysis by using equal sample size and same coverage of each population with different populations combinations (Figure S14). These results also consistent with our inference. Third, in our PSMC analysis, we cutting the coverage of Beijing individuals into the same with Shaoxing individuals. They show the same curve with Shaoxing ducks (Figure S17). So, we propose that all domestic ducks should have the same PSMC curves.

Cutting sequence coverage was performed by samtools[23] (samtools view -bhs).

4.3 $\partial a \partial i$ simulation

There are two ways to compare the two phylogenetic trees (spot-billed, (mallard, domestic)) and (domestic, (mallard, spot-billed)). First, as some literates did, simulated the three populations split model and calculated the likelihoods to compare the fit of each model [24].

We think this is unsuitable to our date. Because this would take more parameters into model that would significantly increase the difficult to simulate and is very time consuming. Furthermore, the likelihoods are influenced by multiple parameters. We hardly give a property demographic model and constraint the parameters, since there is lack of reliable evidence (beside genetic data) about the real evolutionary history of ducks. Moreover, both PCA and structure analysis shows that wild duck populations are structured that is challenging with $\partial a \partial i$ because the dadi model treats populations

each as distinct homogenous entities. So, the best way is simplifying the model and making it easy to be get convergent parameters.

We used an alternated way by simulating paired divergence and comparing the distribution of divergence time. If (Figure 2e left) is the correct evolutionary relationship, then we expected longer divergence time between mallard duck and spot-billed than the ones between the two wild ducks and the domestic ducks. Otherwise if (Figure 2e right) is the correct tree topology. We would expect the divergence time between two wild duck populations and domestic population were similar and longer than the one between mallard duck and spot-billed duck.

As we only focus on divergence time, we first used an extreme simple model with only two parameters (Figure 3a left). This way would eliminate the influence between different parameters. Especially taking migration rate into simulation would have a significant impact in divergence time which could be purely due to mathematical property of *∂a∂i* software.

The distribution of each estimated parameter and the confident intervals were presented in Figure S15 and Table S12, respectively. And the comparison of divergence time between the domestic and mallard/spot-billed duck, mallard duck and spot-billed duck were shown in Figure 3b left.

We then add migration into model (Figure 3a right). The distribution of each estimated parameter was presented in Figure S16 that show bimodality distributions rather than normal, so we don't calculate the confident intervals. And the comparison of divergence time between the domestic and mallard/spot-billed duck, mallard duck and spot-billed duck under this model were shown in Figure 3b right. The maximum likelihood *T_s* between domestic and mallard/spot-billed duck, mallard duck and spot-billed duck are 333,504 years ago, 334,738 years ago and 74,915 years ago respectively.

Combined the two models' result, we inferred that the ancestor of wild duck diverged from domestic duck about 100Kyr~300Kyr, and subsequently diverged into mallard duck and spot-billed duck about 70Kyr.

In conclude, *dadi* simulation on both models, considering or not considering the migration, are support the phylogenetic tree that is inferred from our previous population structure analysis. Although *dadi* software cannot distinguish short-divergence with low migration from long divergence with high-migration, this usually cause the estimate *T_s* not convergent or convergent into two peaks (smaller *T_s* accompanied by high migration or bigger *T_s* coupled with low migration rate). In our second simulations (with migration), estimate *T_s* does increased, when compared with first model, due to introduce migration. But the *T_s* still convergent into two peaks, higher estimated *T_s* with higher estimated migration rate whereas shorter *T_s* with lower migration rate. a reasonable explanation could be: although *dadi* have difficult to determine longer or shorter *T_s*, thus having simulation on both situation and produce two peaks of estimate *T_s*, the *T_s* between mallard and spot-billed is still significant smaller than *T_s* between wild and domestic one in any case.

In additional, we add parameters to allow population change exponentially (IM_2 model). We found that many of simulations with IM model can't get the convergent estimates or get estimates with some parameters hit the boundary, especially for

divergence between mallard duck and spot-billed duck. This may be because as the number of parameters (2 for “split_no_M” model, 4 for “split_M” and 7 for “IM_2” model) increased, the correlation between parameters (which is a purely mathematical property of) will significantly increase the difficulty to getting the correct estimates. And the two wild populations are highly structured and exist many fixed difference sites which is particular challenging with $\partial a \partial i$. Furthermore, the distribution of each parameters shown two or more peaks. So, we hardly estimated results. But we found that 29 of 39, 53 of 68 bootstrap fits yield increased domestic population size change after they split with mallard duck/spot-billed duck. All bootstrap fits yield increased wild duck population size change.

(data not show)

```
def split_nm(params,ns,pts):
```

```
    s,Ts=params
```

```
    xx=dadi.Numerics.default_grid(pts)
```

```
    phi=dadi.PhiManip.phi_1D(xx)
```

```
    phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
```

```
    phi=dadi.Integration.two_pops(phi,xx,Ts,nu1=s,nu2=(1-s))
```

```
    fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
```

```
    return fs
```

model code1: split without migration

```
def split_m(params,ns,pts):
```

```
    s,Ts,m12,m21=params
```

```
    xx=dadi.Numerics.default_grid(pts)
```

```
    phi=dadi.PhiManip.phi_1D(xx)
```

```
    phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
```

```
    phi=dadi.Integration.two_pops(phi,xx,Ts,nu1=s,nu2=(1-s),m12=m12,m21=m21)
```

```
    fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
```

```
    return fs
```

model code2: split with migration

```
def IM_2(params,ns,pts):
```

```
    s,nu1,nu2,TS,m12,m21=params
```

```
    xx=dadi.Numerics.default_grid(pts)
```

```
    phi=dadi.PhiManip.phi_1D(xx)
```

```
    phi=dadi.PhiManip.phi_1D_to_2D(xx,phi)
```

```
    nu1_func=lambda t: s *(nu1/s)**(t/TS)
```

```
    nu2_func=lambda t: (1-s) *(nu2/(1-s))**(t/TS)
```

```
    phi=dadi.Integration.two_pops(phi,xx,TS,nu1=nu1_func,nu2=nu2_func,m12=m12,m21=m21)
```

```
    fs=dadi.Spectrum.from_phi(phi,ns,(xx,xx))
```

```
    return fs
```

model code2: split with migration and population exponentially change.

full program is provided in <https://github.com/vitogump/lifelit>.

4.5 Selection analyses

As the long time from the early selection occurrence in wild lineage, the accumulation of recombination, mutation and the other population effects may obscure the selection region. We use a relative narrow window width to detect early stage selection. We filtered variation that passed the following criteria:

- 1) No CpG sites
- 2) At least 5 individuals were observed in each population or 10 reads were observed for pool populations
- 3) At least 20 reads were observed when the site was fixed in a population
- 4) Derived allele was detected in both wild and domestic lineage

For windows with less than 10 variation sites were discarded.

In Hp calculation, we didn't require min SNP numbers because the region with extreme little SNP could be Runs of homozygosity (ROH) that is also our interest.

For the regions under early stage selection in wild lineage or under recent selection in domestic lineage. We performed more detail analysis to visualize the genetic variations in Figure S23, S24.

The comparison between the details of selection regions in Figure S23, S24 reflect that the early stage selections mostly couldn't be visualized by current variations directly. But regions under recent selection in domestic lineage showed clear trace.

GO enrichment analysis were performed by using Fisher's Exact Test (see <https://github.com/vitogump/lifelit>).

In the df calculation, we required all sampled individuals were sufficient covered for each df sites.

Whole genome alignment to chicken were performed by LASTZ[25].

4.6 select candidate casual mutation to verify its functional activity

For the sake of simplicity, we catalog the selection regions by its overlap with CDS, UTR, intron, intergenetic region. And rank SNPs by the ΔAF between wild lineage and domestic lineage. we selected the SNPs that located in UTR, intron, intergenetic region and closed to the genes that may play important functional in the duck evolution or domestication.

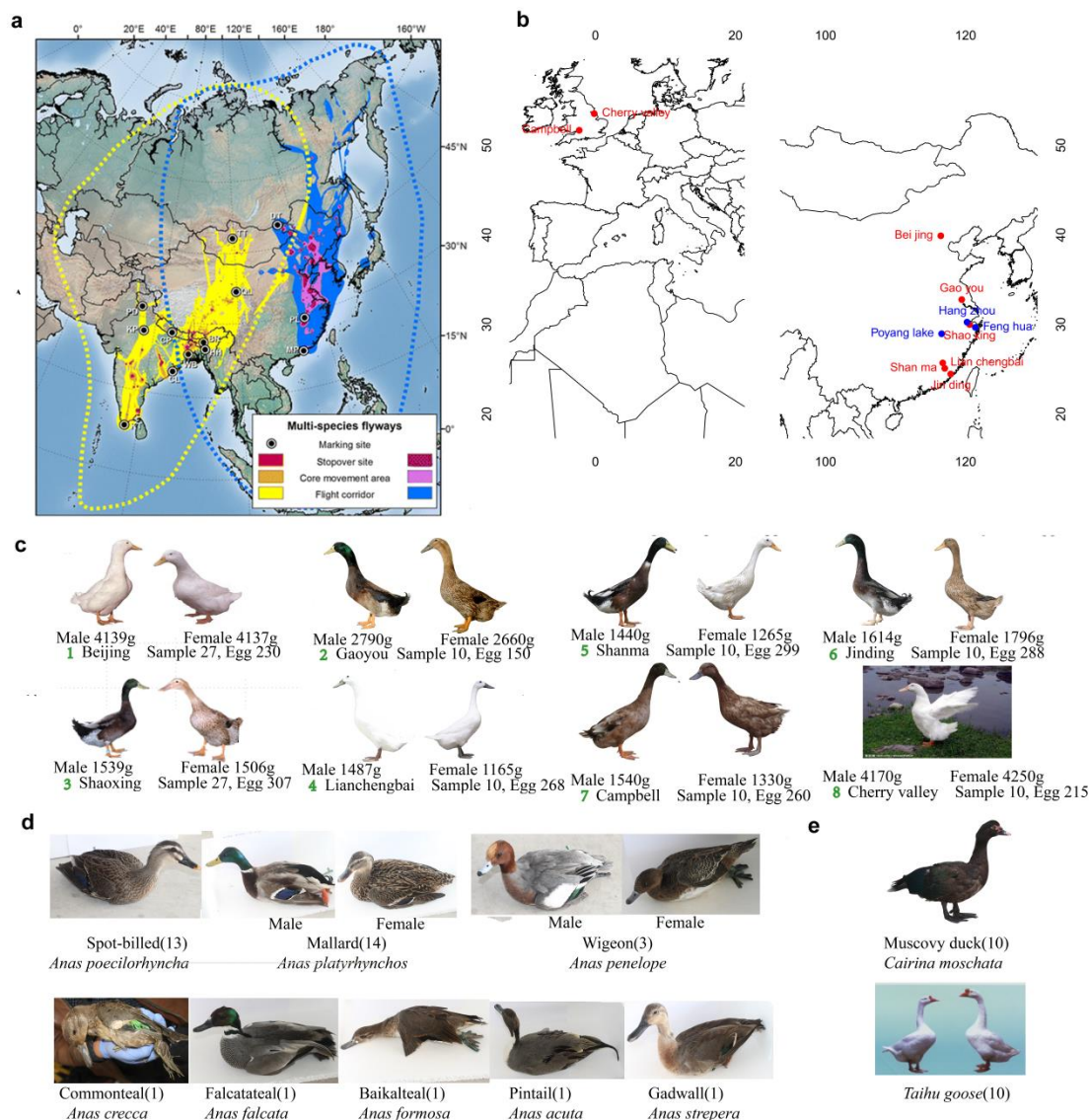


Fig. S1. Sample distribution and phenotypic variation in domestic and wild ducks. Photos were taken from the sequenced samples except Taihu goose and Cherry valley being download from web. **(a)** Estimated migration routes of Anatidae in Central Asian Flyway (CAF) and East Asian-Australasian Flyway (EAAF)[26]. Regions in the flyway of CAF displayed in yellow-red and for EAAF displayed in blue-purple. From darkest to lightest, colors represent 50%, 75% and 99% cumulative probability contours. **(b)** Geographical distribution of wild duck and domestic duck samples. Blue dots represent the place where wild duck samples collected from. Compare to **a**, our wild duck samples were collection from the main migration route of the Anatidae flyway. **(c)** Phenotype of the eight domestic duck breeds. Number of individuals, average weight (male/female), average eggs production per year were marked. **(d)** Photos and sample numbers of the eight wild duck species. **(e)** Photos and sample number of Muscovy duck and Taihu goose used as outgroup.

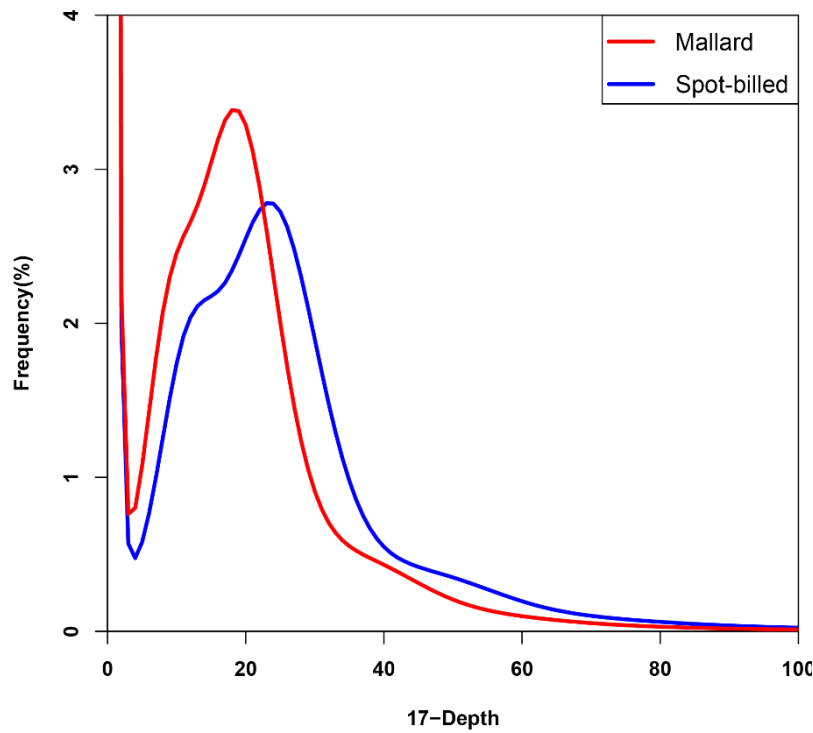


Fig. S2. Distribution of 17-mer frequency in the corrected pair-end reads of the mallard and the spot-billed genome.

Only the reads from short-size libraries (< 500 bp) were included in this analysis. We identified 22,312,838,068 kmers using 21-fold data of mallard and 28,137,616,208 kmers using 27-fold data of spot-billed. The D2B and D2L genome size can be estimated as $G = \frac{K_number}{K_depth}$ (the volume peak), which is estimated to be 1.14Gb and 1.15Gb, respectively.

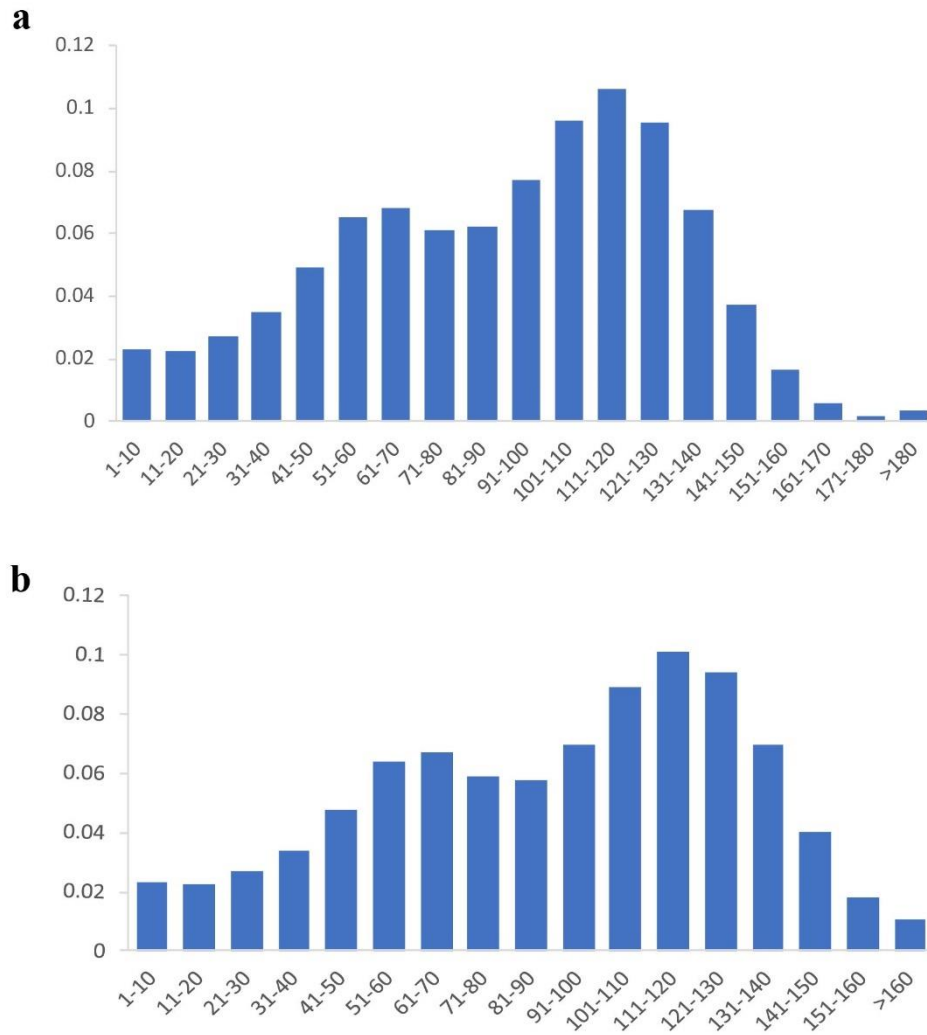


Fig. S3. Distribution of the sequencing depth of the mallard and the spot-billed assemblies.
 (a) the mallard assembly. (b) the spot-billed assembly.

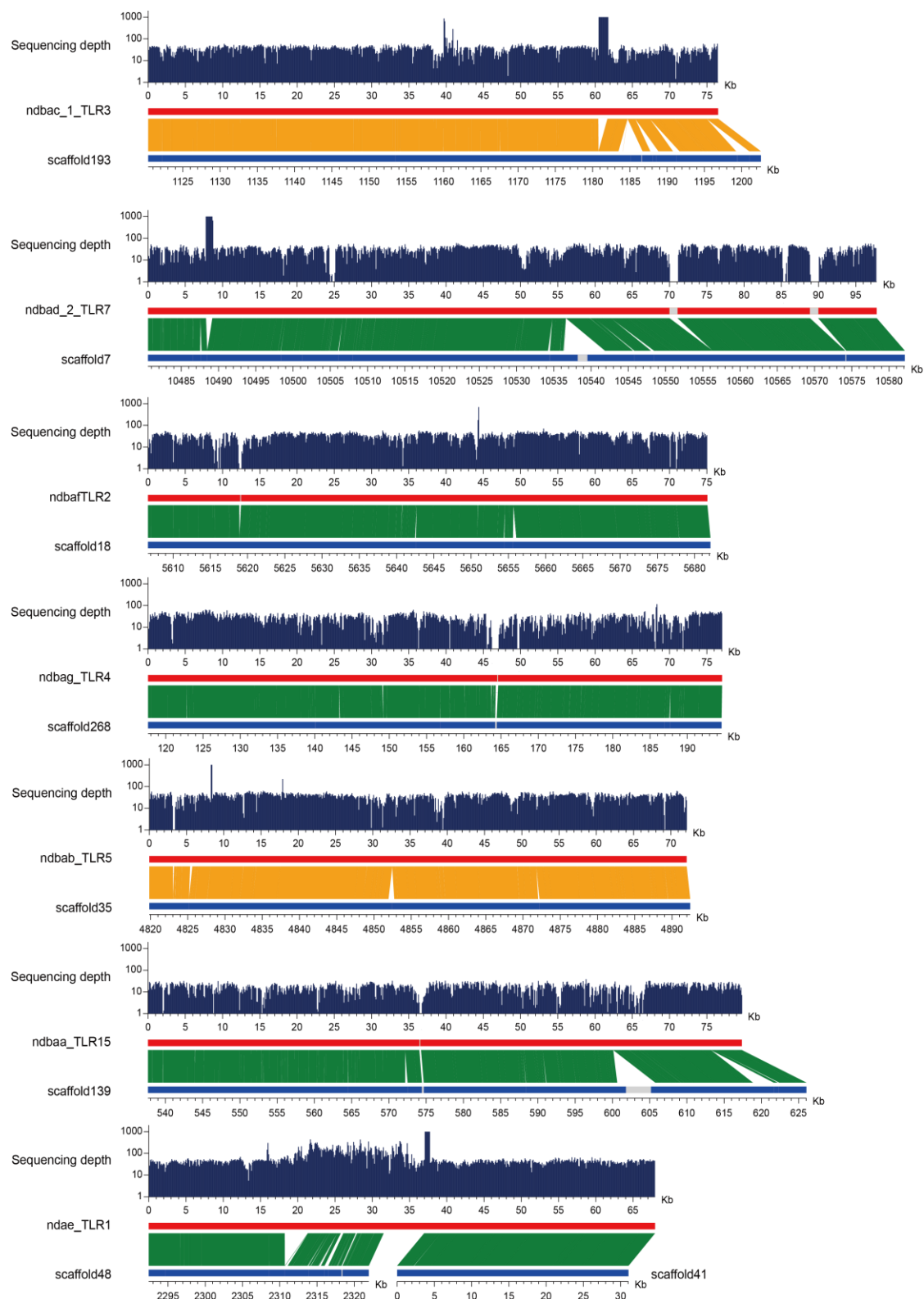


Fig. S4. Evaluation of the mallard assembly with sequences of seven BACs. Seven BACs were from chromosome 1, 3 or 4. The BAC sequences and assembly scaffolds are shown in red and dark blue, respectively. Orange represents positively and green was reversely aligned, the remaining unclosed gaps on the scaffolds are marked as white blocks.

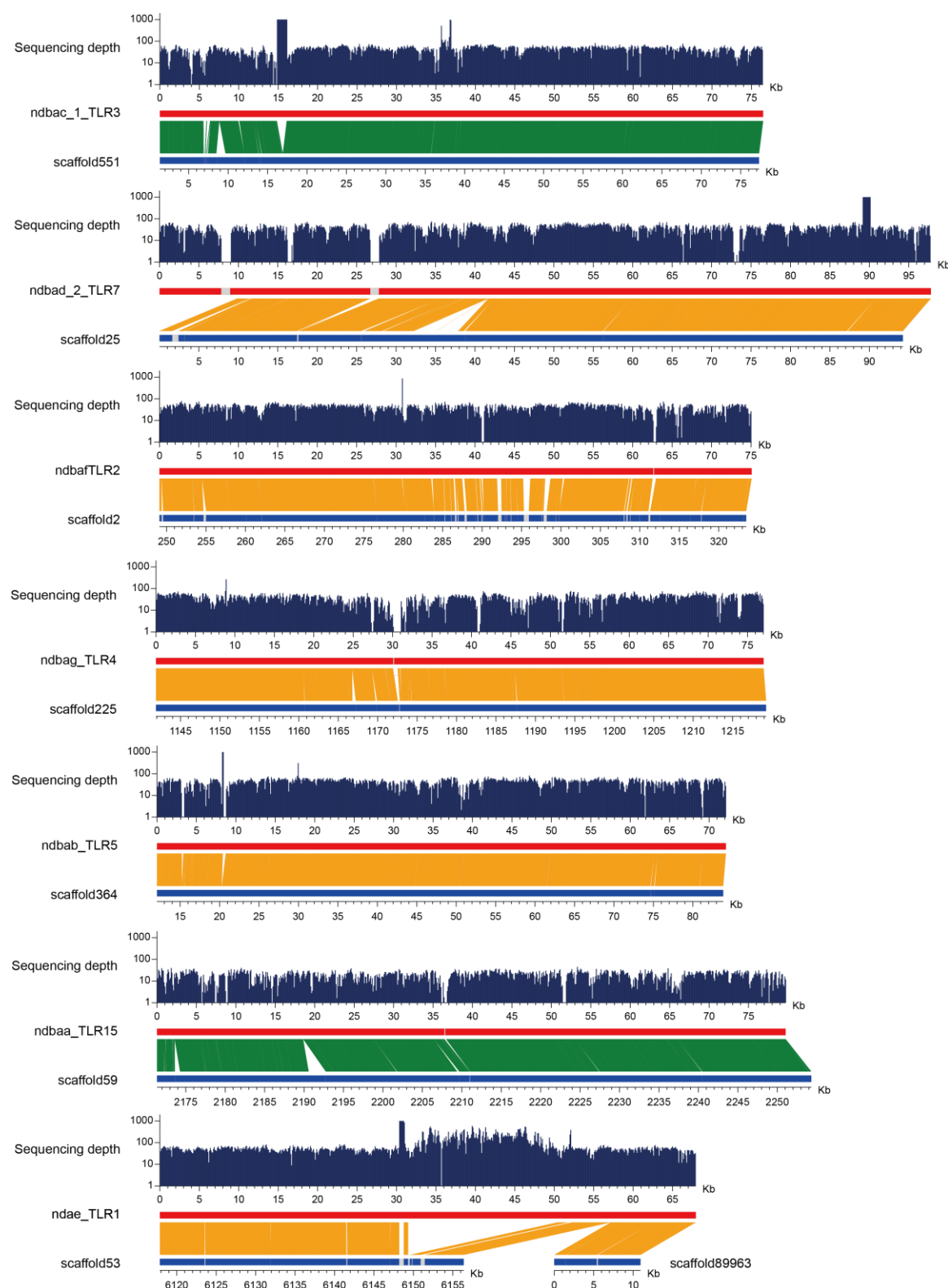


Fig. S5. Evaluation of the spot-billed assembly with sequences of seven BACs.
 Seven BACs were from chromosome 1, 3 or 4. The BAC sequences and assembly scaffolds are shown in red and dark blue, respectively. Orange represents positively and green was reversely aligned, the remaining unclosed gaps on the scaffolds are marked as white blocks.

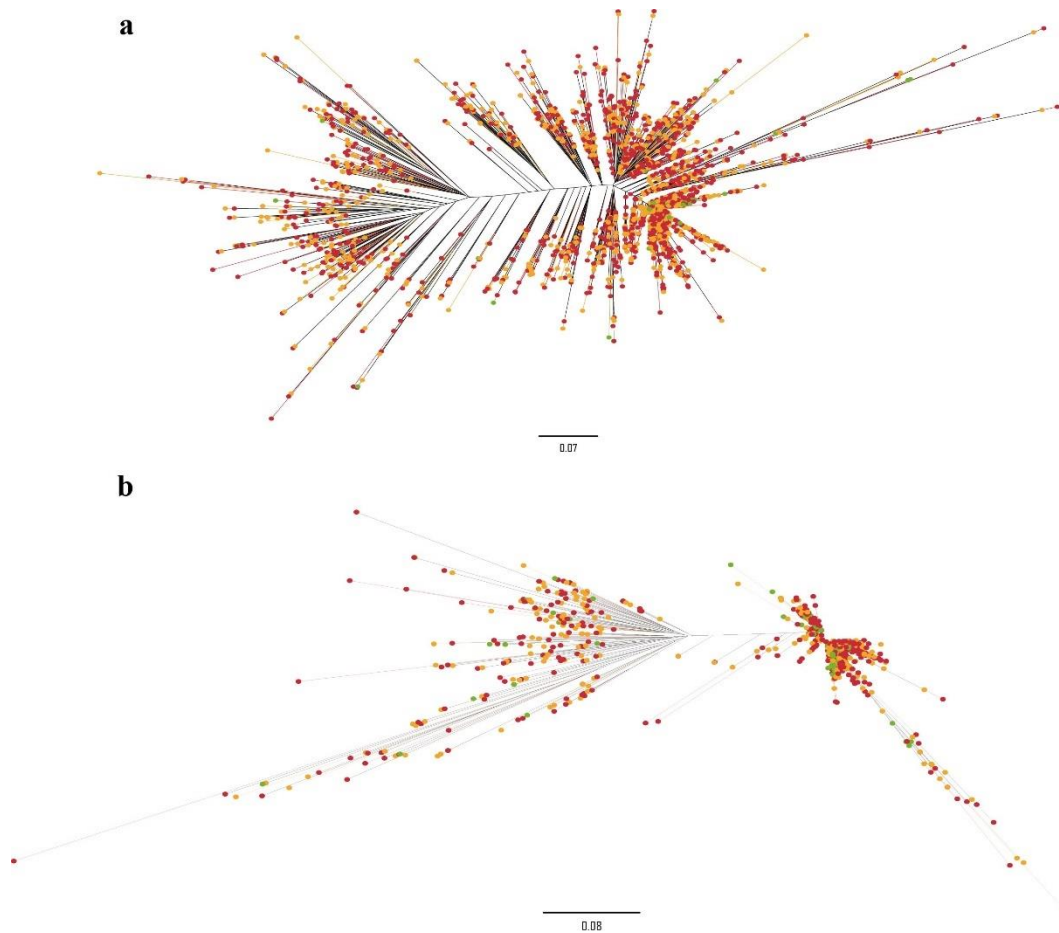


Fig. S6. Phylogeny of two CR1 retrotransposon clade in four bird genomes.
Green, red and yellow represent Beijing duck, mallard and spot-billed, respectively.

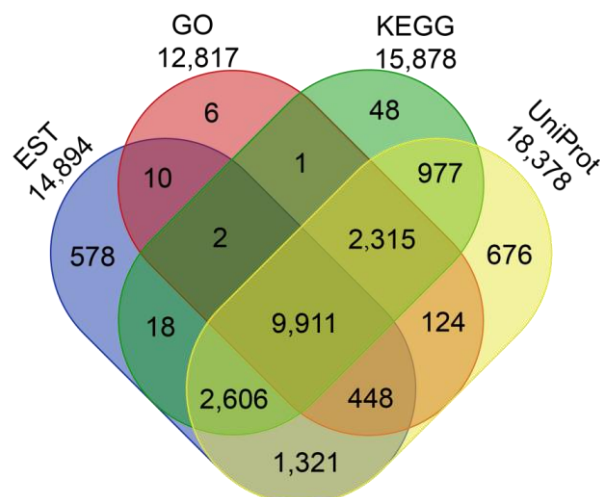


Fig. S7. Venn diagram showing the mallard reference genes annotated using three databases or supported by EST.

Total 21,056 mallard genes supported by 319,996 ESTs of Beijing duck and functioned annotation with three databases. And 19,041 genes were supported by one of them.

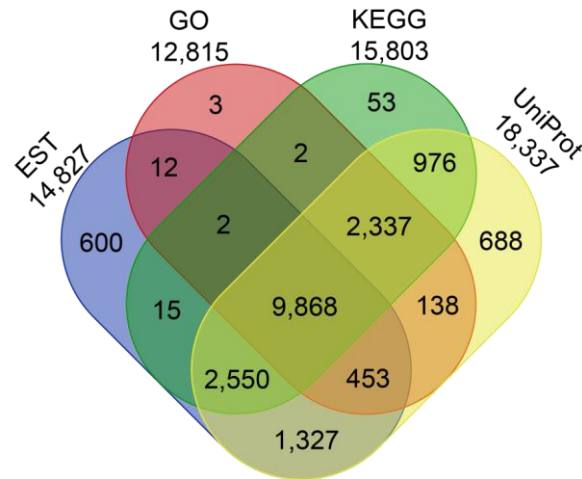


Fig. S8. Venn diagram showing the spot-billed reference genes annotated using three databases or supported by EST.

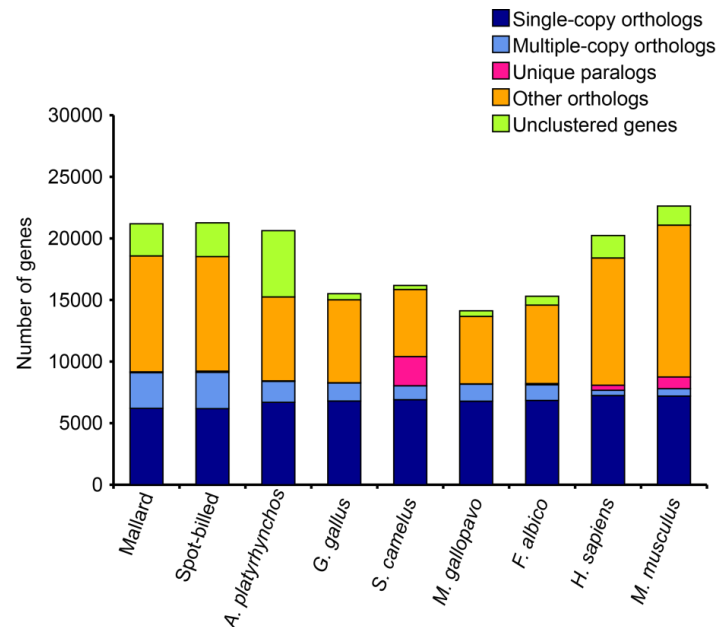


Fig. S9. Protein homologous comparison among the reference gene sets of nine species by TreeFam.

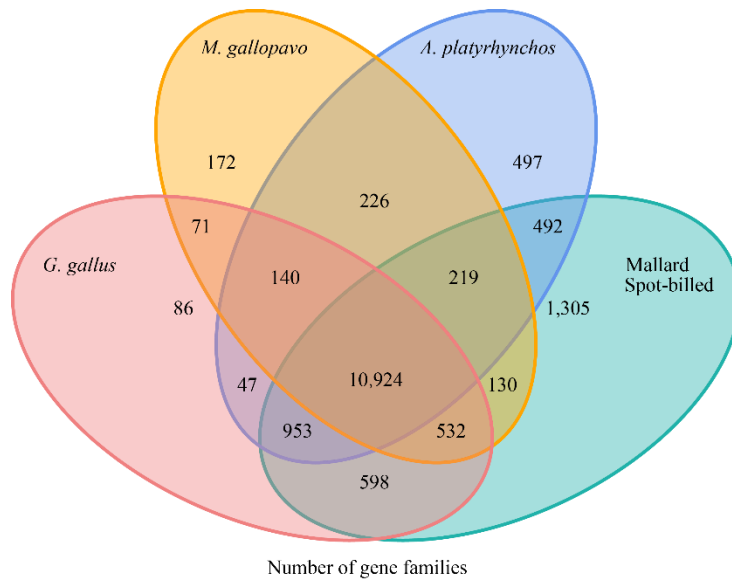


Fig. S10. Venn diagram shows gene family clusters in five species by TreeFam.

The number of unique and shared gene families among five genomes.

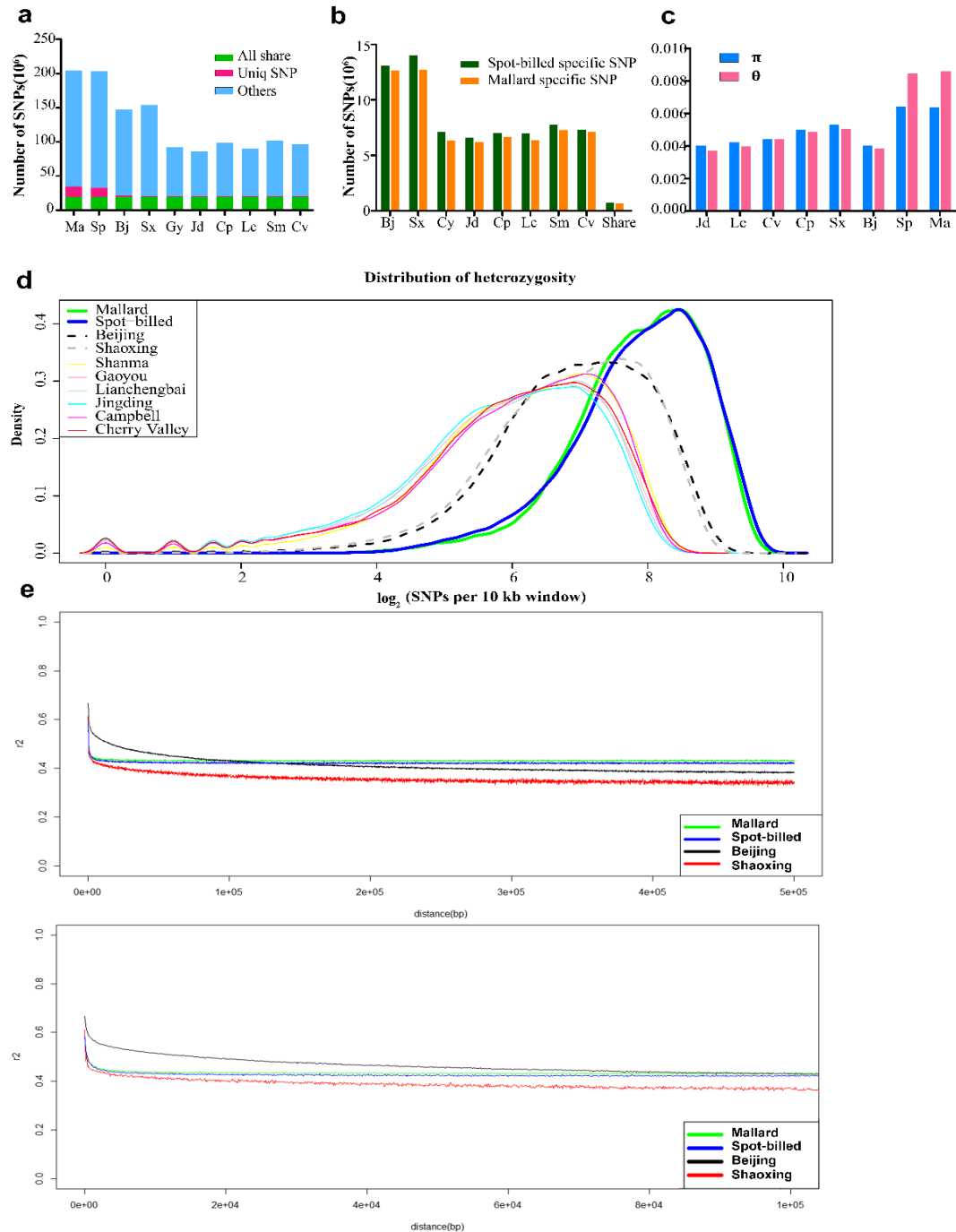


Fig. S11. Distribution of genetic variation and characteristics for two wild and eight domestic duck populations.

Mallard, Spot-billed, Beijing, Shaoxing, Gaoyou, Jinding, Campbell, Lianchengbai, Shanma and Cherry valley are represented by Ma, Sp, Bj, Sx, Gy, Jd, Cp, Lc, Sm, Cv. (a) Total number of SNPs. (b) Total number of mallard and spot-billed specific SNPs segregating in each domestic population. (c) Genetic diversity was measured by nucleotide diversity π and Watterson's estimator Θ . (d) Distribution of heterozygosity. (e) LD decay measured by r^2 estimator against distance (50 kb up and 10 kb down) between the SNPs in mallard, spot-billed, Beijing and Shaoxing populations.

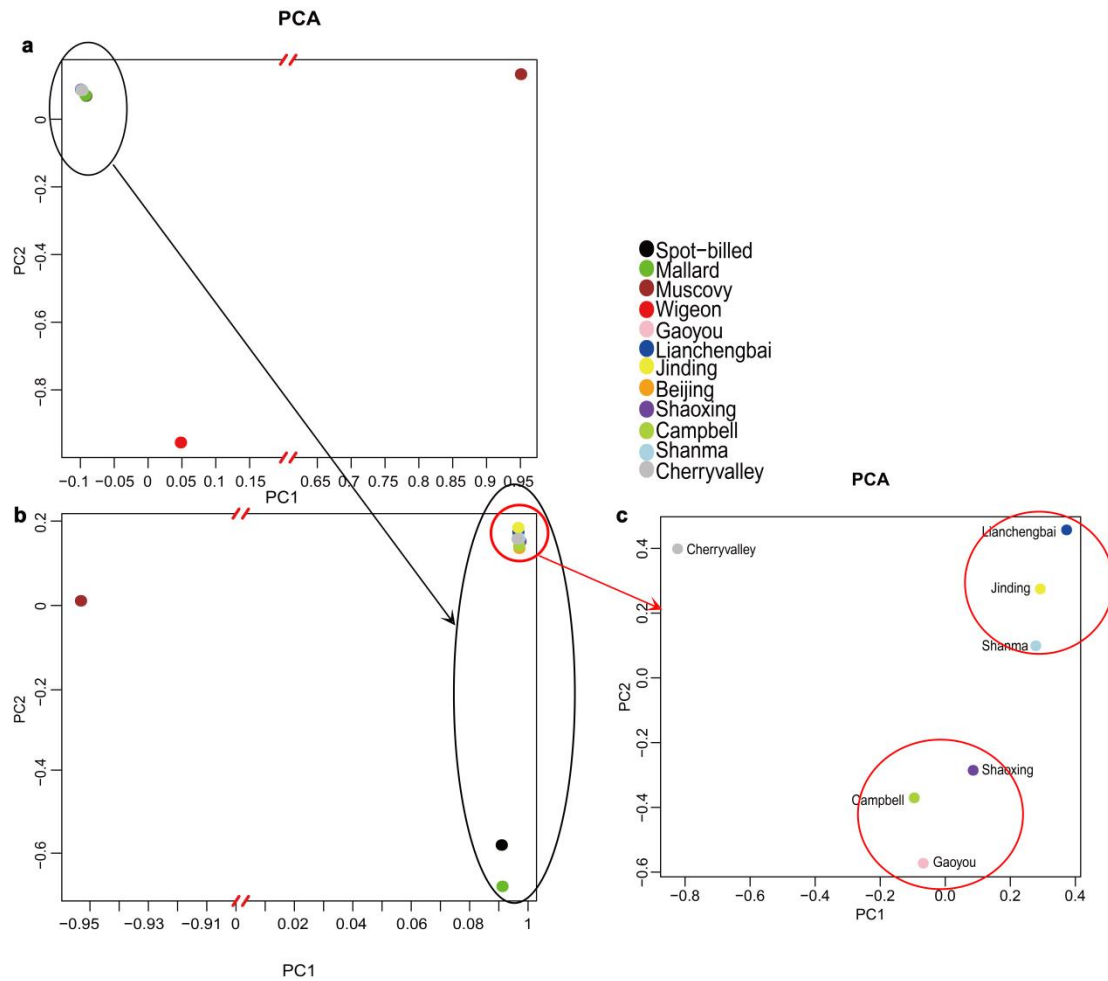


Fig. S12. Principal component analysis of wild and domestic ducks.

(a) PCA analysis on 12 populations. Two wild duck (mallard and spot-billed) and eight domestic duck populations were clustered in the top left, while the wigeon populations (consist of 3 individuals) and the muscovy duck population showing large distance with eight domestic duck populations. (b) PCA analysis on eight domestic duck populations and two wild duck (mallard and spot-billed) populations along with one muscovy duck population. It shows that 8 domestic duck populations cluster together and clearly separated from Mallard and Spot-billed in the second component. (c). PCA analysis on seven domestic duck populations. It shows that these populations clustered according to their geographical distributions.

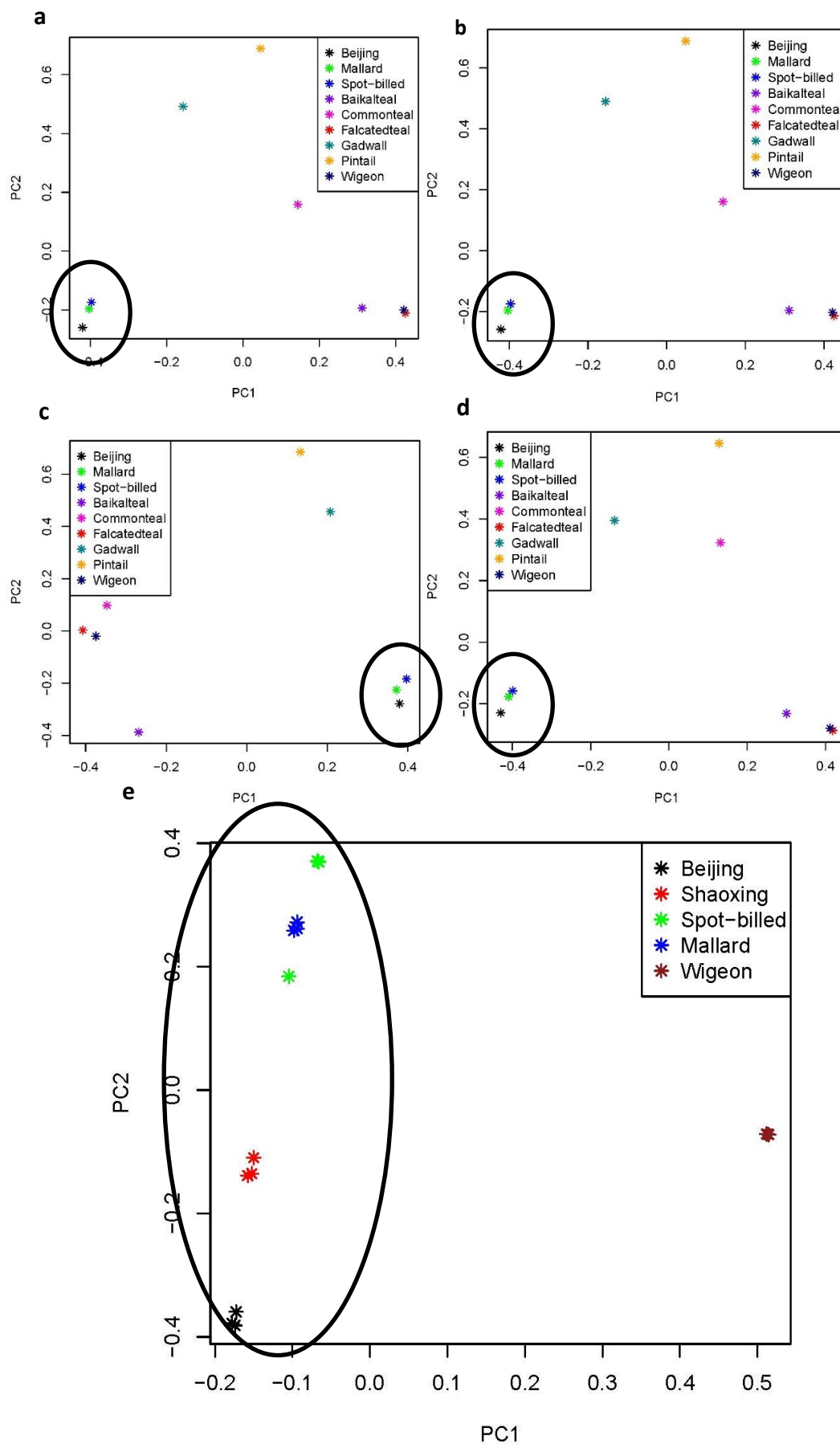


Fig. S13. Principal component analysis of wild and domestic duck with same

coverage and sample size for each population.

(a-d) shows PCA analysis on eight wild species (one individuals each, random sample reads into 12× that is equal to Beijing individual) and a Beijing individual with different data process strategy. a) using LD pruned SNPs from all scaffolds of Beijing assemble. b) using LD pruned SNPs from the scaffolds regions which were homologous to chicken autosomes. c) using LD pruned SNPs from the scaffolds regions which were homologous to chicken Z chromosome. d) using LD pruned SNPs from all scaffolds of mallard duck genome assemble. e) 3 individuals for each of the 5 populations (mallard duck, Spot-billed, Beijig, Shaoxing and wigeon, random sample reads into 3× for each individuals).

Black cycles indicate spot-billed, mallard duck and domestic duck are clustered together.

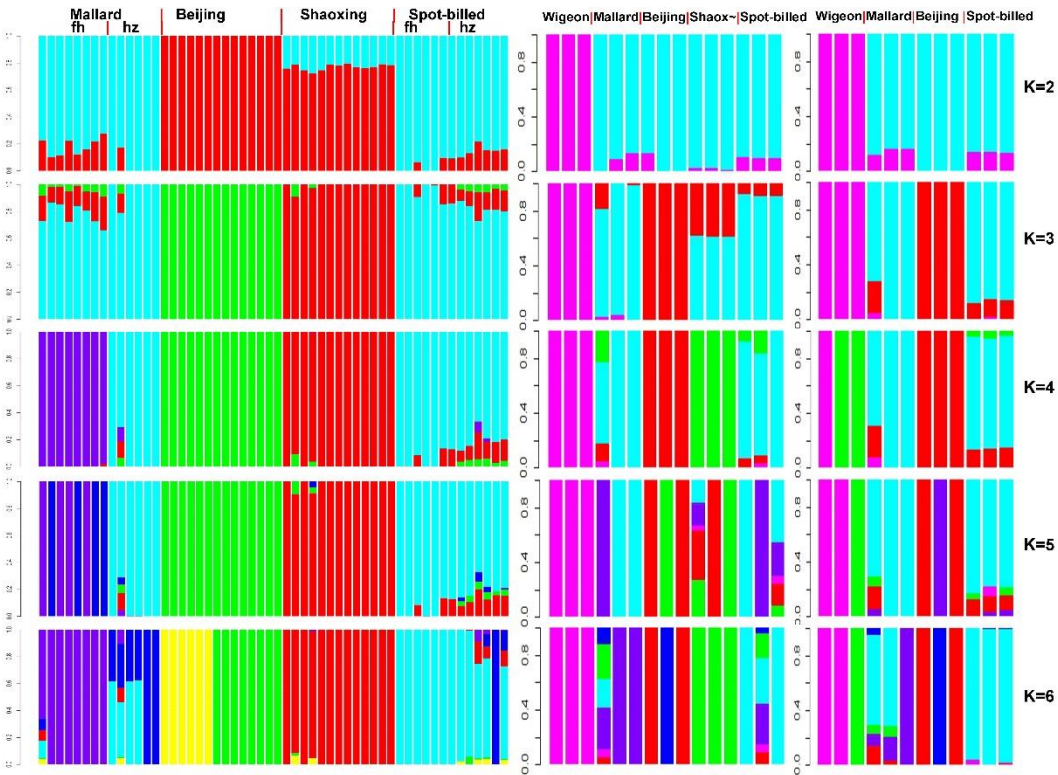


Fig. S14. Genetic structure analysis with equal sample size and same coverage of each population for different populations combinations.

Samples in left panel were random sampled to 3-fold coverage. Beijing duck samples were from a designed population, which were from a same population and formed a high growth line and low growth line after several generations selection. Samples in middle panel were random sampled to 3-fold coverage and three samples for each of the 5 populations. Samples in the right panel were random sampled to 12-fold coverage and three samples for each of the 4 populations. K range from 2 to 6.

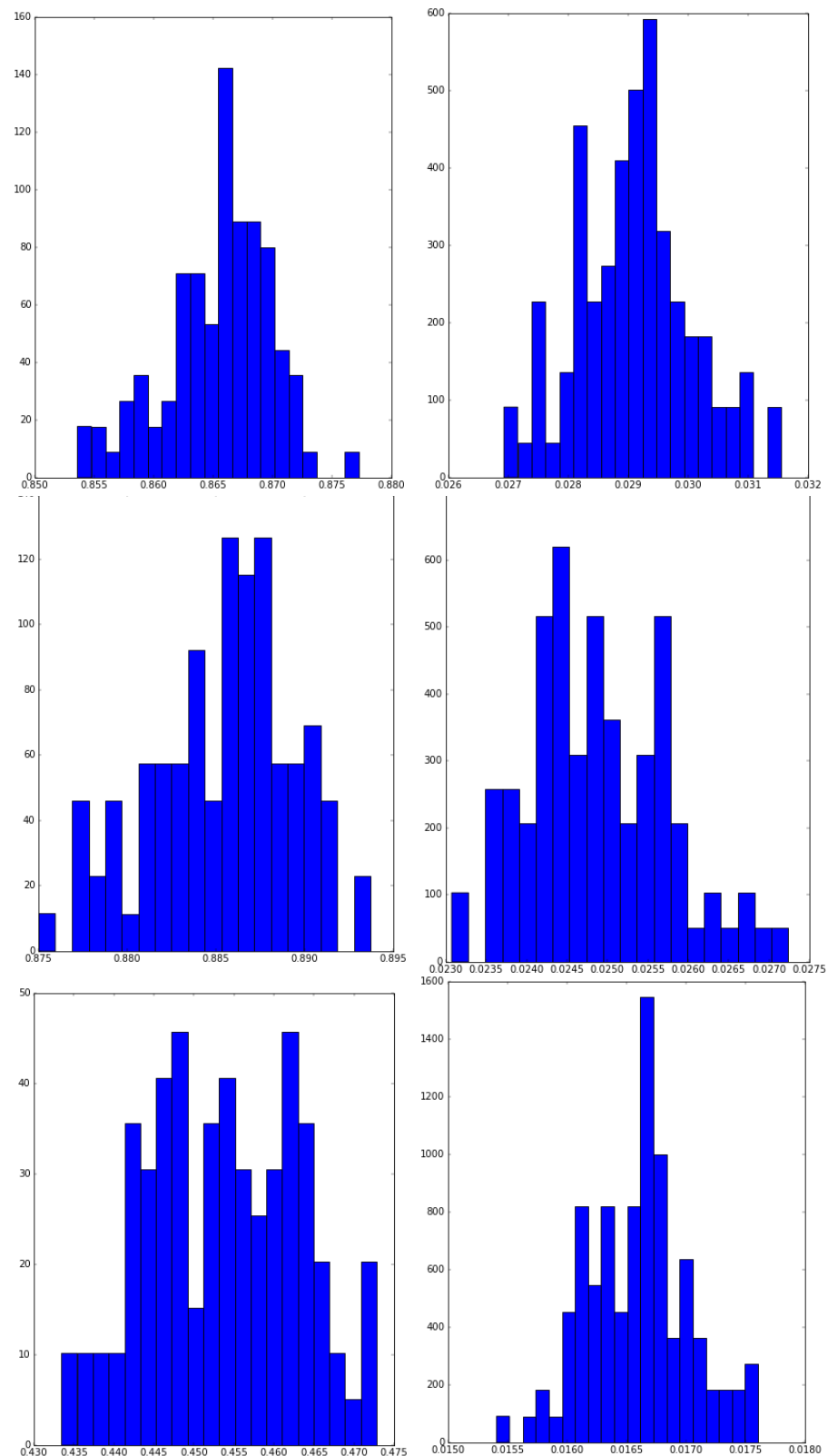


Fig. S15. Split without migration model conventional bootstrap results.

Panels from up to down corresponding to simulations of mallard and domestic duck divergence, spot-billed and domestic duck divergence, mallard duck and spot-billed duck divergence, separately. Left picture of each panels shows the distribution of s , whereas right one shows the distribution of T_s .

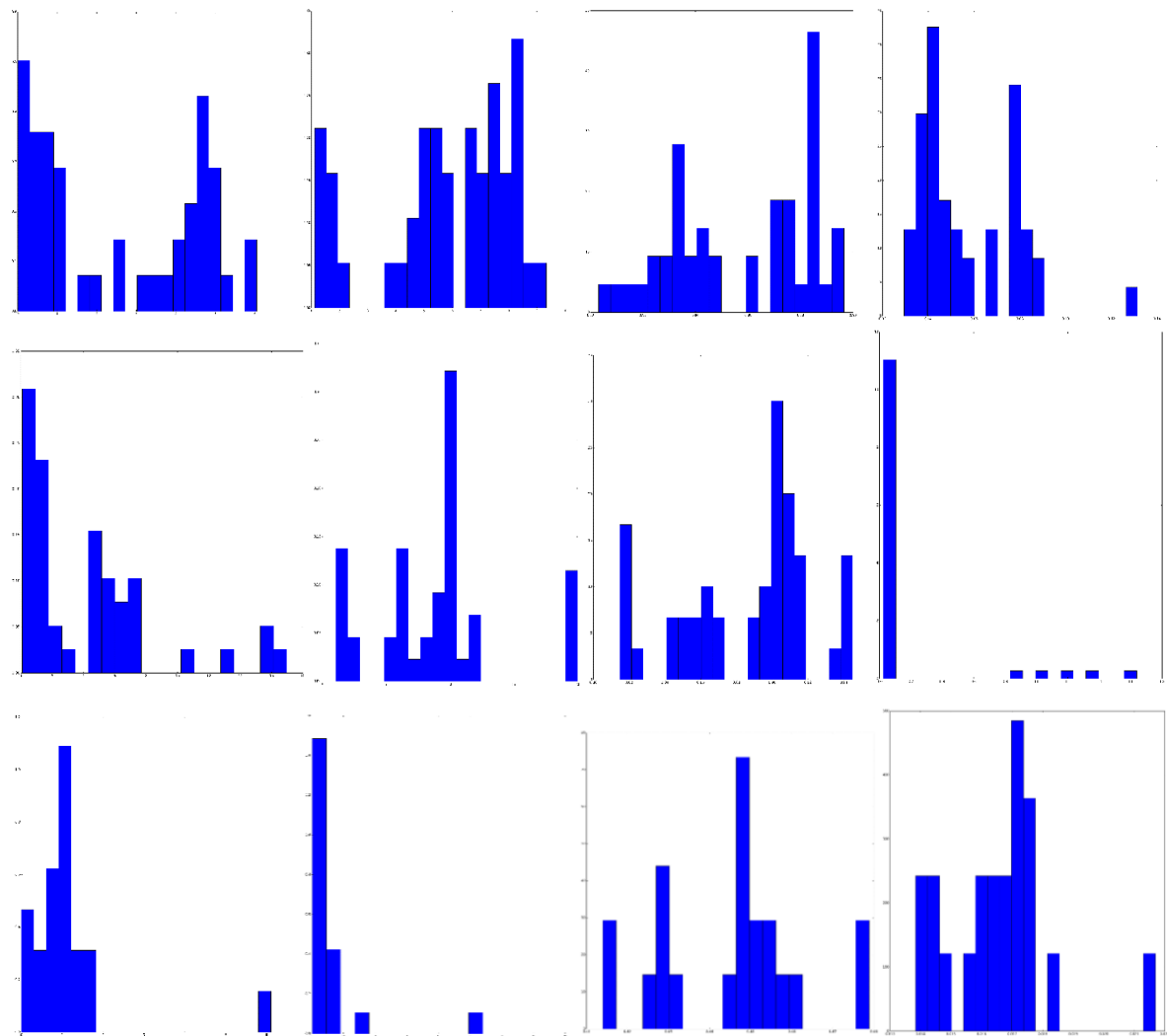


Fig. S16. Split with migration model conventional bootstrap results.

Panels from up to down corresponding to simulations of mallard and domestic duck divergence, spot-billed and domestic duck divergence, mallard duck and spot-billed duck divergence, separately. Columns from left to right corresponding the distributions of m12, m21, s, Ts parameters.

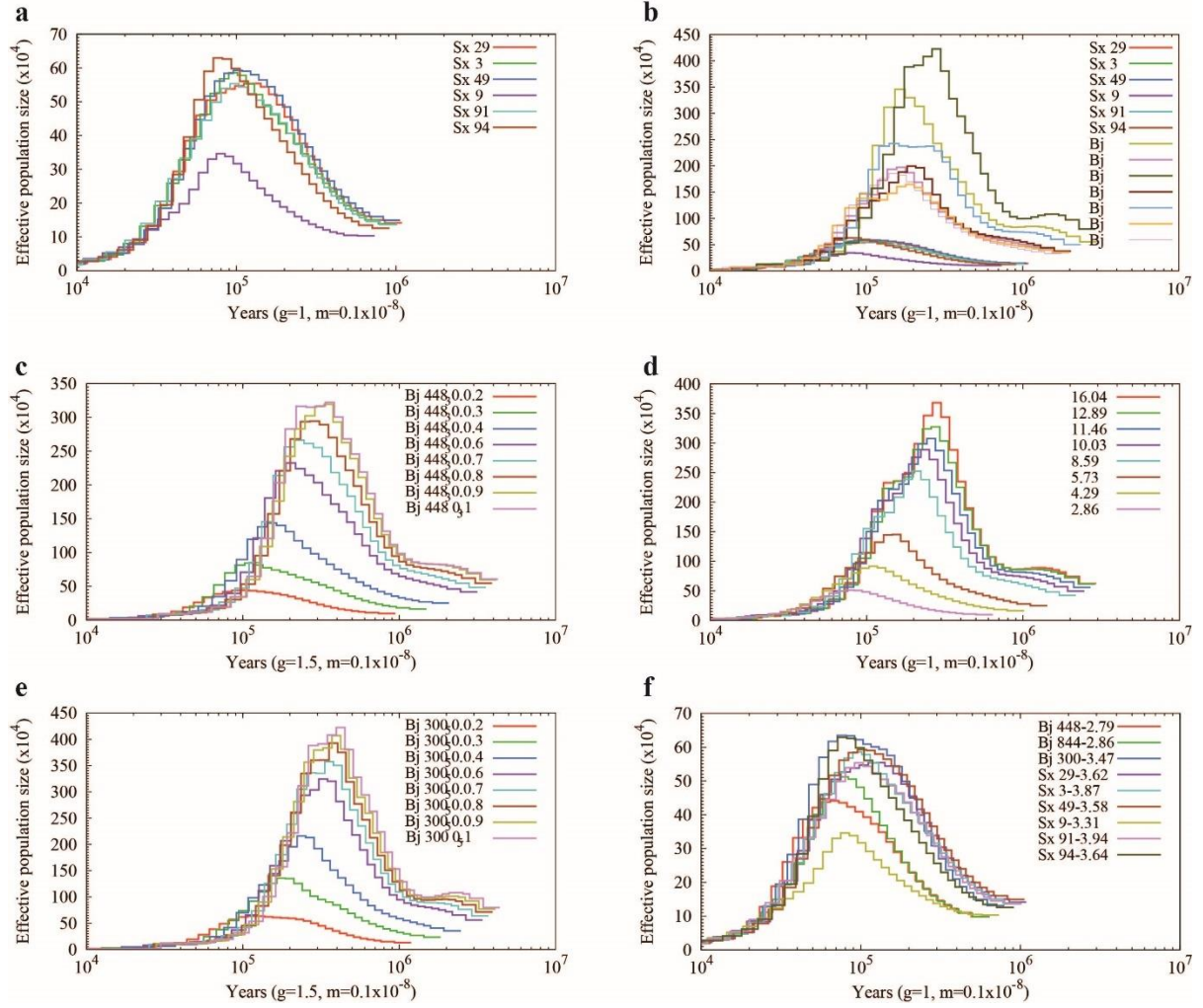
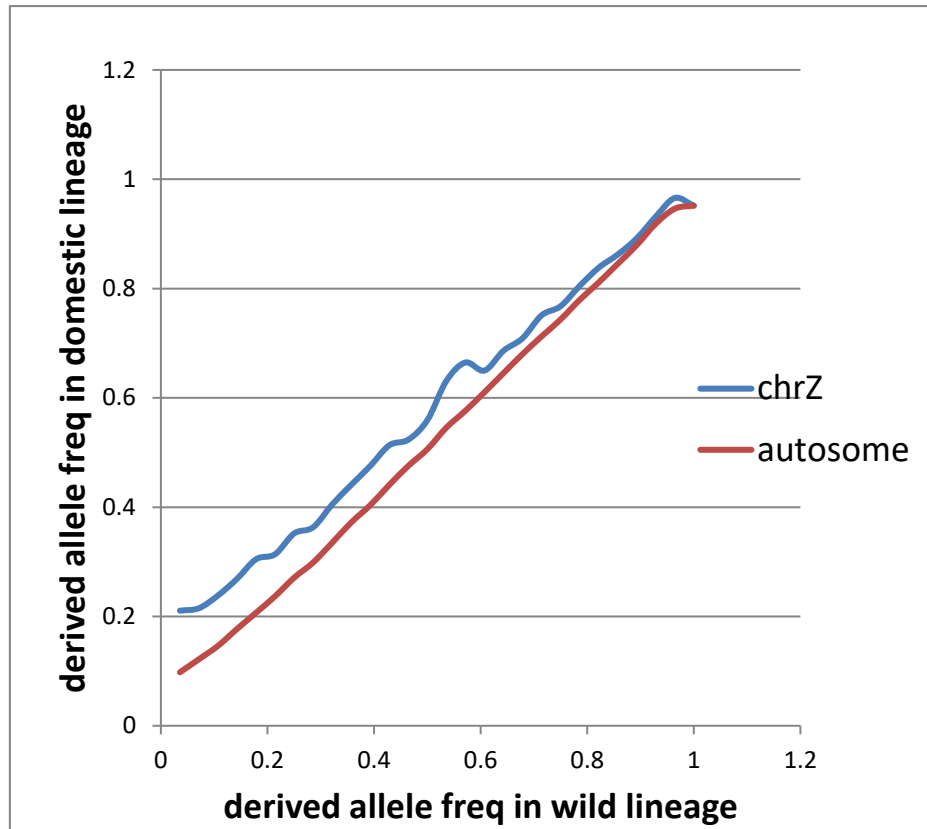


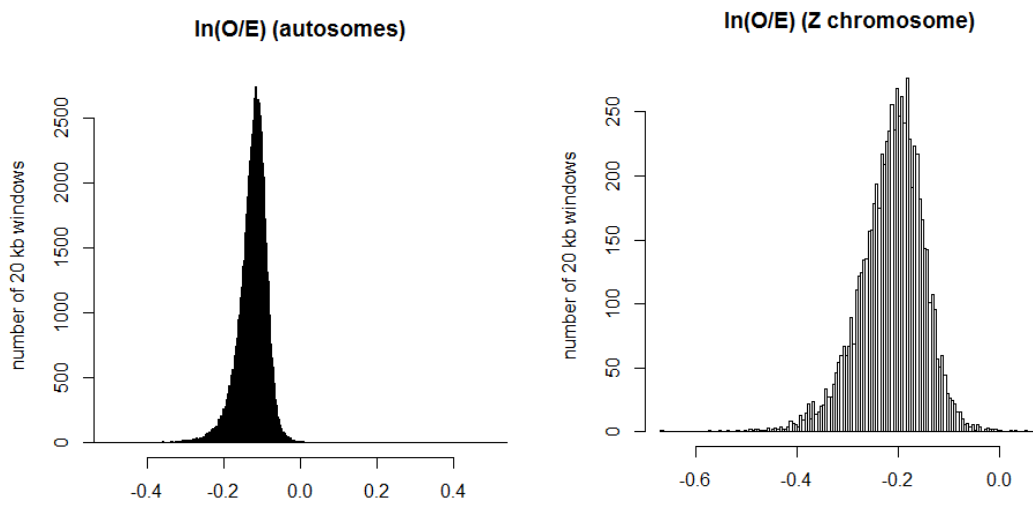
Fig. S17. Comparison of PSMC between Beijing (Bj) and Shaoxing (Sx) ducks with different coverage data.

(a) Demographic history of six Shaoxing ducks with a coverage of 2- to 3.5-fold. (b) Comparison demographic history of six Shaoxing (2- to 3.5-fold) and Beijing (13.5-fold) ducks. (c-e) PSMC analyses of Beijing or Shaoxing ducks based on different coverage of random sample reads. Sample ID in c, d and e are 44448_30, 43844_32, 41300_50, respectively. Coverage are represented in different color lines in legend. (f) PSMC curve of Beijing and Shaoxing ducks inferred with comparable coverage reads are similar.

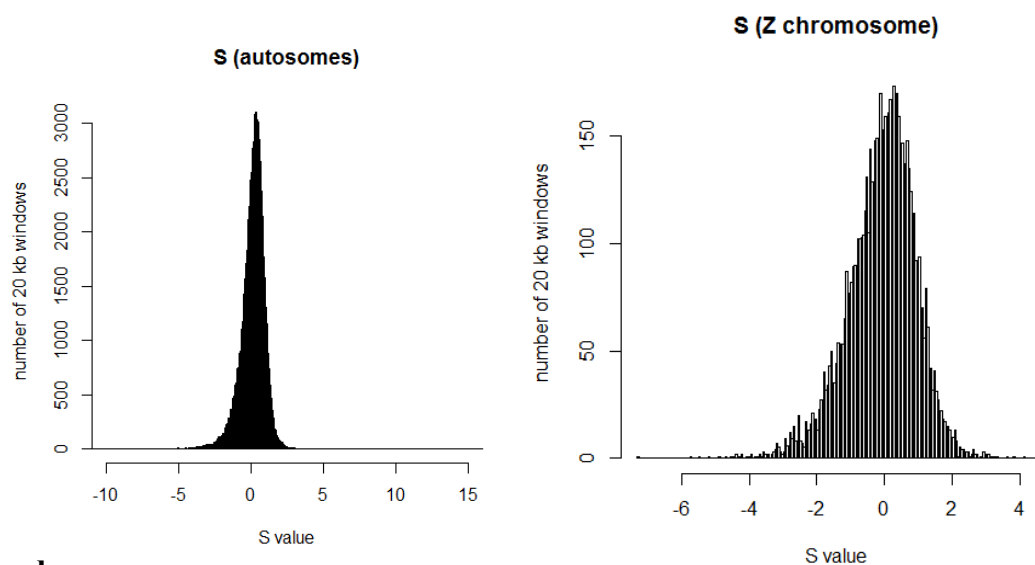
a



b

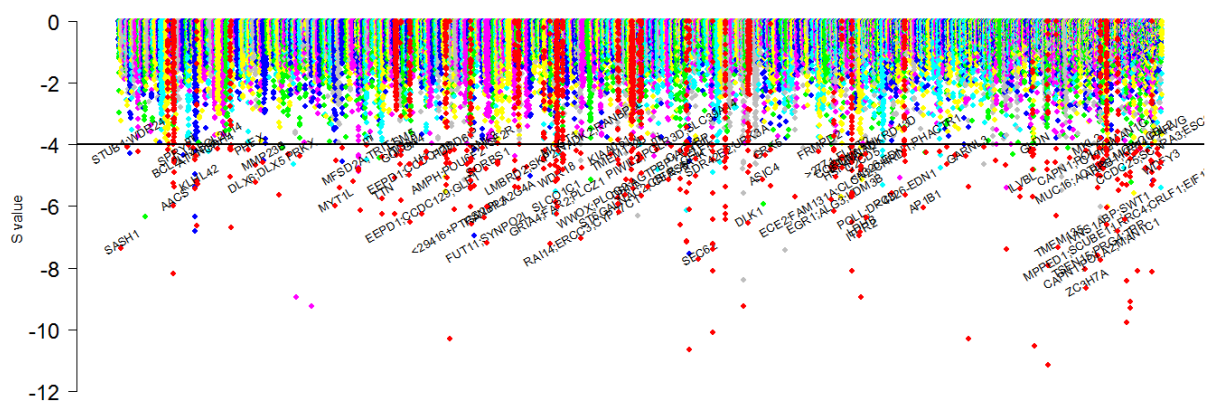


c



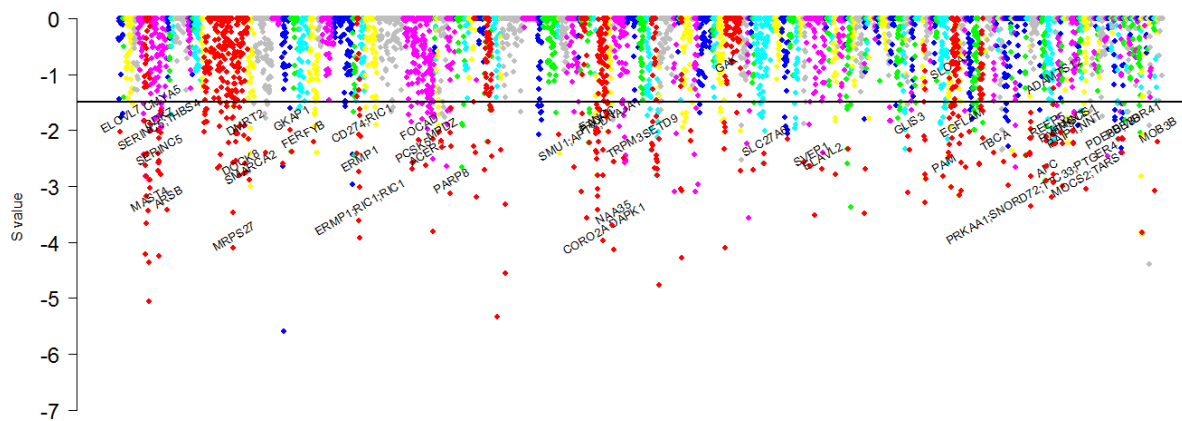
1476

d



1477

e



1478

1479

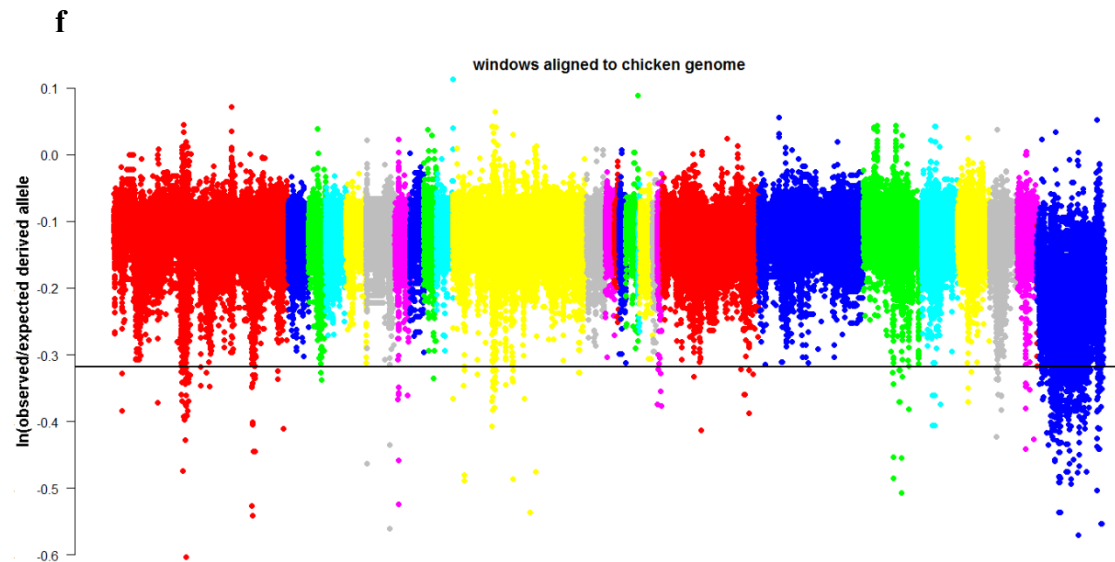
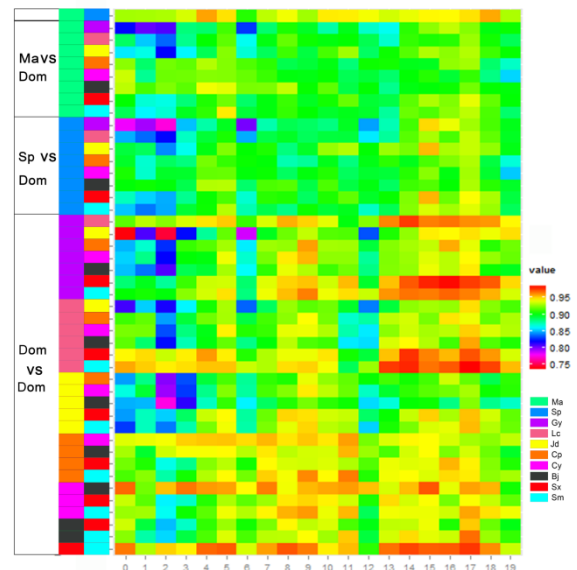
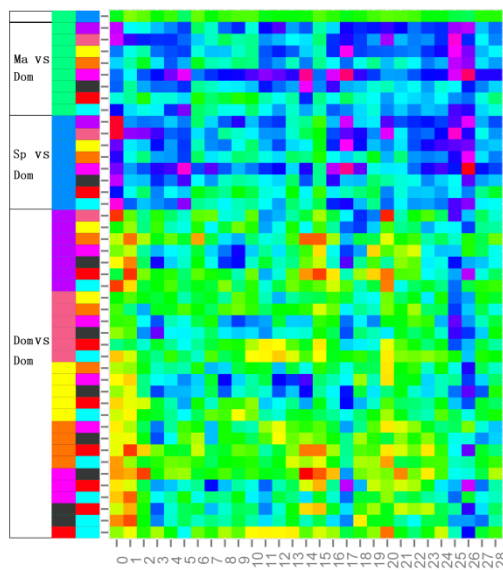
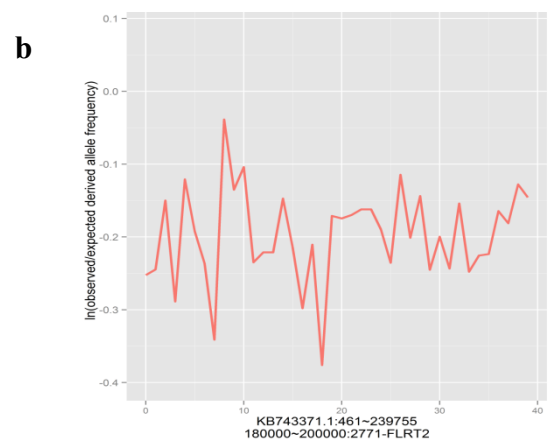
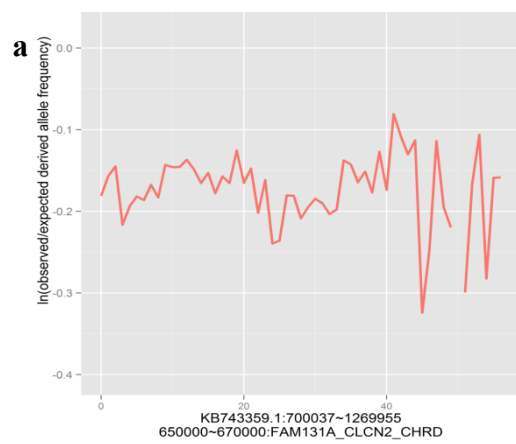


Fig. S18. Whole genome screen for S score in wild duck with 20 kb window sliding by 10 kb.

(a) Correlation of observed derived allele frequency between wild (combine mallard and spot-billed) and domestic duck (combine all 8 domestic duck breeds) lineage. (b) Distribution of the $\ln(\text{observed/expected derived alleles frequency})$ for autosomes and Z-chromosome. (c) Distribution of the S score for autosomes and Z-chromosome. (d) The negative end of the S score distribution presented along duck autosomes. Genes that located in or near selected regions were showed its symbol. (e) The negative end of the S score distribution presented along duck Z-chromosome. (f) The S score distribution presented along scaffolds that is arranged according to their alignment to the chicken genome.



1493

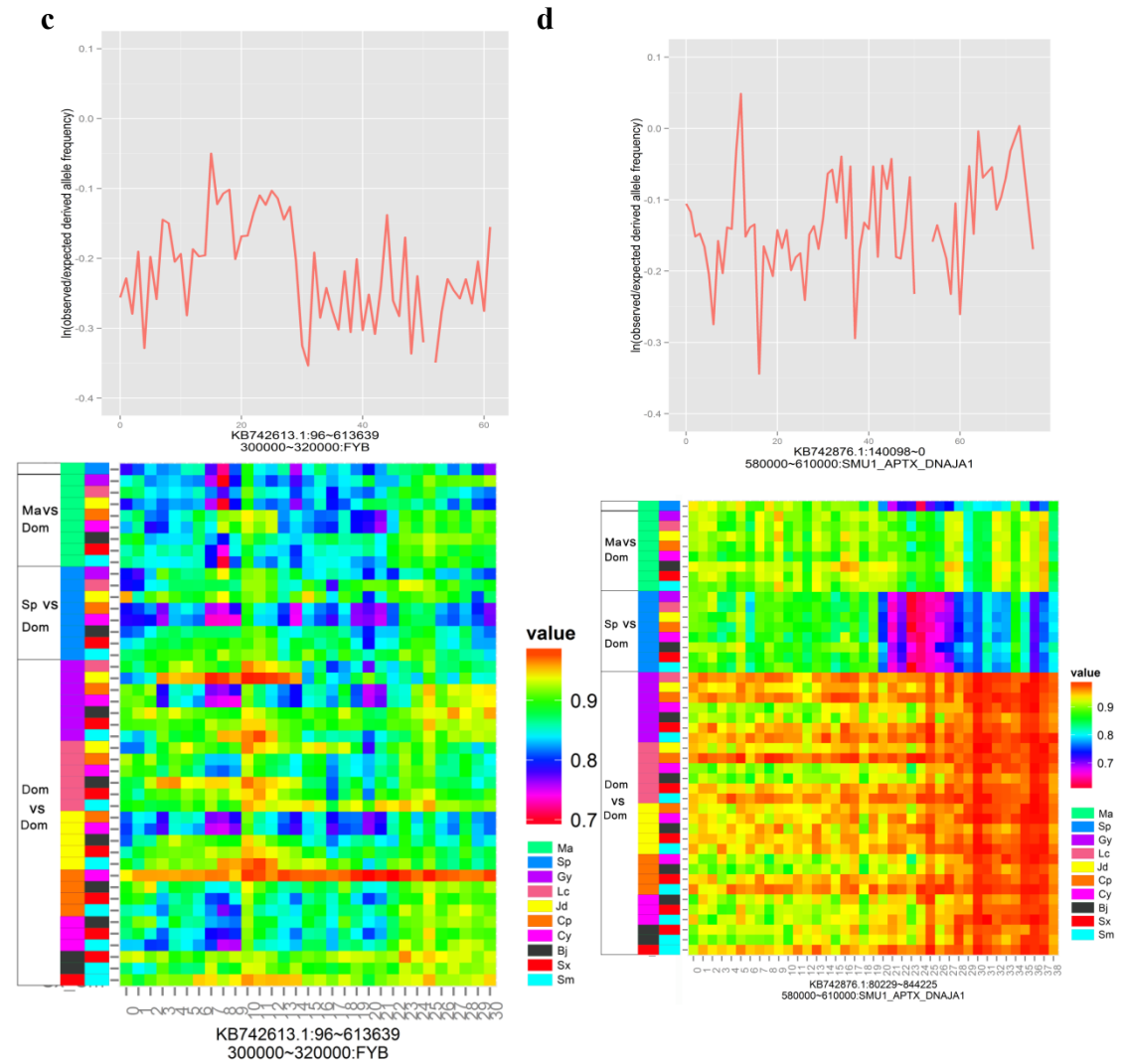


Fig. S19. Details of four putative selective sweep regions in the early stage of the wild duck lineage.

Each panel shows the distribution of the log-transformed value of the ratio for the observed/expected derive allele frequency using non-overlap 10 kb window consist (up) and scores between pairwise comparisons in 20 kb non-overlap windows (low). (a) *CHRD* locus. (b) *FLRT2* locus. (c) *FYB* locus. (d) *DNAJA1* locus.

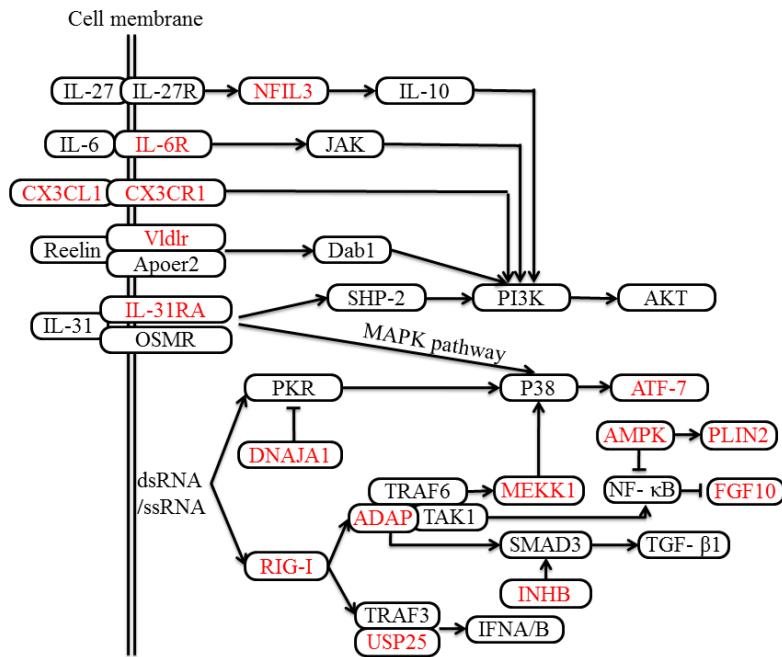


Fig. S20. The RIG-I pathway preferring to be under positive selection.

Sixteen immune genes detected to be under positive selection were in red. Three of them (RIG-I, DNAJA1 and ADAP, MAP3K1) are involved in the host immune response to influenza A viruses.

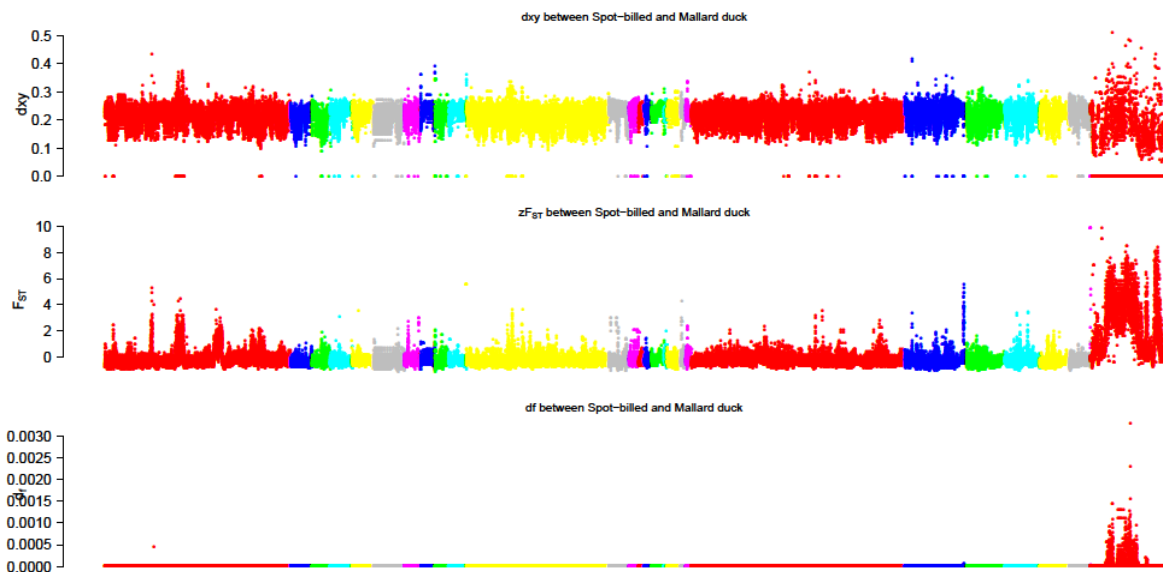
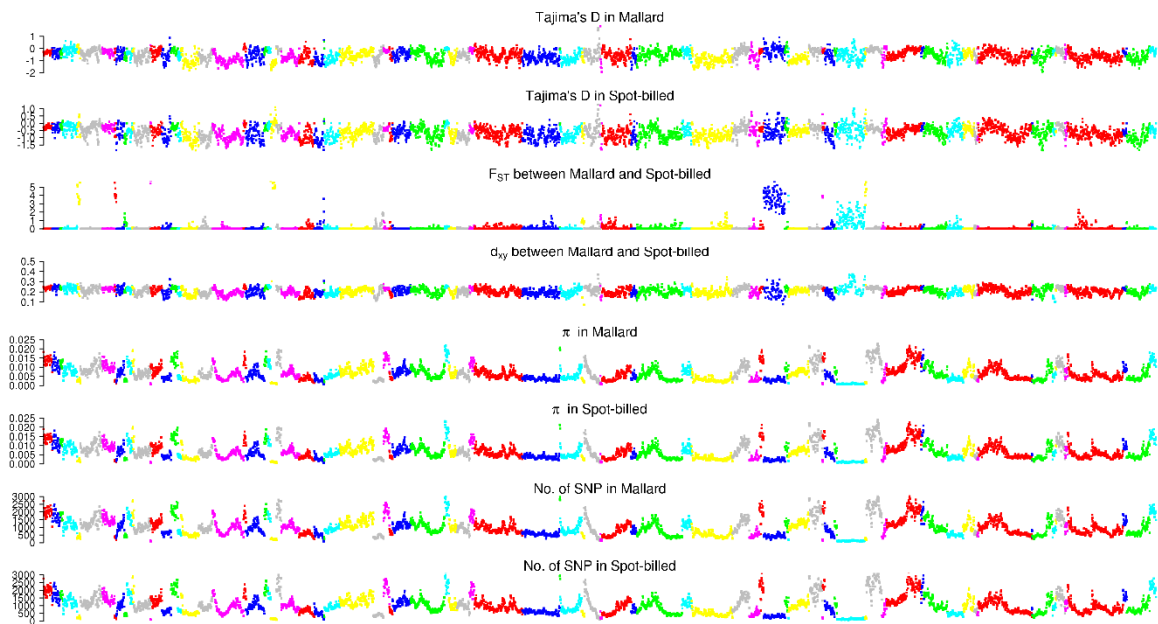


Fig. S21. Distribution of F_{ST} (fixation index), dxy and dfs (fixed difference sites) along Beijing duck scaffolds between mallard duck and spot-billed.

F_{ST} and dfs were counted with 40 kb window by step-size of 20 kb. The duck scaffolds that is shown in red and arranged at the right most are homologous to chicken Z chromosome and referred as duck Z chromosome.

1512

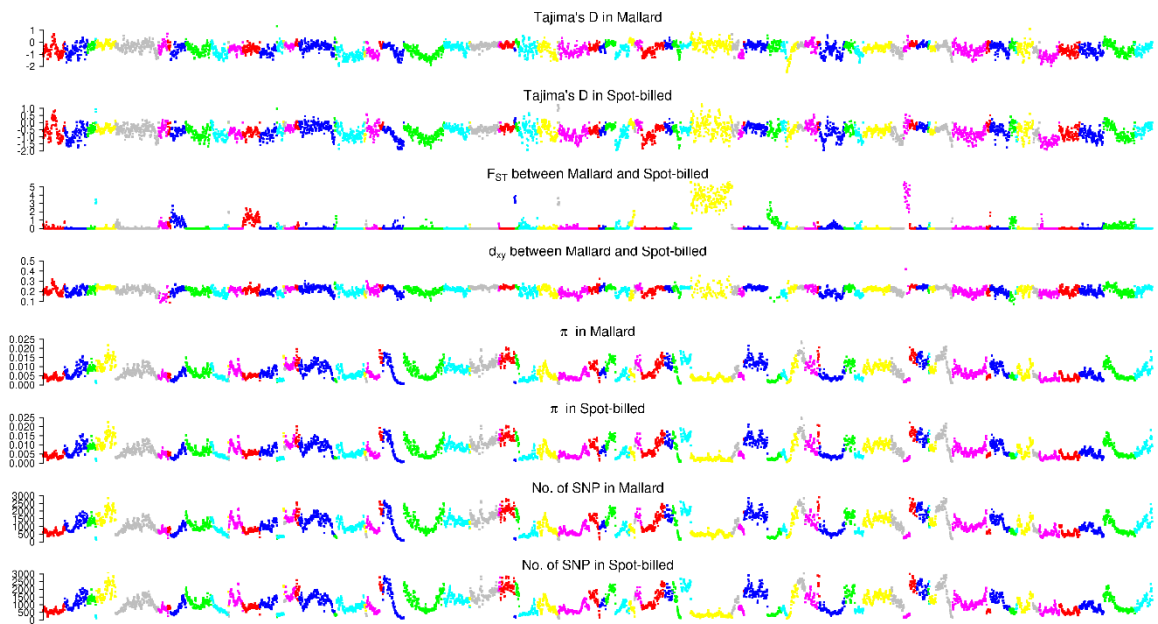
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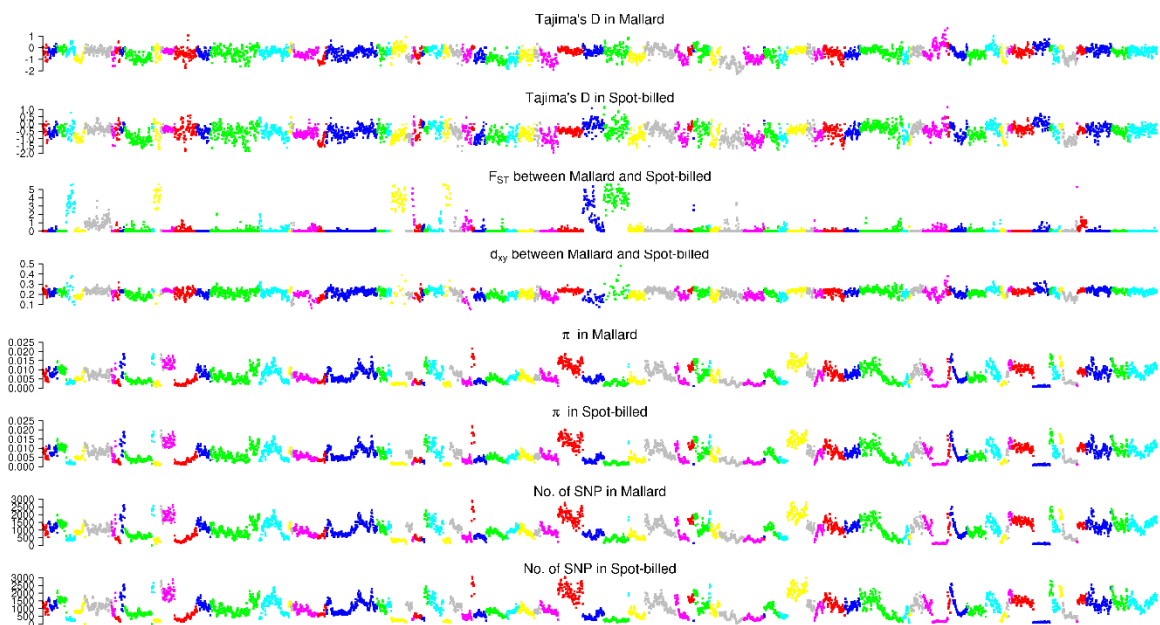
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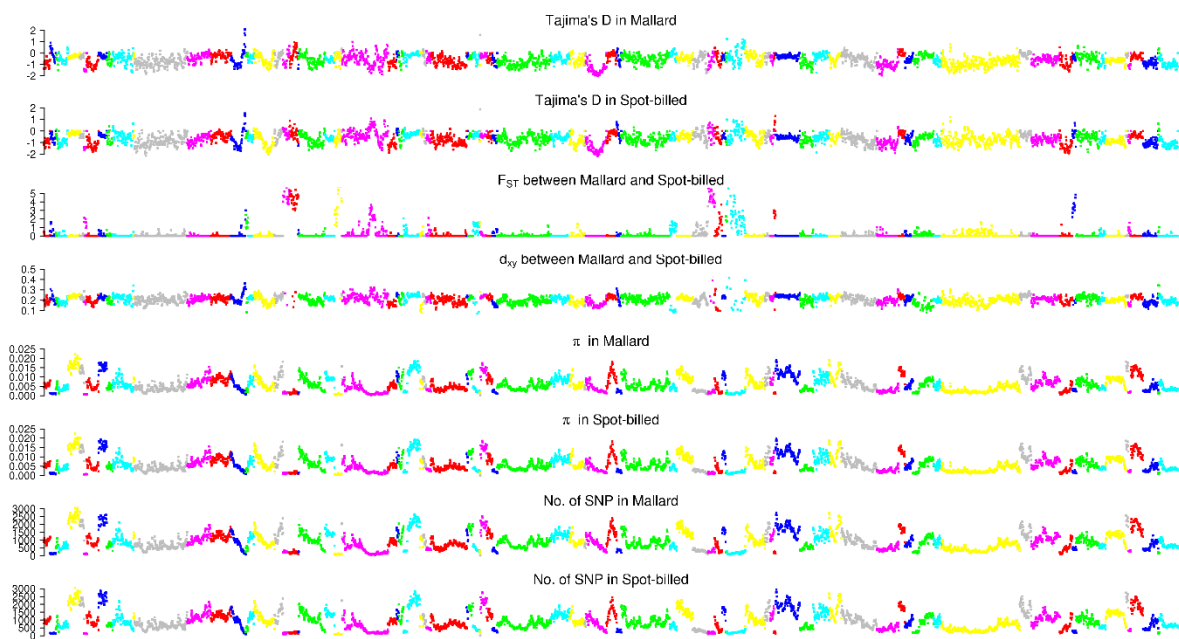
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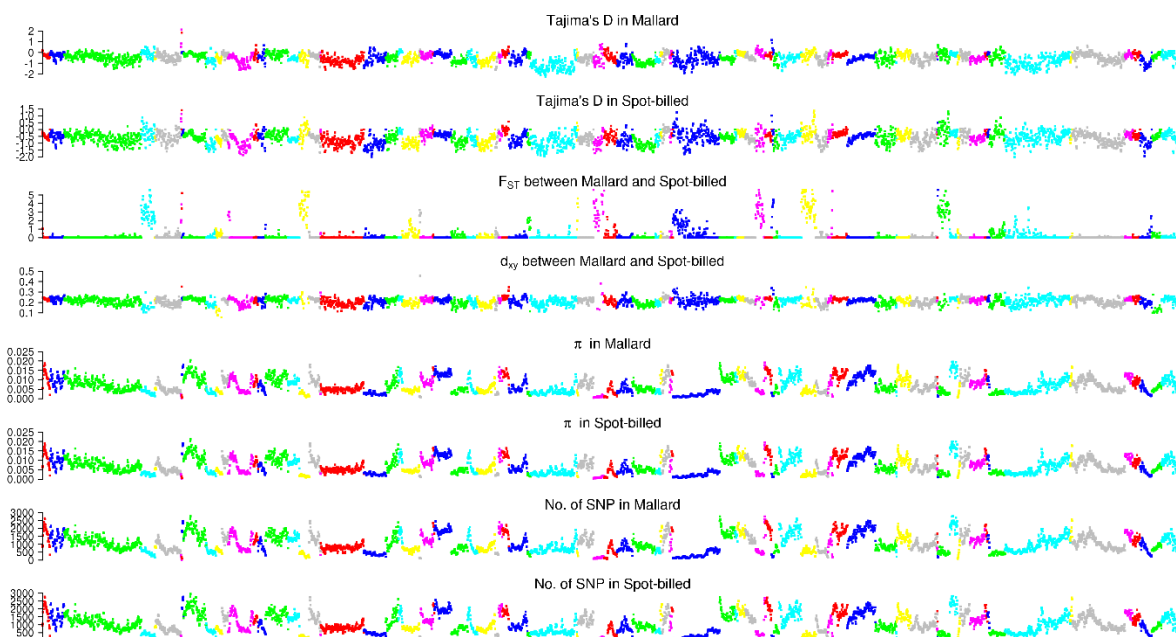
Scaffolds from KB742608.1 to KB742730.1



Scaffolds from KB742733.1 to KB742856.1



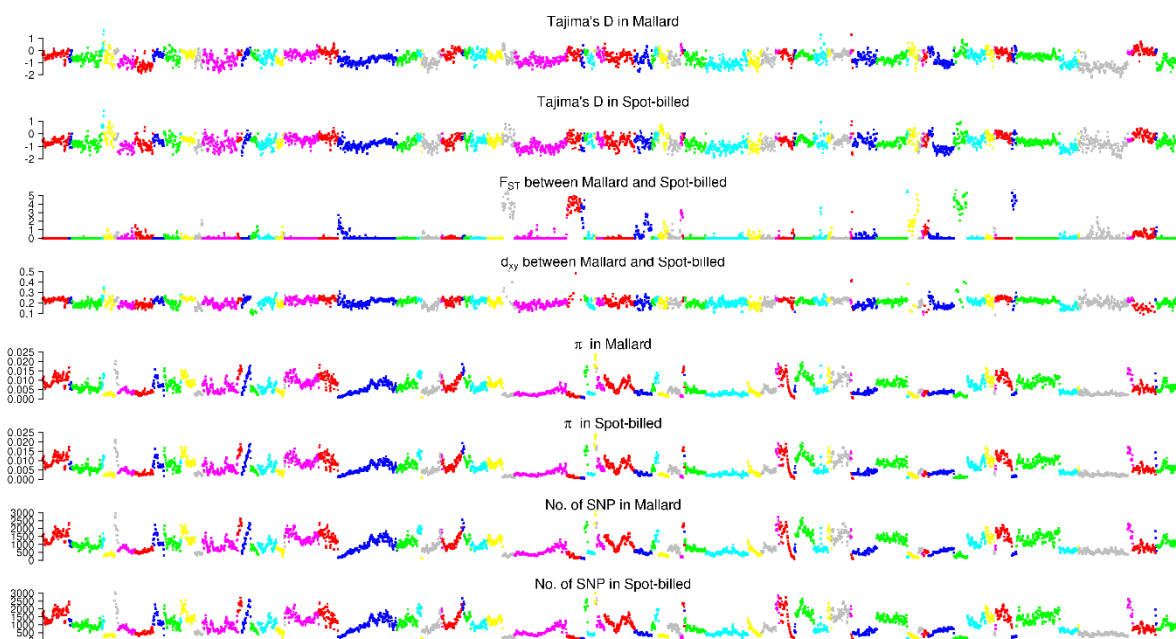
Scaffolds from KB742860.1 to KB742974.1



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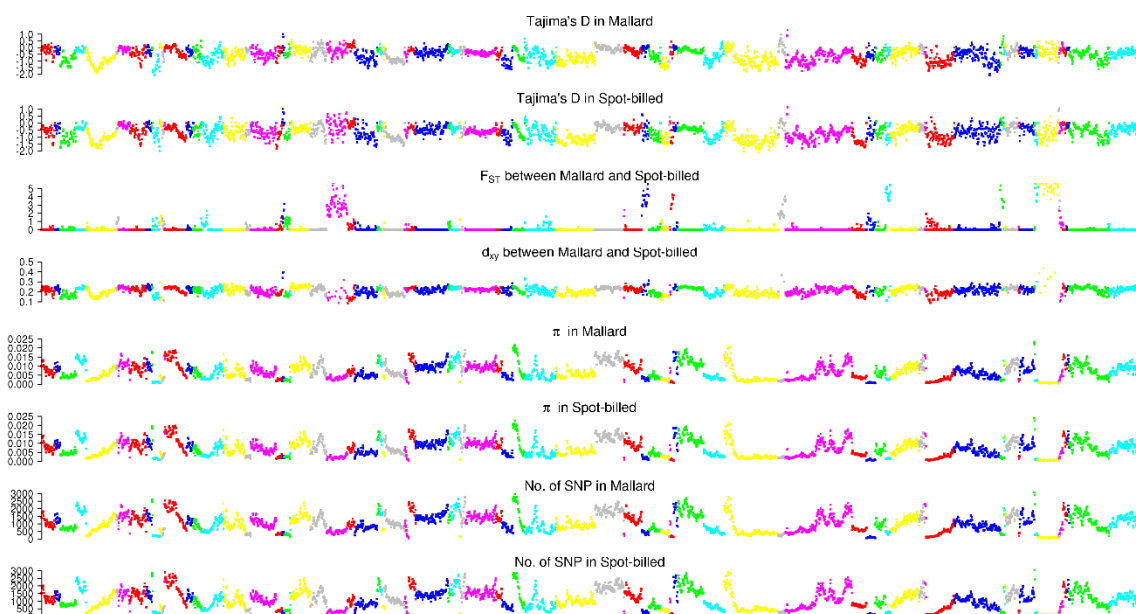
Scaffolds from KB742976.1 to KB743103.1



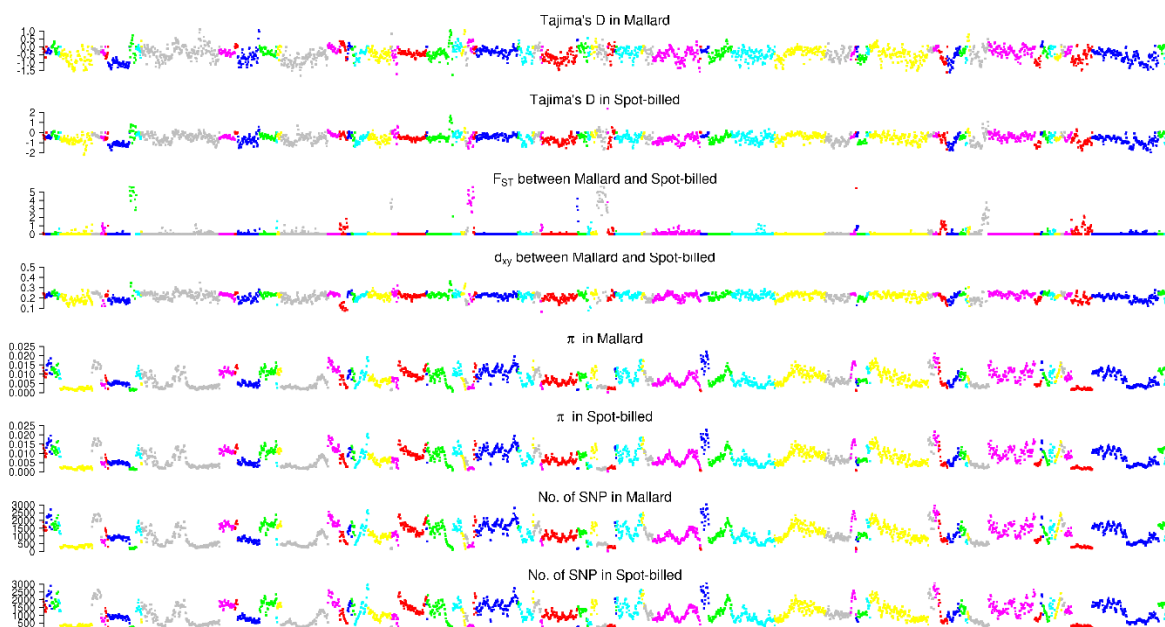
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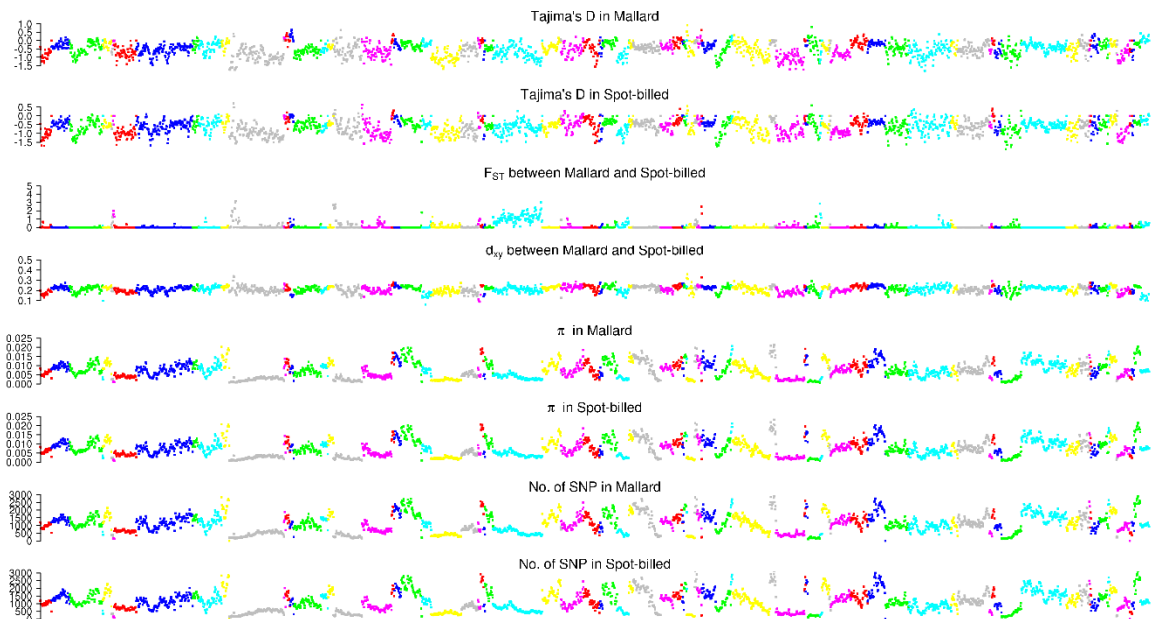
Scaffolds from KB743106.1 to KB743245.1



Scaffolds from KB743248.1 to KB743387.1



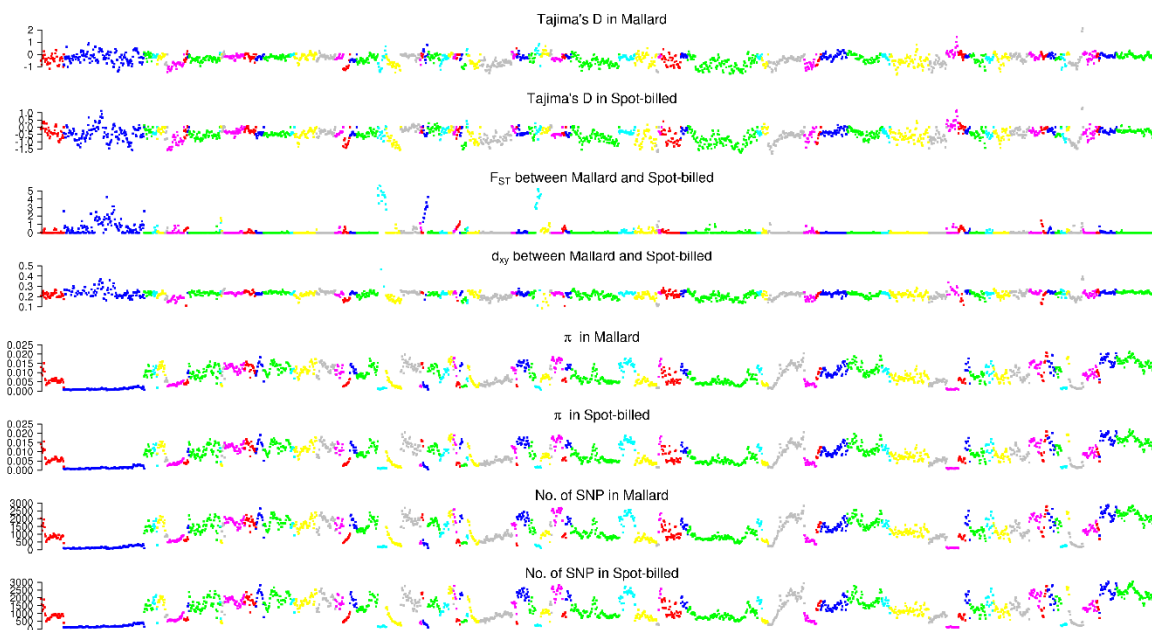
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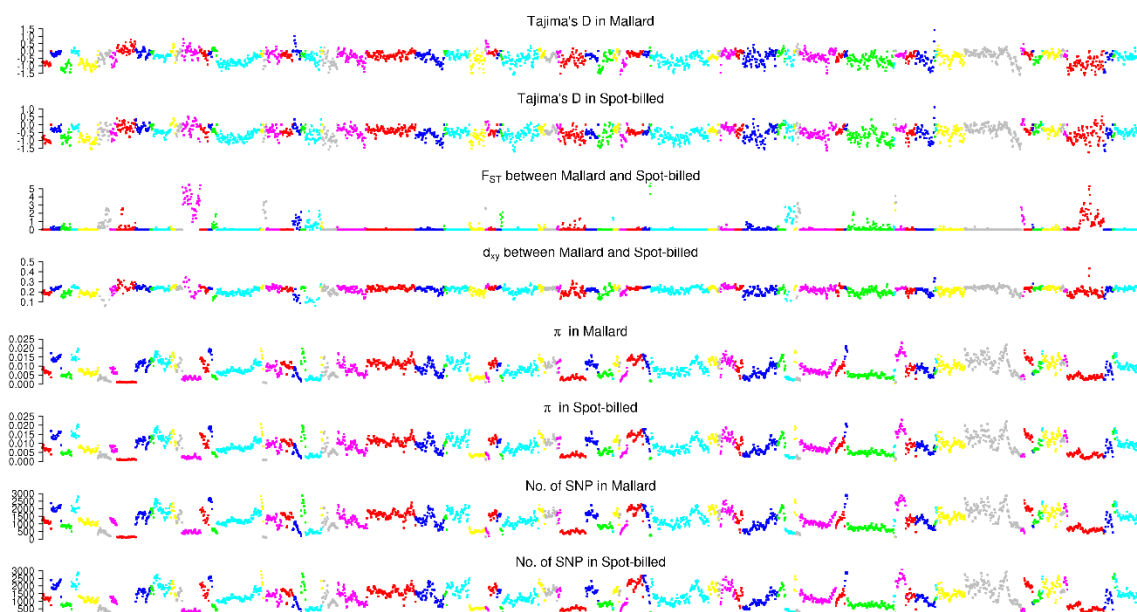
Scaffolds from KB743527.1 to KB743672.1



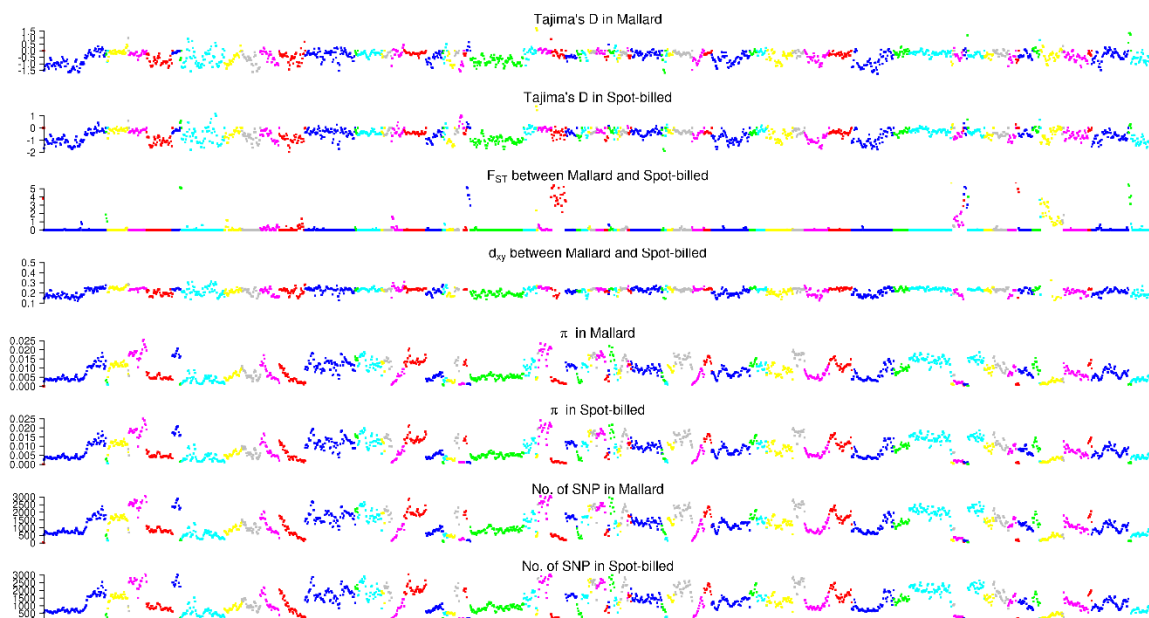
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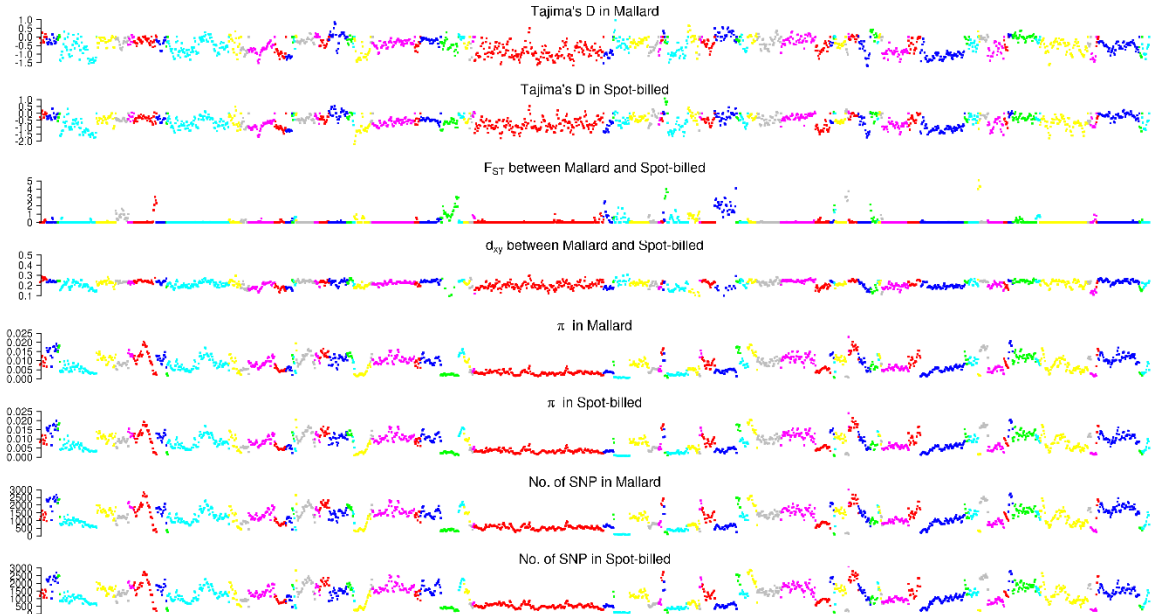
Scaffolds from KB743674.1 to KB743833.1



Scaffolds from KB743839.1 to KB744029.1



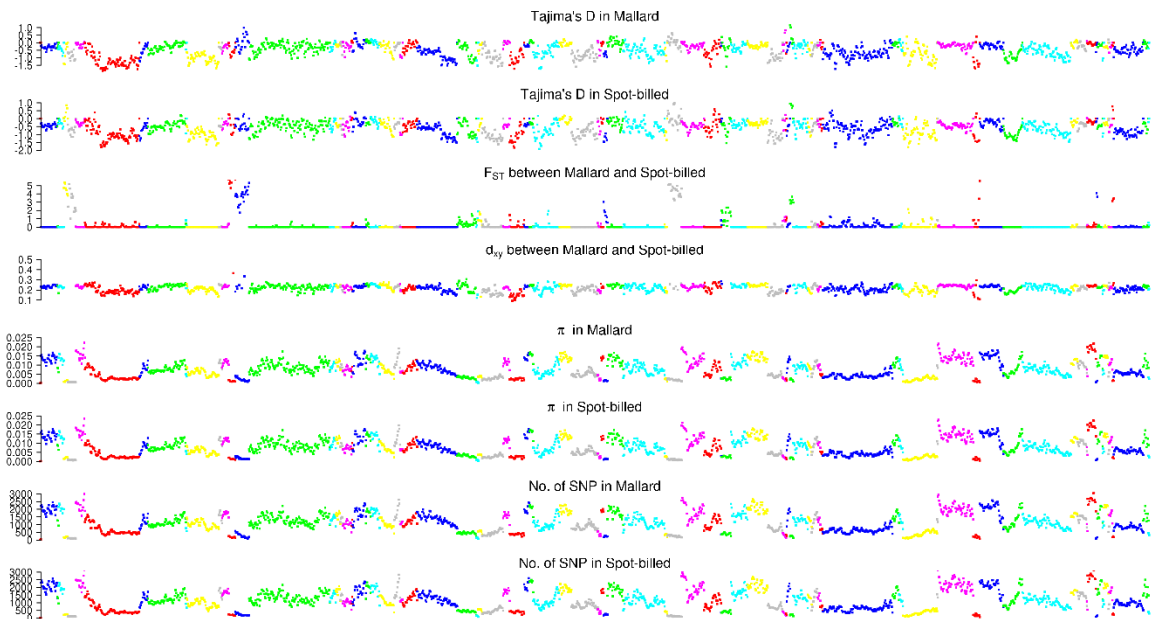
Scaffolds from KB744032.1 to KB744232.1



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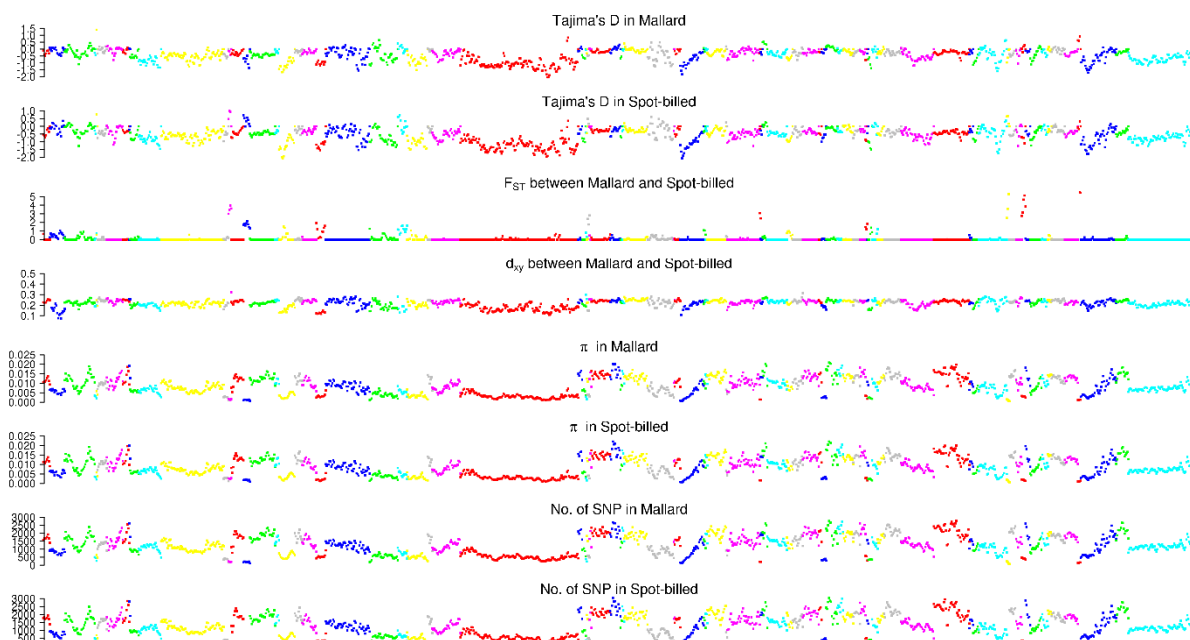
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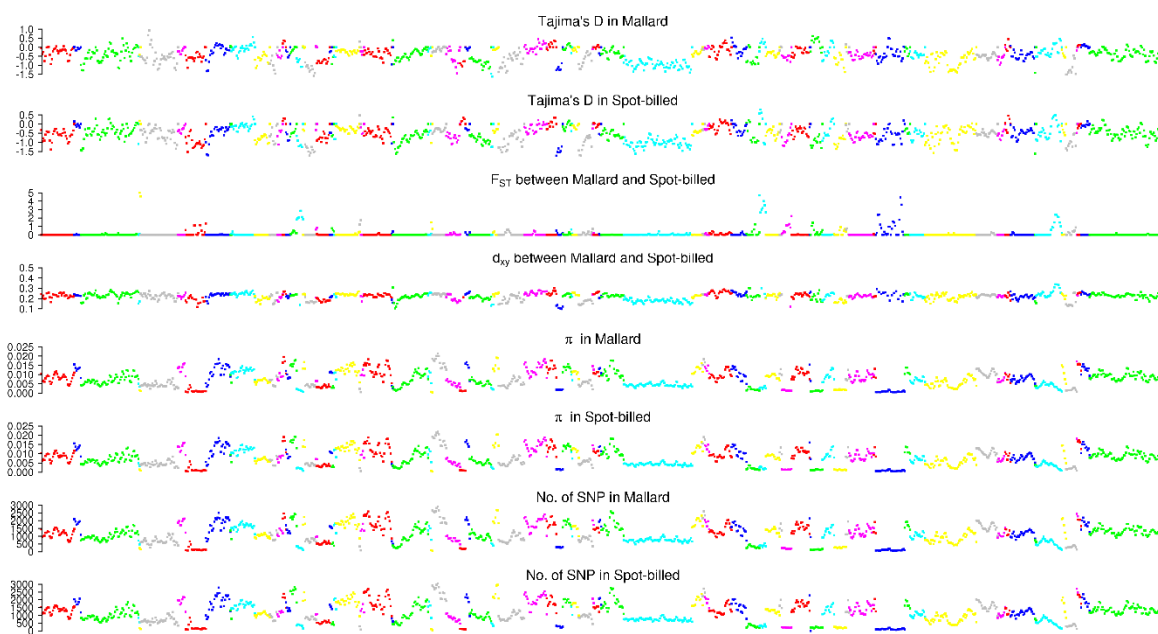
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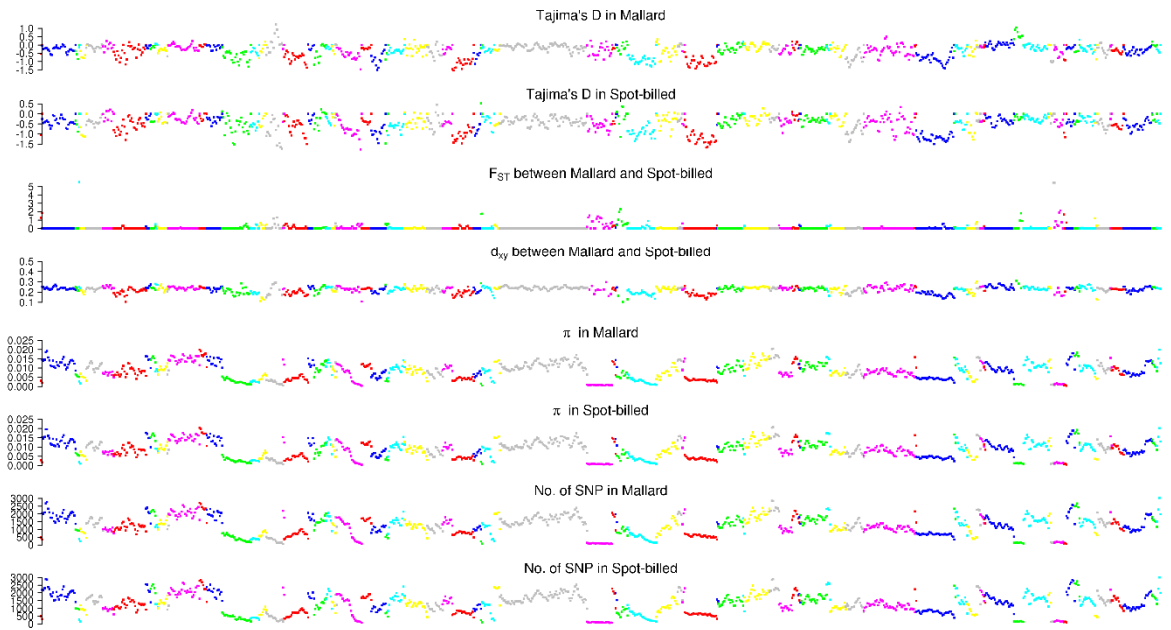
Scaffolds from KB744475.1 to KB744721.1



Scaffolds from KB744725.1 to KB745078.1



Scaffolds from KB745085.1 to KB745764.1



Scaffolds from KB745806.1 to KB747284.1

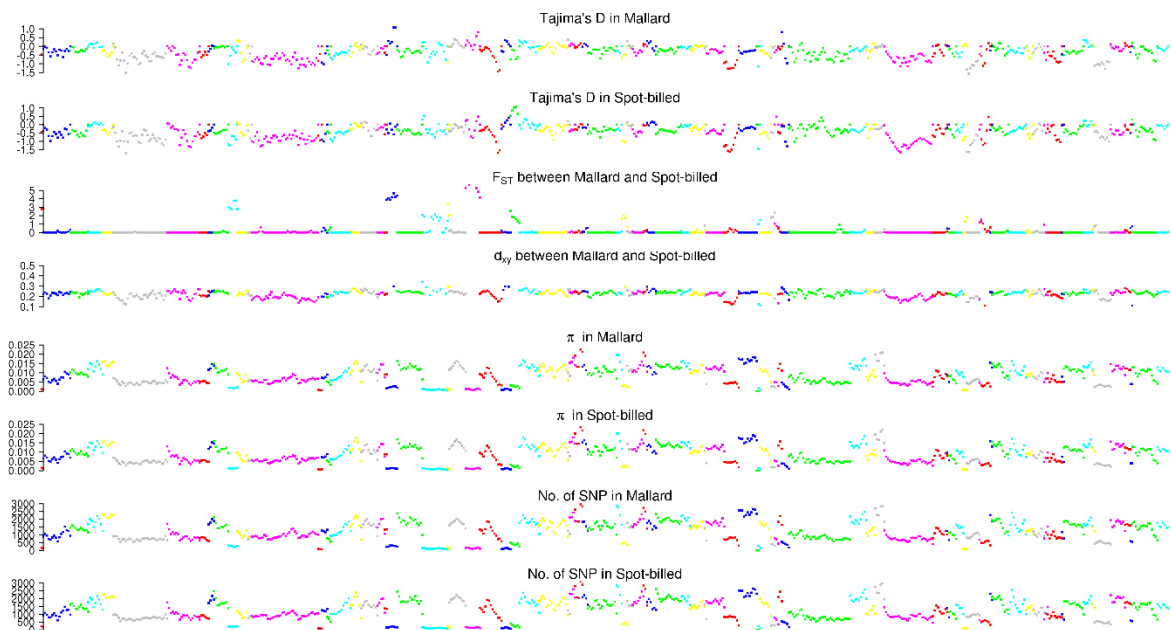


Fig. 22. Distribution of Tajima'D, positive end of zF_{ST} , d_{xy} , number of SNPs per window, nucleotide diversity (π) and zHp along scaffolds in ducks.

These numbers were counted with 40 kb window by step size of 20 kb.

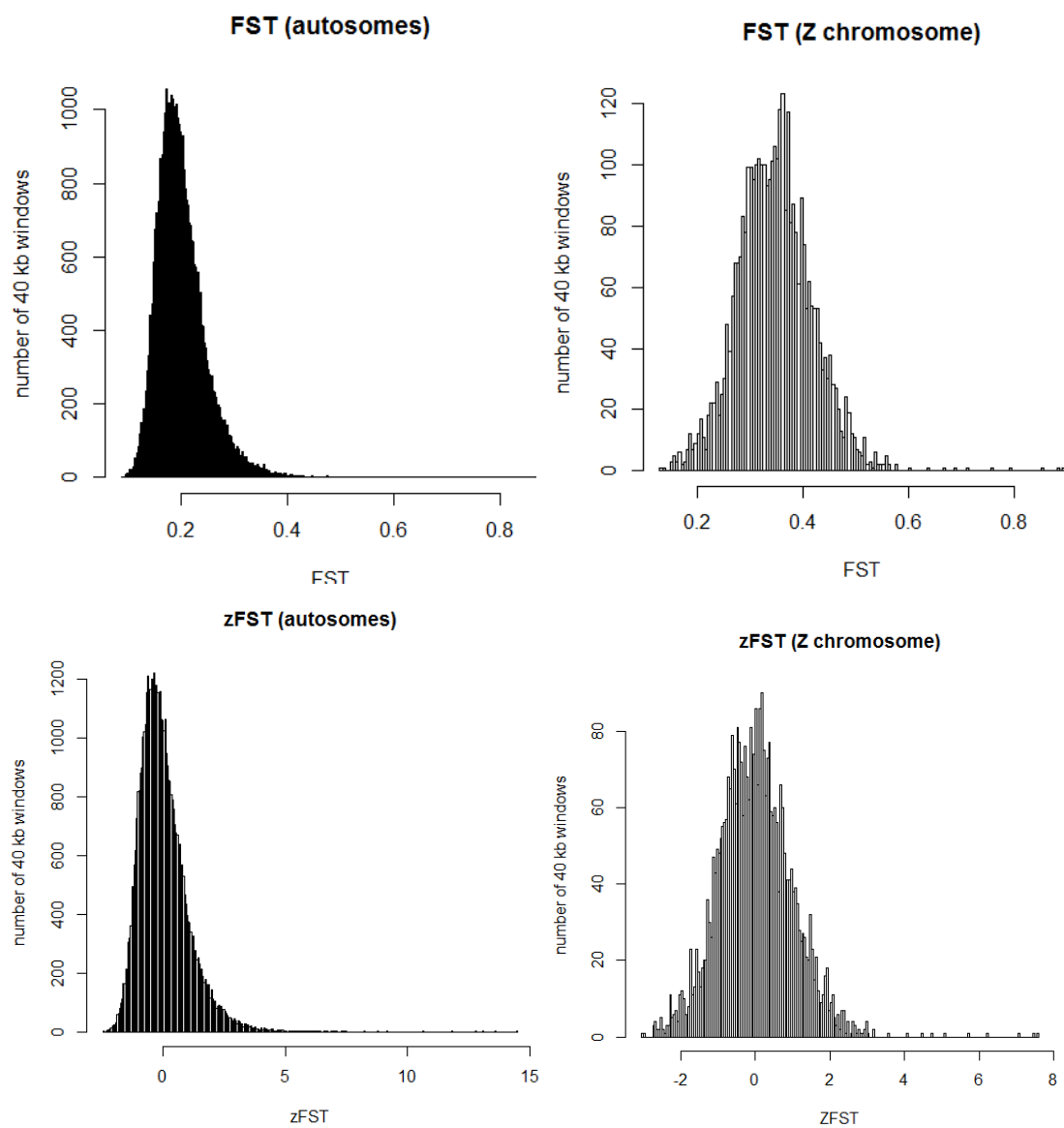


Fig. S23. Distribution of F_{ST}/zF_{ST} between the wild and the domestic duck lineage.

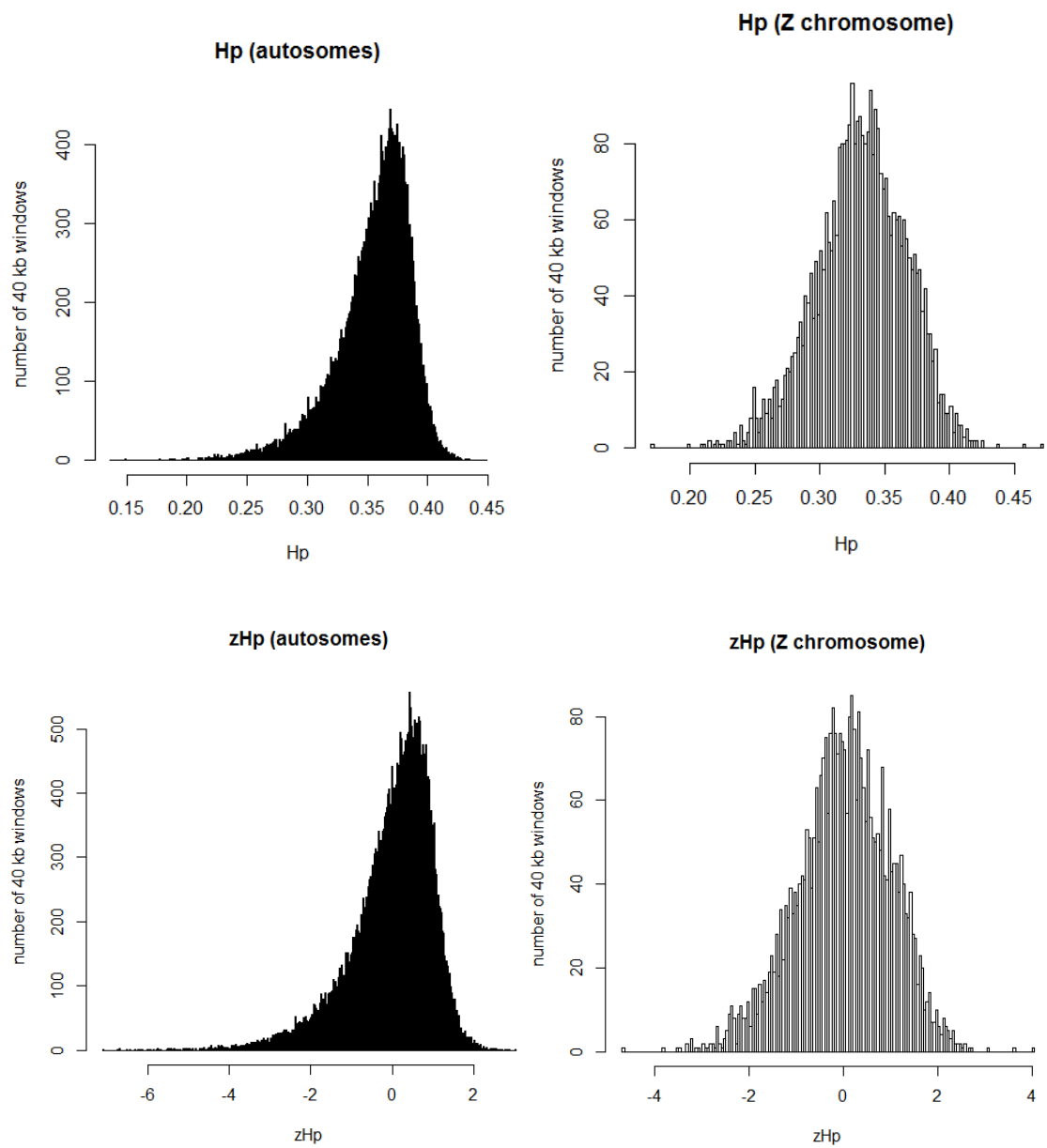


Fig. S24. Distribution of Hp/zHp in the domestic duck lineage.

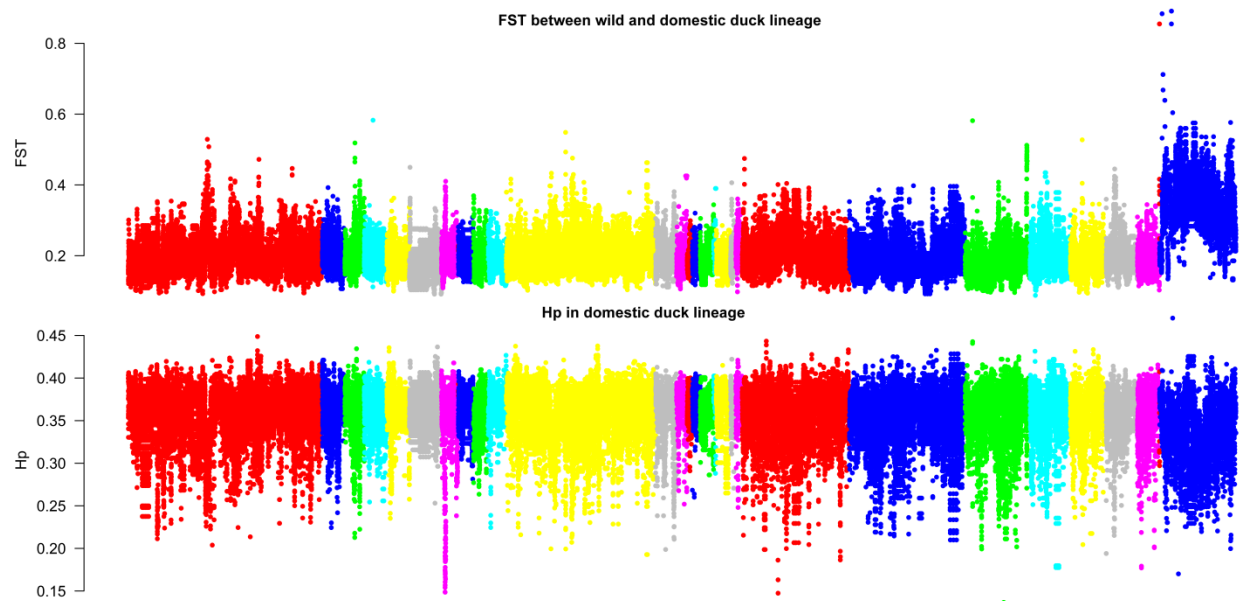


Fig. S25. Distribution of F_{ST} between the wild and the domestic duck lineage, and H_p in the domestic duck lineage.

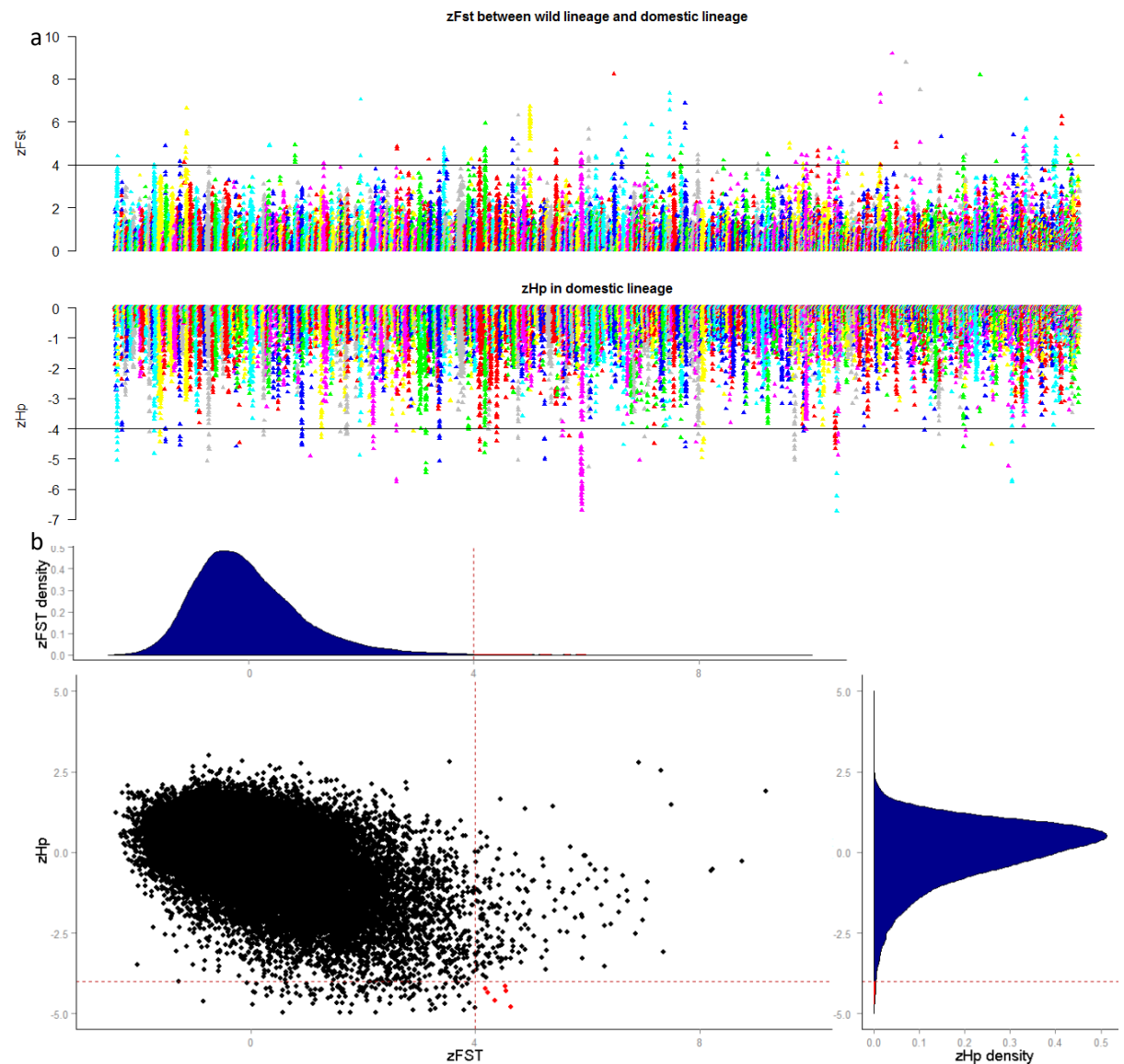


Fig. S26. Distribution of zF_{ST} and zHp in ducks.

zF_{ST} and zHp were counted with 40 kb window by a step size of 20 kb. (a) The positive end of the zF_{ST} between wild lineage and domestic lineage and negative end of zHp in domestic lineage distribution plotted along scaffolds. (b) Distribution of zF_{ST} and zHp . Red dash line indicate threshold and the windows identified as selection regions by both zHp and zF_{ST} were show in red dots. (scaffolds homologous to chicken Z chromosome are not shown)

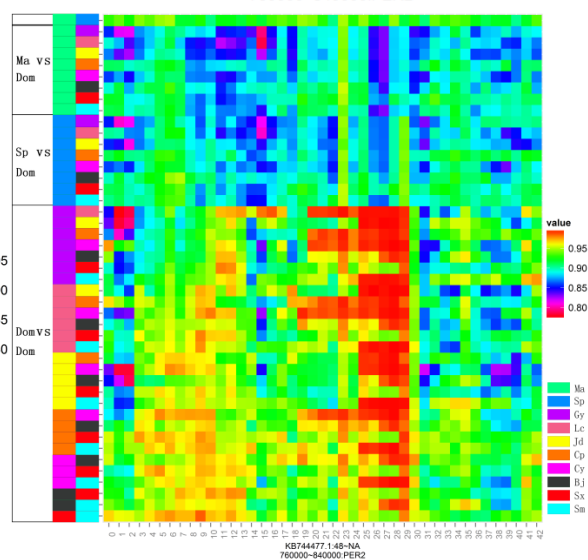
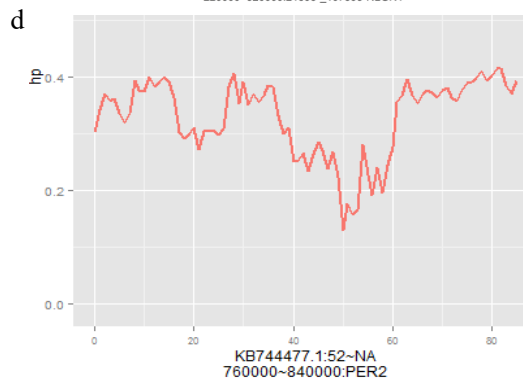
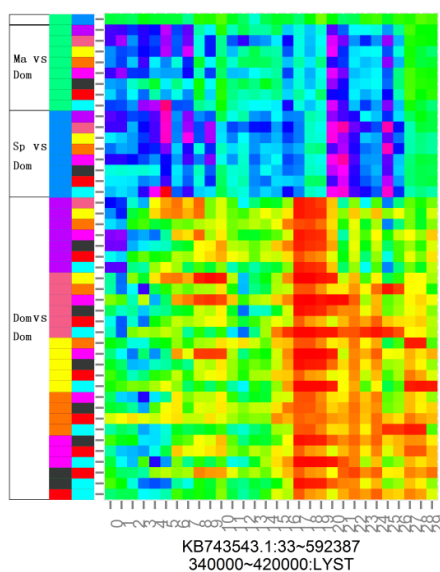
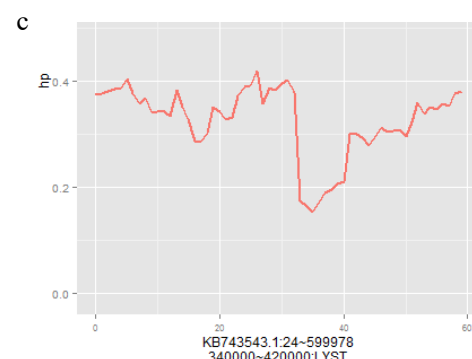
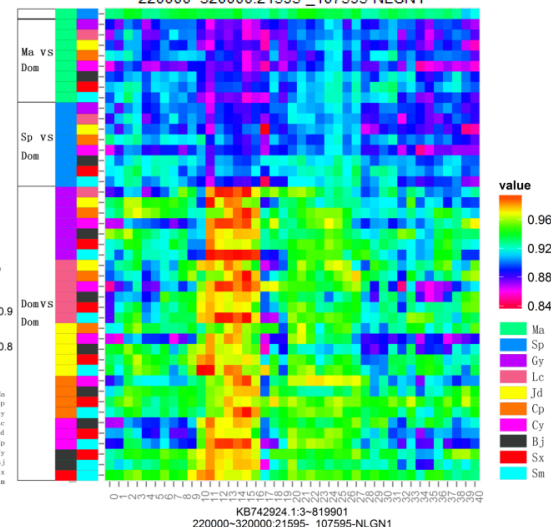
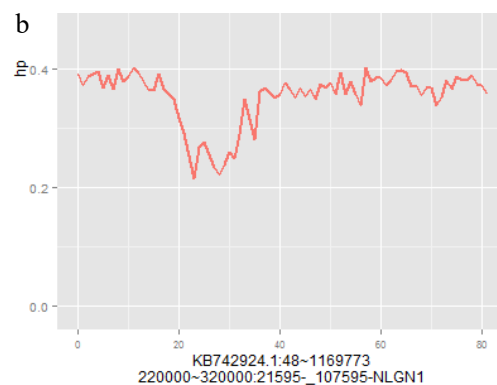
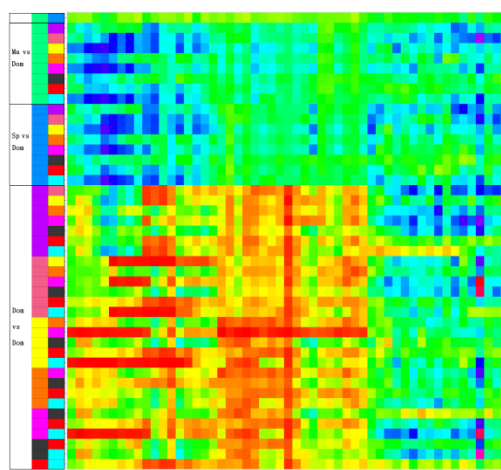
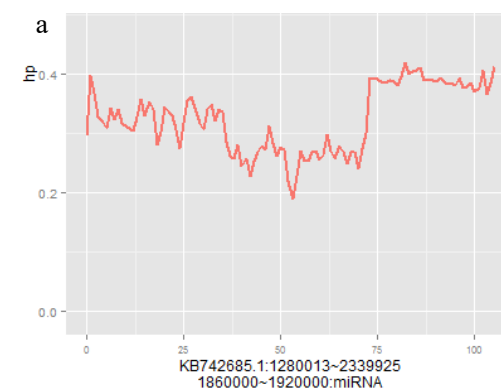


Fig. S27. Details of four putative selective sweep regions in the domestic duck lineage.

Each panel shows distribution of the hp value using non-overlap 10 kb window (upside) and scores between pairwise comparisons in 20 kb non-overlap windows (downside).
(a) *miR-3598-5p* locus. (b) *NLGN1* locus. (c) *LYST* locus. (d) *PER2* locus.

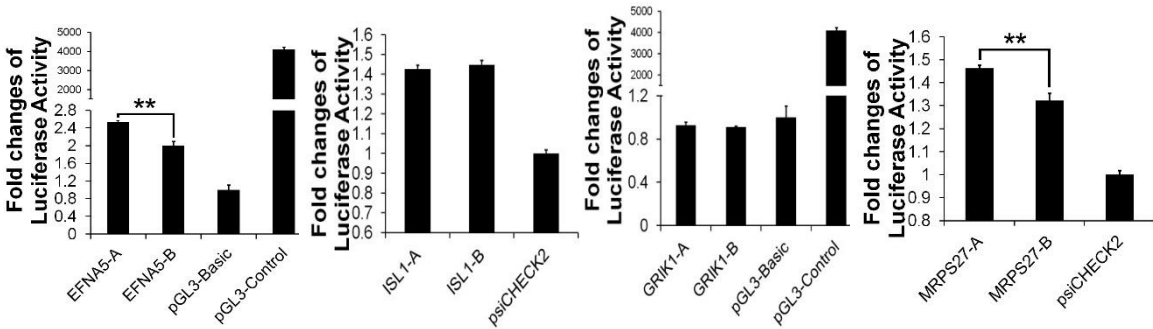
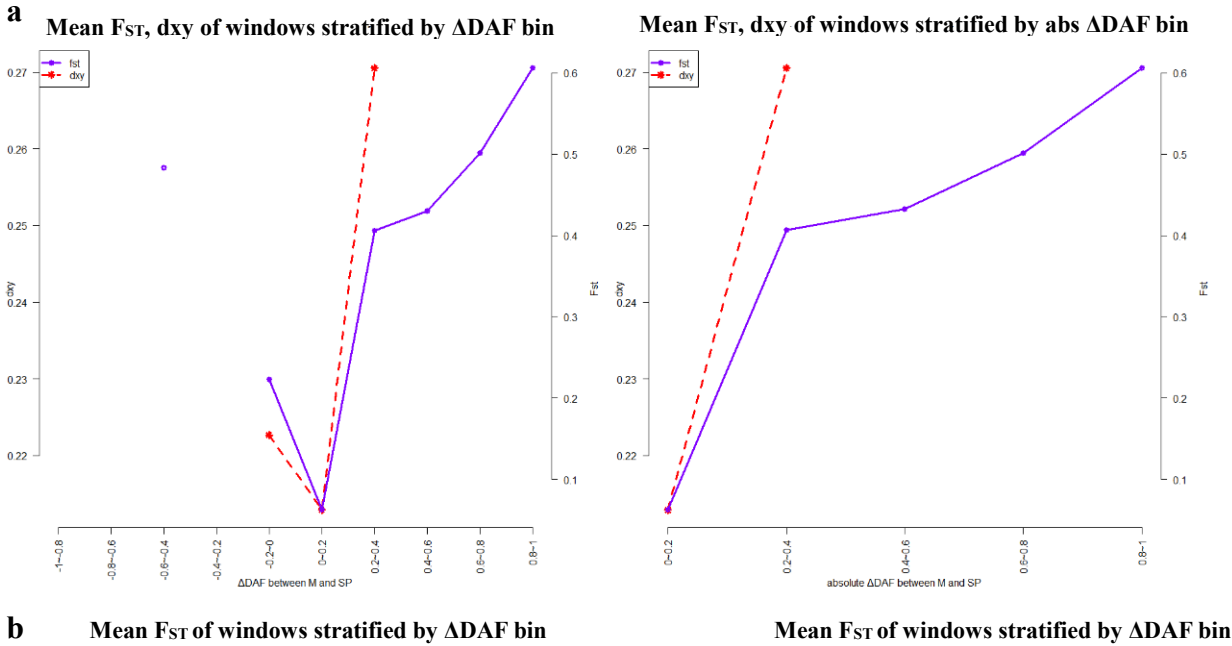


Fig. S28. Effects of casual candidate mutation on luciferase reporter activity.

The *ISL1* was under positive selection in early stage of the wild lineage. The *GRIK1* and *LYST* were under positive selection in the domestic lineage. “-A”, “-B” represent wild mutation types and domestic mutation types respectively.



b Mean F_{ST} of windows stratified by ΔDAF bin

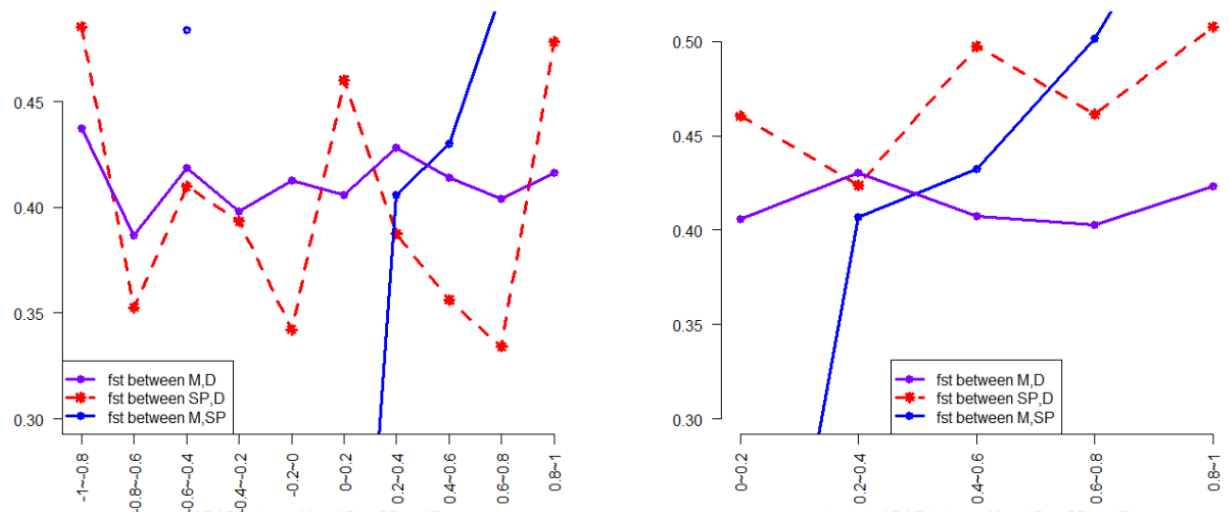
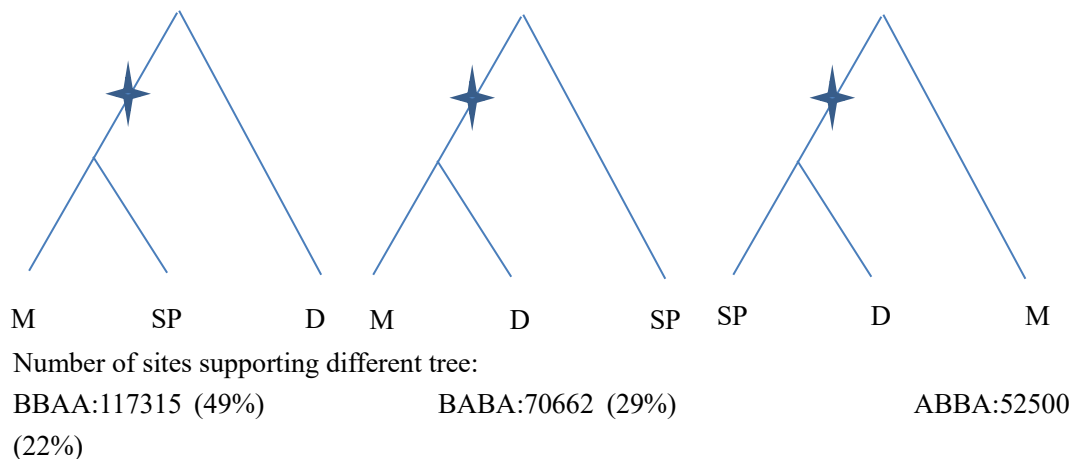


Fig. S29. Observation on genomic divergence values stratified by Δ DAF provide evidence of introgression between domestic and wild ducks.

(a) Mean F_{ST} (purple lines), d_{xy} (red dash line) of windows stratified by Δ DAF (absolute Δ DAF in right) bins between mallard and spot-billed duck populations. d_{xy} is not available in higher Δ DAF bins due to fixed SNPs are exclude in d_{xy} calculation.

(b) Mean F_{ST} between domestic and wild duck populations decreased in high Δ DAF bins may due to introgression. Blue lines (corresponding to purple line in (a)) indicate no evidence of introgression between mallard and spot-billed in compare with the measurement between wild (red dash line: spot-billed, purple line: mallard) and domestic ducks. “M”, “SP” and “D” indicate mallard, spot-billed and domestic duck, respectively.



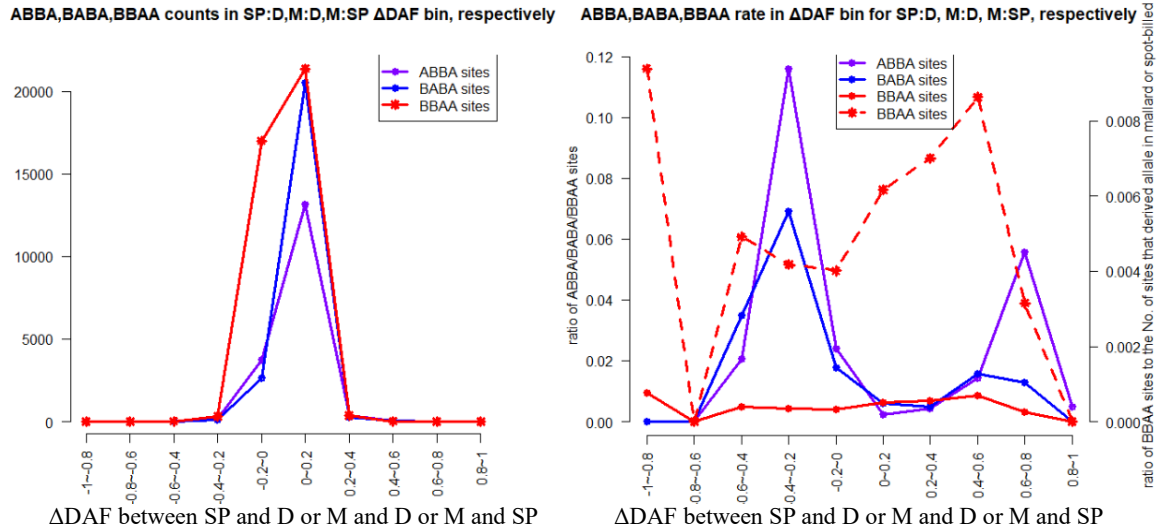


Fig. S30. Absent evidence of introgression between mallard duck and spot-billed duck population based on ABBA-BABA analysis of mallard, spot-billed and domestic duck. (a) Number of sites supporting different trees is indicated both as a percentage and as actual numbers. **(b)** Left panel, number of ABBA, BABA and BBAA sites were stratified by Δ DAF between spot-billed: domestic (purple line), mallard: domestic (blue line) and mallard: spot-billed (red line), respectively. Right panel, BBAA rate (red line) were measured as the BBAA counts divided by number of sites that derived allele present in either mallard or spot-billed population in each bins (ABBA (purple line), BABA (blue line) similarly). The red dash line is corresponding to right y-axis. “M”, “SP” and “D” indicate mallard, spot-billed and domestic duck, respectively.

1620 **Supplementary Tables**

1621 **Table S1: Statistics of the reads in mallard and spot-billed genome assemblies**

Paired-end libraries (bp)	Paired-end insert size	Number of libraries	Average read length (bp)	Total data (Gb)	Sequence coverage (X)	Physical coverage (X)
mallard	250bp	3	150_150	89.76	70.81	59.01
	500bp	2	100_100	46.18	36.94	92.35
	800bp	2	100_100	57.4	45.92	183.68
	2K	3	49_49	27.37	21.9	446.94
	5K	2	49_49	16.32	13.06	666.33
	10K	3	49_49	11.67	9.34	953.06
	20K	2	49_49	13.75	11	2,244.90
Total	/	17	/	262.4 5	208.97	4,646.27
spot-billed	250bp	3	150_150	95.64	76.51	63.76
	500bp	2	100_100	63.39	50.71	126.78
	800bp	2	100_100	45.31	36.25	145
	2K	3	49_49	26.2	20.96	427.76
	5K	2	49_49	17.11	13.69	698.47
	10K	3	49_49	15.37	12.3	1,255.10
	20K	2	49_49	12.17	9.74	1,987.76
Total	/	17	/	275.1 9	220.16	4,704.63

1622 **Table S2: Statistics of the three *de novo* ducks assemblies**

Category	Contig			Scaffold		
	Beijing Duck	Mallard	Spot-billed	Beijing Duck	Mallard	Spot-billed
Total length(kb)	1,069,961	1,176,969	1,187,333	1,105,049	1,270,808	1,317,385
Total number	227,597	172,189	194,934	78,487	61,591	67,685
Longest size(bp)	263,737	484,574	673,122	5,998,093	17,658,973	25,320,044
N50 size(bp)	26,138	38,271	33,061	1,233,631	2,489,142	2,054,876
N50 number	11,193	7,876	9,194	268	125	154
N90 size(bp)	3,085	5,381	4,386	195,458	145,390	102,224

N90 number	53,850	38,114	45,269	1,097	870	1,035
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1624 **Table S3: Comparison in length and coverage of the mallard and the spot-billed**
1625 **genome using seven Beijing duck BAC sequences**

BACs name	Length of BACs (bp)	mallard coverage	spot-billed coverage
ndae_TLR1	67,954	85.91%	70.03%
ndbaa_TLR15	79,704	98.50%	98.25%
ndbab_TLR5	72,128	98.41%	98.32%
ndbac_1_TLR3	76,489	97.53%	96.49%
ndbad_2_TLR7	97,799	95.64%	87.51%
ndbae_TLR2	75,080	98.82%	93.59%
ndbag_TLR4	77,042	97.79%	97.98%

1626 **Table S4: Evaluation of the mallard genome coverage using duck EST sequences**

Dataset	Number	Total length (bp)	Covered by assembly (%)	with >90% sequence in one scaffold		with >50% sequence in one scaffold	
				Number	Percent (%)	Number	Percent (%)
>0bp	319,996	98,257,350	99.02	294,845	92.14	315,689	98.65
>200bp	144,337	73,939,509	99.53	135,799	94.08	143,128	99.16
>500bp	43,896	43,155,260	99.84	41,595	94.76	43,687	99.52
>1000bp	13,652	22,939,026	99.99	12,831	93.99	13,613	99.71

1627 **Table S5: Evaluation of the spot-billed genome coverage using duck EST**
1628 **sequences**

Dataset	Number	Total length (bp)	Covered by assembly (%)	with >90% sequence in one scaffold		with >50% sequence in one scaffold	
				Number	Percent (%)	Number	Percent (%)
>0bp	319,996	98,257,350	99.01	294,324	91.98	315,566	98.62
>200bp	144,337	73,939,509	99.53	135,565	93.92	143,053	99.11
>500bp	43,896	43,155,260	99.85	41,531	94.61	43,670	99.49
>1000bp	13,652	22,939,026	99.99	12,802	93.77	13,606	99.66

1629 **Table S6: Type and proportion of transposable elements(TEs) in the Mallard,**
1630 **Spot-billed, Beijing duck and chicken genome**

TE type	mallard length (bp)	% genome	Spot-billed length (bp)	% genome	Beijing duck length (bp)	% genome	Chicken length (bp)	% genome
DNA	9,479,699	0.75	9,438,606	0.71	5,590,903	0.51	13,872,990	1.33

LINE	98,060,422	7.71	101,177,399	7.68	63,146,176	5.71	116,778,031	11.15
SINE	1,389,649	0.1	1,274,941	0.09	892,886	0.08	573,610	0.05
LTR	52,195,974	4.1	48,381,075	3.67	21,255,312	1.92	47,374,647	4.53
Other	21,136	0	12,303	0	3,201	0	3,338	0
Unknown	1,100,812	0.08	3,402,377	0.25	4,600,535	0.41	3,351,756	0.32
Total	136,134,153	10.71	139,051,537	10.55	80,740,993	7.31	149,208,388	14.25

1631 **Table S7: Gene annotation of the mallard and spot-billed reference gene sets using**
1632 **five protein databases**

Database	Mallard		Spot-billed	
	Number	Percent (%)	Number	Percent (%)
Total	21,056	100	21,123	100
InterPro	15,335	72.83	15,308	72.47
GO	12,817	60.87	12,815	60.67
KEGG	15,878	75.41	15,803	74.81
Swissprot	17,551	83.35	17,531	82.99
TrEMBL	18,371	87.25	18,327	86.76
All annotated	18,467	87.70	18,426	87.23
Unannotated	2,589	12.29	2,697	12.77

1633

1634 **Table S8: Gene family of two mammals and seven birds clustering by TreeFam**

Species	Total genes	Unclustered genes	Families	Unique families	Ave. genes per family
mallard	21,056	2,478	14,541	30	1.28
Spot-billed	21,123	2,604	14,530	42	1.27
Beijing duck	20,629	5,381	13,498	18	1.13
<i>G. gallus</i>	15,508	493	13,351	6	1.12
<i>S. camelus</i>	16,178	326	11,579	105	1.37
<i>M. gallopavo</i>	14,123	455	12,414	7	1.10
<i>F. albico</i>	15,303	716	13,284	33	1.10
<i>H. sapiens</i>	20,238	1,825	15,036	117	1.22
<i>M. musculus</i>	22,627	1,549	15,407	139	1.37

1635 **Table S9: Over represented GO terms (p-value <0.001) of wild duck expanded**
1636 **gene families**

1637 See file "Table_S9.xlsx"

1638 **Table S10: Summary for whole-genome resequencing**

Group	Species/breeds	Sample number	Sequence depth	Total data (Gb)	Read length	Sequence strategy	SNP number
Wild duck	Mallard	14	14.1×	218.2	125	Individual	32,982,331
	Spot-billed	13	14.5×	209.5	125	Individual	33,581,513

	Wigeon	3	20×	75.6	125	Individual	21,393,824
	Gadwall	1	20×	17.0	125	Individual	10,756,767
	Pintail	1	20×	25.7	125	Individual	8,551,170
	Falcata teal	1	20×	22.3	125	Individual	7,658,519
	Common teal	1	20×	22.1	125	Individual	9,146,292
	Baikal teal	1	20×	23.2	125	Individual	8,500,280
Indigenous Chinese duck breeds	Beijing	27	13.5×	400.7	125	Individual	14,748,500
	Shaoxing	27	2~3.5×	95.1	100	Individual	15,414,890
	Gaoyou	10	4.7×	56.9	100	Pooling	9,190,980
	Shanma	10	4.0×	50.4	100	Pooling	10,185,166
	Lianchengbai	10	4.7×	57.6	100	Pooling	8,978,662
	Jinding	10	4.4×	55.5	100	Pooling	8,617,344
Commercial breeds	Cherry valley	10	7.5×	94.5	100	Pooling	9,615,871
	Campbell	10	4.5×	57.3	100	Pooling	9,821,965
Outgroup	Muscovy	10	5.24×	57.9	100	Pooling	--
	Taihu goose	10	3.5×	39.7	100	Pooling	--

Table S11: Tests for population mixture of mallard, spot-billed, Beijing and Shaoxing ducks

Source 1	Source 2	Target	f3	Standard error	Z	SNPs
Mallard	Spot-billed	Beijing	0.007886	0.000748	10.546	9,997
Mallard	Spot-billed	Shaoxing	0.008662	0.000814	10.643	9,943
Mallard	Spot-billed	Mix	0.007279	0.000305	23.834	2,930,914
		Beijing, Shaoxing				
Mallard_hz	Spot-billed_hz	Beijing	0.008656	0.001076	8.045	9,975
Mallard_hz	Spot-billed_hz	Shaoxing	0.010200	0.000683	14.930	9,907
Mallard_fh	Spot-billed_fh	Beijing	0.007320	0.000603	12.147	9,986
Mallard_fh	Spot-billed_fh	Shaoxing	0.007380	0.000988	7.470	9,921

Considering the extensive substructure exist in wild ducks which reflect those individuals may trace their recent ancestry to different populations even species, we assume different clusters combinations to source 1 and source 2, and combine different domestic breeds to one target population, to minimize the population effect. All the Z were significantly greater than zero.

Table S12: Divergence without migration inferred parameters.

combinations	Parameter ^a	Conventional bootstrap 95% confidence interval	Conventional bootstrap 95% confidence interval (convert to real)	Maximum likelihood
Domestic, mallard duck	s (Nmallard)	0.85658-0.874169	1567694-1637111	1595689
	1-s(Ndomestic)	0.125831-0.143419	233297-265236	-
	Ts	0.0272-0.031008	100818-114733	102427
Domestic, spot-billed duck	s(Nspot-bill)	0.8776821-0.893130	1660302-1731619	1659603
	1-s(Ndomestic)	0.10687-0.122315	205404-233549	-

	Ts	0.02321-0.02653	89220-101346	91700
Spot-billed, mallard duck	s(Nmallard)	0.4352336-0.472079	949555-1031148	969384
	1-s(Nspot~)	0.52792-0.564766	1146439- 1239046	-
	Ts	0.015755-0.0174258	68583-76294	69432

^a see model code above.

Table S13: Summary of selective sweep regions in the wild duck lineage

These selective sweep regions occurred during their population in conjunction with or shortly after, their population divergence from domestic lineage ($S < -4$ for autosomes, $S < -1.5$ for Z-chromosomes). See file “Table_S13.xlsx”

Table S14: Enriched gene ontology of positively selected genes in the wild duck lineage.

See file “Table_S14.xlsx”

Table S15: Enriched gene ontology in biological process terms of positively selected genes in wild duck lineage.

See file “Table_S15.xlsx”

Table S16: Mean values of population genomic parameters for Z-chromosome and autosomes in mallard and spot-billed

Index		Autosomes		Z-chromosome	
		mean	SD	Mean	SD
Pi	Mallard	0.007520	0.004199	0.003397	0.002495
	Spot-billed	0.007607	0.004321	0.003020	0.002625
SNPs	Mallard	1,123	572	494	348
	Spot-billed	1,150	600	440	376
Tajima's D	Mallard	-0.525593	0.435471	-0.257150	0.445641
	Spot-billed	-0.625759	0.420070	-0.348016	0.492465
F_{ST}		0.049303	0.034864	0.259623	0.150430

Table S17: Mean values of genomic parameters inside/outside high F_{ST} (between mallard and spot-billed ducks) regions in autosomes

Species		pi (π)		No. of SNPs		Tajima's D	
		mean	sd	mean	sd	mean	sd
Mallard	Inside	0.002421	0.00217898	360	305	-0.61100	0.579419
	Outside	0.007595	0.00418171	1,134	569	-0.52064	0.433172
Spot-billed	Inside	0.002331	0.00196792	334	280	-0.47211	0.656325
	Outside	0.007683	0.00430549	1,161	597	-0.6228	0.415501

Table S18: Summary of domestic candidate regions ($zHp < -4$ for autosomes, $zHp < -2$ for Z-chromosomes)

See file “Table_S18.xlsx”

Table S19: Summary of high F_{ST} regions between the wild and domestic duck lineage.

See file “Table_S19.xlsx”

Table S20: Enriched gene ontology of positively selected genes in/close to domestic candidate regions.

See file “Table_S20.xlsx”

Table S21: Enriched gene ontology in biological process terms of positively selected genes in/close to domestic candidate regions.

See file “Table_S21.xlsx”

Table S22: Information of variant sites being performed luciferase assay to evaluate the activity.

Gene symbol	Variant position	Catalog of variant position	ΔAF	AF in wild	AF in domestic
<i>EFNA5</i>	KB744645.1:114292,114293,114301	10 kb*	0.949,0.916,0.740	0,0.916,0	0.949,0,0.741
<i>EFNA5</i>	KB744645.1:58363	1 th intron	0.809	0.835	0.026
<i>ISL1</i>	KB743873.1:25803 (indel)	3' UTR	0.875	0.875	0
<i>MRPS27</i>	KB742554.1:2151394,2151434,2151445	3' UTR	0.960,0.419,0.417	0,0.419,0.4	0.960,0.419,0.4
<i>GRIK1</i>	KB742951.1:543269	2 th intron	0.730	0.151	0.881
<i>LYST</i>	KB743543.1:403502	5 th intron	0.611	0.685	0.074

Note: 10 kb* means 10 kb downstream of the gene.

Table S23. Averaged all ΔDAF (absolute ΔDAF) of windows inside/outside high differentiation regions.

Mean value across SNPs	high differentiation regions	Genomic background
absolute ΔDAF	0.0385672	0.03672404
ΔDAF	0.0370516	0.03497989

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